The Cost of Reliable Near-Optimal Set Maintenance with Arriving Arms

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Abstract

We study reliable maintenance of all relatively near-optimal arms while collecting rewards when new arms arrive exogenously and remain available. Reward laws are stationary, but arrivals change the membership boundary. Reporting progress is measured by cumulative unresolved arm time outside a prescribed boundary band, with simultaneous correctness of all labels. One uncertain arrival can obstruct many earlier certifications. A common-reference lower bound quantifies this effect, and matching worst-schedule area bounds yield an arrival-growth boundary for joint sublinear performance. On specified repeated-update families, temporary membership tasks impose rapid-maintenance regret costs, while separate constructions attain deferred and intermediate service levels. We develop Flexible Certificate-Cost Planning (F-CCP) as a tunable implementation. It combines sequential evidence and local certificate planning with a chosen reward kernel and separate observable budgets for reward and certification. Its guarantees include label reliability, unconditional expected-regret protection, and service-rate area bounds that retain budget rejections. Fixed-checkpoint and activation-budgeted constructions remain distinct analytical witnesses. Frozen paired experiments show lower mean costs in many finite-budget comparisons, together with exceptions, heavier regret tails, and nonzero reporting failures. These results do not characterize a general regret–maintenance frontier.

Keywords: sequential experimentation; arriving arms; near-optimal set maintenance; dynamic regret; confidence sequences.

1 Introduction

Sequential experiments often acquire new candidates while earlier ones remain available. An allocation rule must learn which candidates yield high rewards. An experimenter may also need to know which of all available candidates are within a specified tolerance of the current best. These

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objectives need not favor the same observations. A candidate can be sufficiently inferior to receive little reward-driven exploration and still be difficult to classify relative to the near-optimality boundary.

We study these objectives together under permanent exogenous arrivals and stationary reward laws. If $r^*(t)$ is the largest mean among the arms available at time t, an arm is relatively near-optimal when its mean is at least $r^*(t) - \varepsilon$. Reports certify membership, certify exclusion, or leave the arm unresolved. We require one simultaneous correctness event for all labels at all times. To rule out indefinite abstention, we charge the cumulative unresolved time of arms whose means lie more than η from the membership boundary. The question is how much regret is necessary, and how much is attainable, under this reporting requirement.

New arrivals affect the task in two ways. An uncertain new candidate may be better than the incumbent by enough to invalidate many existing positive reports. Evidence about that one candidate is then relevant to many reports at once. A later improvement can also make an earlier difficult membership question unnecessary: a candidate close to the old boundary may be far below the new one. Its existing upper confidence bound can be combined with a new reference's lower bound to certify exclusion. Neither effect requires the reward law of an old arm to change.

Our observation model and regret comparator have established precedents. Permanent arrivals form a monotone special case of sleeping-bandit availability (Kleinberg et al., 2010); ballooning bandits already compare reward with the best arm currently available (Ghalme et al., 2021). Exogenous arrivals differ from drawing arms on demand from a reservoir (Berry et al., 1997) and from the disappearing actions of mortal bandits (Chakrabarti et al., 2008). Benchmark improvements here arise from new arms, whereas nonstationary bandits allow the reward distributions of existing arms to change (Besbes et al., 2014; Auer et al., 2019). Sequential allocation and adaptive experimentation have a long statistical history (Robbins, 1952; Lai and Robbins, 1985; Lai, 1987); the expanding set of actions retains that allocation problem and adds an evolving inferential target.

For a fixed arm set, identifying every relatively near-optimal arm is the ALL-ε problem (Mason et al., 2020). Joint reward and identification objectives also have established precedents (Degenne et al., 2019; Yang et al., 2026; Cho et al., 2025). Our task requires continuing reports under arrivals, with all-time label reliability and an explicit cumulative cost for unresolved arm time. The distinction concerns the reporting protocol and its cost, rather than a new arriving-arm model. The simultaneous recovery scales used in our lower bounds build on existing information inequalities and multiple-recovery arguments (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018).

We give three results about maintenance cost. First, a common-reference lower bound shows how one uncertain arrival can obstruct many other certifications. Its information queue accumulates cost over time within one reward environment. Second, we obtain matching worst-schedule orders for cumulative unresolved area, including finite-horizon truncation, and derive the polynomial arrival boundary for jointly sublinear area and expected regret. This worst-case lower bound already occurs in an initial static batch; the common-reference result describes a separate consequence of continuing arrivals. Third, on a specified repeated-update family, rapid resolution of temporary

membership tasks has matching inverse-square regret costs. Allowing a constant-factor larger area budget admits a logarithmic regret upper bound. A nonvanishing-probability lower bound holds in one fixed environment under the same cumulative-area requirement. The deferred endpoint has only an upper bound. A separate activation-budgeted witness covers an intermediate interval on a more strongly separated family; the general causal frontier under overlapping arrivals remains open.

Fixed checkpoints make attainable costs explicit, but their quotas can consume a moderate budget before useful reports appear. F-CCP uses every observation to update valid reporting evidence and plans one certificate observation at a time. A predictable priority controls the proposed service rate. Separate budgets limit the virtual-index cost of reward choices and of certification, so exploration by the reward kernel does not exhaust the certification account. The resulting guarantees concern reliability, regret protection, and conditional service bounds; the local planning proxy is not a solution of the full statistical allocation problem. The fixed-checkpoint witnesses are retained for their original guarantees in the supplement.

The components have precedents. Betting processes provide time-uniform inference for bounded means (Waudby-Smith and Ramdas, 2024); our particular mixture and its bounded-reward validity are proved explicitly. Thompson sampling supplies a flexible reward rule (Thompson, 1933). MOTS obtains fixed-arm minimax guarantees by modifying sampled scores (Jin et al., 2021). Our protection instead constrains the virtual-index cost of the actual selected action under arrivals, and does not transfer MOTS's theorem to a cold-temperature Beta rule. Conservative bandits control performance against a fixed baseline (Wu et al., 2016); the two accounts here control index deficits and are not realized shortfalls from a separately run baseline. The spectral accounting and MC-MOSS concentration tools are inputs from the companion work (Temporal Geometry), not a new characterization claimed here.

Information about future task expiration also matters. A schedule-specific comparator can defer a temporary question in a way that one causal policy facing an unknown schedule cannot reproduce uniformly. We retain that information distinction and an activation-budgeted intermediate-regime result alongside the general maintenance bounds. The latter uses its own window conditions and is not an optimality claim for F-CCP.

Section 2 fixes the joint task and separates task parameters from allocation preferences. Section 3 presents F-CCP and its guarantees. Sections 4 and 5 give the general costs, temporary-task bounds, and information limits. Section 6 evaluates frozen configurations, including mean improvements, tail costs, failed reports, and the corrected same-controller comparison.

2 Model, reliable reports, and maintenance cost

2.1 Observation model and regret

Let $\mathcal{A}$ be a possibly infinite set of arms. Arm i arrives at time τ_i, after which it remains available. At every physical time $t = 1, 2, \ldots,$

$$\mathcal{A}_t = \{i : \tau_i \le t\}, \qquad 1 \le K_t = |\mathcal{A}_t| < \infty.$$

Arbitrary finite batches are allowed, including $K_T > T$ at an evaluation endpoint. Each arm has an unknown stationary reward law on $[0,1]$ with mean r_i. Its successive potential observations $(Y_{i,n})_{n\geq 1}$ are independent and identically distributed, and the streams of different arms are independent. Arrivals are exogenous and independent of both these streams and policy randomization. Unless stated otherwise, each result is for a fixed deterministic arrival schedule.

At time t, the policy first receives all current arrivals, then selects one arm $I_t \in \mathcal{A}_t$, observes one reward X_{t,I_t}, and finally issues its report. It uses the observed history

$$H_{t-1} = (\mathcal{A}_s, I_s, X_{s,I_s})_{s=1}^{t-1}$$

and independent randomization. Means, future arrivals, and true membership labels are unavailable to the policy. With $N_i(t) = \sum_{s\leq t} \mathbf{1}\{I_s = i\}$, the physical observation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. Write

$$\hat{r}_i(n) = \frac{1}{n}\sum_{s=1}^{n} Y_{i,s}, \qquad r^*(t) = \max_{i\in\mathcal{A}_t} r_i, \qquad \Delta_i(t) = r^*(t) - r_i \quad (i \in \mathcal{A}_t).$$

At a deterministic endpoint T, cumulative dynamic regret is

$$R_T(\pi) = \sum_{t=1}^{T} \Delta_{I_t}(t).$$

This is a sum of mean gaps, random through the policy's arm selections; $\mathbb{E}R_T$ is its expectation. The quantity does not compare realized reward sequences. Physical time always counts actual observations.

2.2 Reliable reports and cumulative unresolved time

For $0 < \varepsilon < 1$, the target set is

$$\mathcal{G}_t^{\varepsilon} = \{i \in \mathcal{A}_t : \Delta_i(t) \leq \varepsilon\}.$$

After the current observation, a report partitions all of $\mathcal{A}_t$ into $\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t$: certified members, certified nonmembers, and unresolved arms. Define

$$E_{\text{lab}} = \{\text{for all } t:\ \mathcal{P}_t \subseteq \mathcal{G}_t^{\varepsilon} \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}.$$

A policy is δ-reliable if $\mathbb{P}(E_{\text{lab}}) \geq 1-\delta$ in every allowed reward environment. A positive report may be withdrawn after a new arrival. On the correctness event, a negative report remains valid because $r^*(t)$ is nondecreasing and the arm's mean is stationary.

Reliability permits all arms to remain unresolved. For a boundary width $0 < \eta < 1$, define

$$\begin{aligned} d_i^{\varepsilon}(t) &= |\Delta_i(t) - \varepsilon|, \\ U_t(\eta) &= \sum_{i\in\mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, d_i^{\varepsilon}(t) > \eta\}, \qquad \mathcal{U}_T(\eta) = \sum_{t=1}^{T} U_t(\eta). \end{aligned} \tag{1}$$

The count includes newly arrived arms with no observations and old arms that become unresolved again. Every such arm–time pair is charged; the loss is not limited to the time before a first decision. Within the boundary band, unresolved time is exempt, but any label issued there must still be correct. The true distance $d_i^{\varepsilon}(t)$ is used only by the evaluator and in the analysis.

Task parameters are fixed within a comparison. Allocation preferences change the strategy, rather than the reporting target. The parameters used below have the following roles.

Parameter	Role
$\varepsilon, \eta \in (0,1)$	Membership tolerance and exempt boundary width
$\delta \in (0, 1/2)$	Simultaneous reporting error budget
$\lambda_t \in [0,1]$	Predictable preference for certification
$p_0 \in (0,1),\ k_0 > 0,\ \nu > 0$	Service floor, backlog scale, and planning price floor
$\mathsf{B},\ \theta > 0$	Reward-score kernel and its temperature
$\beta_R, \beta_C \in [0, \infty]$	Fixed reward and certification guard coefficients
ρ	Auxiliary checkpoint precision; F-CCP uses $\eta/4$

The fixed-checkpoint witnesses may prescribe other precisions before observations. Preference switching does not change $\varepsilon, \eta, \delta$, create observations, or reset accumulated evidence or budget accounts.

The main requirement at a fixed T is

$$\mathbb{E}_\vartheta R_T \le r, \qquad \mathbb{P}_\vartheta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le a\} \ge 1 - \delta. \tag{2}$$

It combines an expected regret bound with a joint high-probability label–area guarantee. In particular, $\mathcal{U}_T = o(T)$ requires the time-average number of unresolved separated arms to vanish, rather than only a small unresolved fraction of all arms. The area guarantee is for each deterministic endpoint; it need not hold for every endpoint on one common area event.

For a performance family Θ, let $\mathfrak{R}_\Theta(a)$ be the infimum of $\sup_{\vartheta \in \Theta} \mathbb{E}_\vartheta R_T$ over policies reliable on the full bounded model and satisfying the probability condition in (2) for every $\vartheta \in \Theta$. The value is $+\infty$ if the feasible class is empty. Full-model label reliability and a family-specific area guarantee are distinct requirements. The lower bounds remain valid when reliability is required only on the lower-bound subfamily.

2.3 How arrivals change the reporting problem

For a fixed old arm, $\Delta_i(t)$ is nondecreasing, so its true membership can change only from member to nonmember. Its boundary distance $|\Delta_i(t) - \varepsilon|$ need not be monotone. A new candidate can first create uncertainty about the current best mean, obstructing positive certification of many old arms even when those arms remain truly near-optimal. A sufficiently good new reference can later move the boundary past earlier candidates, making their fine classification unnecessary. Sections 4 and 5 quantify these two effects under the same reporting protocol.

Arms with close means need not be close to the relative boundary $r^*(t) - \varepsilon$. Conversely, a candidate near that boundary can require fine evidence even when its reward gap is appreciable.

The algorithm's observable work set is therefore defined from confidence intervals, not from the evaluator's true gaps or unresolved area.

3 Flexible certificate-cost planning

The reporting task need not receive a fixed fraction of every sampling budget. Evidence obtained while collecting rewards can already certify some arms, and a small amount of additional work can complete other reports. Flexible Certificate-Cost Planning (F-CCP) uses one set of lifetime statistics for both purposes. Its service preference controls how often certification is attempted; two separate budgets protect the reward allocation and reserve capacity for certification. The resulting guarantees concern reliable labels, regret order, and service completion. They do not identify the optimal allocation for every joint regret–area constraint.

3.1 Evidence and current reporting work

Order arms by arrival, with a fixed order within each batch. Write $n_i = N_i(t-1)$ and Σ_i for the actual count and reward sum available before the action at time t. Initially $n_i = \Sigma_i = 0$ and $[\ell_i, u_i] = [0, 1]$. For $\alpha_i = \delta/\{2i(i+1)\}$, invert the beta-mixture process

$$\mathsf{M}_{i,n}(m) = \frac{\mathrm{B}(\Sigma_{i,n} + 1/2, n - \Sigma_{i,n} + 1/2)}{\mathrm{B}(1/2, 1/2) m^{\Sigma_{i,n}} (1-m)^{n-\Sigma_{i,n}}}, \qquad m \in [0, 1], \tag{3}$$

where B is the beta function, distinguished from the optimal mean $r^*(t)$. The interval is the intersection of $\{m : \mathsf{M}_{i,n}(m) \le 1/\alpha_i\}$ over all observed positive prefixes. At the actual count

$$\rho = \eta/4, \qquad q_i = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil, \tag{4}$$

intersect it also with $[\hat{r}_i(q_i) - \rho/2, \hat{r}_i(q_i) + \rho/2] \cap [0, 1]$. Continue the mixture updates after this checkpoint. The construction is valid for bounded rewards, including nonbinary rewards; its supermartingale argument is given in Supplement S2. Sequential betting and mixture confidence sequences are established tools (Waudby-Smith and Ramdas, 2024).

For a nonempty confidence rectangle define

$$g_i^- = \max\{0, \max_{j \ne i} \ell_j - u_i\}, \qquad g_i^+ = \max\{0, \max_{j \ne i} u_j - \ell_i\}. \tag{5}$$

Both bounds are zero when there is only one arm. They are the exact minimum and maximum of the gap over this rectangle, including its self-exclusion constraint. After observing a reward, report

$$\mathcal{P}_t = \{i : g_i^+ \le \varepsilon\}, \quad \mathcal{N}_t = \{i : g_i^- > \varepsilon\}, \quad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t). \tag{6}$$

A negative report does not remove an arm from future reward allocation. Among unresolved arms let

$$Z_t = \{i \in \mathcal{V}_t : [g_i^-, g_i^+] \subseteq [\varepsilon - \eta, \varepsilon + \eta]\}, \qquad W_t = \mathcal{V}_t \setminus Z_t. \tag{7}$$

These formulas are also evaluated before the action, after the current arrivals, when choosing that round's service. An arm in Z_t remains reported as unresolved; the rectangle already shows that it incurs no η-unresolved cost. Recompute the sets after every arrival and observation. An empty interval sets a persistent failure flag: all subsequent reports are unresolved and designated service stops. Reward allocation, its guard, and the actual statistics continue.

3.2 A local certificate plan

The plan assigns observations a positive price and asks which current pairwise certificate is cheapest to pursue. Its centers and radii are

$$m_i = \Pi_{[\ell_i,u_i]}\frac{\Sigma_i + 1/2}{n_i + 1}, \quad v_i^{\text{plan}} = \max(m_i - \ell_i, u_i - m_i), \quad \bar{n}_i = \max(1, n_i). \tag{8}$$

Here $\bar{n}_i$ is a planning convention; no fictitious observation enters the confidence interval. For each $i \in W_t$, take as its reference j the arm of largest m_j if $\max_k m_k - m_i > \varepsilon$; otherwise take the largest-upper-bound arm other than i. Resolve reference ties by arrival order. Set

$$h_i = .95\max\{|m_j - m_i - \varepsilon|, \eta - |m_j - m_i - \varepsilon|\}, \quad c_k = \nu + (1 - \lambda_t)\zeta_t(k), \quad a_k = v_k^{\text{plan}}\sqrt{\bar{n}_k}, \tag{9}$$

where $\nu > 0$ and the observable index gap ζ_t is defined below. Solve the two-coordinate proxy problem

$$\min_{x_i \geq \bar{n}_i,\, x_j \geq \bar{n}_j} c_i(x_i - \bar{n}_i) + c_j(x_j - \bar{n}_j) \quad \text{subject to} \quad a_i/\sqrt{x_i} + a_j/\sqrt{x_j} \leq h_i. \tag{10}$$

The interior solution is explicit; boundary and zero-radius cases appear in Supplement S2. Among pairs with an available endpoint, use the pair of smallest proxy cost. Within that pair, select the endpoint maximizing $v_k^{\text{plan}}/(\bar{n}_k c_k)$ among those with $n_k < q_k$ and $x_k > n_k$ (using the actual count). Execute only one pull, then replan. If no pair provides an endpoint, select the widest interval among arms with $n_k < q_k$, with arrival order breaking ties. Call this subroutine Plan_t, and its proposed arm j_t.

The optimization in (10) is exact for its continuous proxy. It holds the centers fixed and approximates radius reduction by an inverse square root. Other competitors can remain after one pair is resolved. Thus its solution is a scheduling rule, not an information complexity or a proof of optimal statistical allocation.

3.3 Virtual indices and two budgets

The reward guard uses a virtual, clipped MC-MOSS index. Its arrival clock and spectral comparison come from the companion study (Temporal Geometry). Initialize base $= 0$. After arrivals, if $K_t > 1$ and either base $= 0$ or $K_t - 1 \geq 2\,\text{base}$, set base $= K_t - 1$, $\sigma = t$, and $d_{\text{clock}} = 2\,\text{base}$. Retain all

observations. With $s = t - \sigma + 1$, define

$$J_i(t) = \begin{cases} 1, & n_i = 0, \\ \min\{1, \hat{r}_i(n_i) + \sqrt{[\log\{s/(d_{\text{clock}} n_i)\}]_+ / n_i}\}, & n_i > 0, \end{cases} \qquad \zeta_t(i) = \max_j J_j(t) - J_i(t). \quad (11)$$

When $K_t = 1$, set $J_i = 1$ and $\zeta_t(i) = 0$ without using an uninitialized clock. The virtual index never forces an observation by itself. In particular, ζ_t is an observable index difference, not realized regret or a difference between two independent runs.

Choose the first initial arm as a counting anchor. Among all other arrived arms, let $k_{\ell,t}$ count current ages $t - \tau_i + 1$ in group 0 (age one), or group $\ell \geq 1$ (ages in $(2^{\ell-1}, 2^\ell]$). The current spectrum and its running maximum are

$$\mathcal{S}_t = \sum_{\ell \geq 0} \min\{2^\ell, \sqrt{k_{\ell,t} 2^\ell}\}, \qquad \overline{\mathcal{S}}_t = \max_{u \leq t} \mathcal{S}_u. \quad (12)$$

The anchor's identity supplies no information about its mean or optimality. Computing (12) uses only arrived arms' current ages, with no endpoint or future-arrival input.

Fix $\beta_R, \beta_C \in [0, \infty]$ before the run. Maintain separate ledgers, initially zero,

$$C_R(t) = \sum_{u \leq t:\, \text{reward}} \zeta_u(I_u), \qquad C_C(t) = \sum_{u \leq t:\, \text{service}} \zeta_u(I_u). \quad (13)$$

Accept a proposed service only if $C_C(t-1) + \zeta_t(j_t) \leq \beta_C \overline{\mathcal{S}}_t$. Otherwise use a reward round and record a denial. On any reward round maximize the selected reward kernel's scores over

$$\mathcal{F}_R(t) = \{i : C_R(t-1) + \zeta_t(i) \leq \beta_R \overline{\mathcal{S}}_t\}. \quad (14)$$

A largest-J_i arm has zero index gap, so this set is nonempty. An infinite budget disables the corresponding check, including when the spectrum is zero; it is implemented as `None`, rather than as budget zero. No ledger is reset at a clock change or preference change.

For example, the default reward scores are one draw per arm,

$$\Xi_i(t) \sim \text{Beta}(1 + \Sigma_i/\theta,\ 1 + (n_i - \Sigma_i)/\theta). \quad (15)$$

For Bernoulli rewards, $\theta = 1$ gives standard Beta$(1,1)$ Thompson sampling (Thompson, 1933); $\theta = .5$ is a colder action rule. For general bounded rewards these fractional-count scores are a sampling heuristic. The evidence always uses the original counts and reward sums. On a reward round draw the full score vector once, then maximize that same vector over (14); do not redraw following rejection. UCB1, Growing-MOSS, MC-MOSS, and KL-UCB can also supply reward scores.

Table 1: F-CCP presets. The default reward kernel is Beta sampling. An inactive service budget is never checked because service is off. The guarantees refer to the conditions stated in this section.

Preset (data ID)	λ/control	θ	β_R	β_C
Reward (F-R)	0/off	.5	16	inactive
Balanced (F-auto-0.6)	.6/adaptive	.5	16	1
Classification (F-area)	1/fixed	.5	16	∞
Compatibility (F-R-TS-exact)	0/off	1	∞	inactive

3.4 Service preference and executable rule

Let $\lambda_t \in [0,1]$ be a predictable service preference. Given a proposed service j_t, write $b_t = |W_t|$ and $\widehat{c}_t = \max_i m_i - m_{j_t}$. The adaptive controller uses

$$w_t = p_0 + (1-p_0)\frac{b_t}{b_t + k_0}\frac{\varepsilon}{\varepsilon + \widehat{c}_t}, \qquad p_t = \begin{cases} 0, & b_t = 0 \text{ or service is disabled,} \\ \dfrac{\lambda_t w_t}{1 - \lambda_t + \lambda_t w_t}, & \text{otherwise.} \end{cases} \tag{16}$$

Fixed-rate control instead uses $p_t = \lambda_t \mathbf{1}\{b_t > 0\}$. Service is disabled when the failure flag is set or $\lambda_t = 0$. The presets use $p_0 = .02$, $k_0 = 4$, and $\nu = .1$. Few or costly proposed certificates tend to reduce the service rate; a large, inexpensive backlog tends to increase it. The controller sees W_t, not the true unresolved area. Neither λ_t nor w_t is an estimated optimal multiplier or a forecast of future arrivals.

Use a phase $\phi_0 = 0$: add p_t each round, and if the result is at least one, attempt one service and subtract one. A denial does not refund this phase. In exact arithmetic the number of attempts through T is $\lfloor \sum_{t \le T} p_t \rfloor$. For fixed $\lambda > 0$, on busy rounds,

$$p_t \ge p_{\min}(\lambda) := \frac{\lambda p_0}{1 - \lambda + \lambda p_0} > 0. \tag{17}$$

There is no positive service lower bound at $\lambda = 0$. Table 1 separates the four presets. Changing only `set_priority(1)` does not disable an existing certification budget. Preference changes occur between completed physical rounds and retain the phase, evidence, counts, and both ledgers.

Within a planned pair, equal endpoint scores favor the candidate i; other ties use the fixed arrival-order scans. The procedure sees no means, hidden identities, future arrivals, or endpoint. Stopping at an arbitrary endpoint leaves the last completed report unchanged. Its straightforward implementation takes $O(K_t^2)$ work per round when pair planning or the self-excluding gap bounds are evaluated by nested scans, and $O(K_t)$ stored arm statistics. Numerical root finding and floating comparisons are described separately in the supplement.

Algorithm 1 Flexible certificate-cost planning (F-CCP)

Require: Fixed $\varepsilon, \eta, \delta$, budgets β_R, β_C, reward kernel, temperature, controller parameters; predictable λ_t.

Initialize empty arm statistics, intervals, and quotas; base $= 0$, $\phi = C_R = C_C = \overline{\mathcal{S}} = 0$; failure $=$ false.
for each physical time $t = 1, 2, \ldots$ **do**
  Admit the current batch in arrival order with $n_i = \Sigma_i = 0$, $[\ell_i, u_i] = [0, 1]$, birth t, and quota (4).
  Update the arrival clock and calculate $J_i(t), \zeta_t(i)$ using (11) (use the singleton convention).
  Update $\overline{\mathcal{S}} \leftarrow \max\{\overline{\mathcal{S}}, \mathcal{S}_t\}$ using current ages.
  Set attempt $=$ false, service $=$ false, $p_t = 0$.
  if not failed and $\lambda_t > 0$ **then**
    Recompute the pre-action reports and W_t from current intervals.
    if $W_t \neq \varnothing$ **then**
      $j_t \leftarrow \mathrm{Plan}_t$ using (8)–(10); calculate p_t.
      $\phi \leftarrow \phi + p_t$.
      if $\phi \geq 1$ **then**
        $\phi \leftarrow \phi - 1$; attempt $=$ true.
      end if
    end if
  end if
  if attempt and a legal j_t exists **then**
    if the service guard is off or $C_C + \zeta_t(j_t) \leq \beta_C \overline{\mathcal{S}}$ **then**
      $I_t \leftarrow j_t$; service $=$ true; $C_C \leftarrow C_C + \zeta_t(I_t)$.
    else
      Record one service denial; retain the decremented phase.
    end if
  end if
  if not service **then**
    Generate the reward score vector once from the current actual history.
    $I_t \leftarrow$ first maximizer of these scores over $\mathcal{F}_R(t)$ (all $\mathcal{A}_t$ when that guard is off).
    $C_R \leftarrow C_R + \zeta_t(I_t)$.
  end if
  Pull I_t once; observe X_{t,I_t}; $n_{I_t} \leftarrow n_{I_t} + 1$, $\Sigma_{I_t} \leftarrow \Sigma_{I_t} + X_{t,I_t}$.
  if not failed **then**
    Intersect the selected arm's interval with its new mixture interval; at $n_{I_t} = q_{I_t}$ also intersect the fixed-prefix interval.
    If any interval is empty, set failure $=$ true permanently.
  end if
  Report (6), or $(\varnothing, \varnothing, \mathcal{A}_t)$ if failed.
end for

3.5 Reliability, reward protection, and service completion

Proposition 3.1 (Reliable reports)**.** *Under the bounded stationary model, every legal predictable choice of preferences, prices, and service actions in F-CCP satisfies* $\mathbb{P}(E_{\mathrm{lab}}) \geq 1-\delta$*. On a common event* $\mathcal{E}$ *of this probability, all confidence intervals cover their means at all observed counts, all reports are correct, and every arm in* Z_t *lies in the boundary band.*

The evidence argument does not require the proxy plan to predict its actual sampling cost. The mixture and checkpoint errors each receive $\delta/2$, and a union bound gives $\mathcal{E}$.

Proposition 3.2 (Compatibility endpoint)**.** *Fix a legal reward kernel* B*. If service is off from the start* ($\lambda_t \equiv 0$)*,* $\beta_R = \infty$*, and the complete kernel state, initialization, and random-number source agree with those of* B*, the two procedures have identical actions, lifetime counts, and regret on every potential reward realization and arrival schedule.*

This is a same-backend pathwise statement. A common seed in different random-number backends does not provide the required coupling. The colder, guarded F-R preset is distinct from this endpoint. Switching off service after using it does not recreate a baseline's independent past.

The regret comparison uses an arbitrary-action inequality,

$$\mathbb{E}R_T \leq \min\left\{T, 300\mathcal{S}_T + \mathbb{E}\sum_{t\leq T}\zeta_t(I_t)\right\}. \tag{18}$$

Supplement S3 proves the bridge from the companion MC-MOSS potential-prefix and arrival-clock bounds to arbitrary actions; the potential-prefix bounds apply to the actual sequence of selected actions.

Theorem 3.3 (Two-budget reward protection)**.** *Fix finite nonnegative* β_R, β_C *and use legal pre-observation actions. For every deterministic schedule and endpoint,*

$$\mathbb{E}R_T \leq \min\{T, (300 + 2\beta_R + 2\beta_C)\mathcal{S}_T\}. \tag{19}$$

If service is off throughout the run, omit $2\beta_C$ *even when its guard is disabled. If* $\beta_R < \infty$ *and* $\beta_C = \infty$*, then, with* $Q_T^{\mathrm{cap}} = \sum_{i\in\mathcal{A}_T} q_i$*,*

$$\mathbb{E}R_T \leq \min\{T, (300 + 2\beta_R)\mathcal{S}_T + Q_T^{\mathrm{cap}}\}. \tag{20}$$

These bounds are unconditional, including reporting failures.

The ledgers satisfy their respective running-spectrum budgets, and $\overline{\mathcal{S}}_T \leq 2\mathcal{S}_T$. Substitution in (18) proves the finite-budget bound. Unlimited certification costs at most Q_T^{cap} actual service pulls. Thus F-R has the conservative bound $332\mathcal{S}_T$, Balanced has $334\mathcal{S}_T$, and Classification retains the extra Q_T^{cap} term. The companion spectral minimax lower bound makes the finite-budget regret order sharp, for fixed coefficients; the spectral characterization itself is an existing input (Temporal Geometry). None of these bounds compares F-CCP's realized regret with an independently run Thompson sampler.

Theorem 3.4 (Area under a positive service rate)**.** *Suppose every pre-action busy round has $p_t \geq p_{\min} > 0$. Let D_T^{srv} be the number of service attempts rejected by the certification budget. On $\mathcal{E}$, simultaneously at every endpoint,*

$$\mathcal{U}_T(\eta) \leq K_T \min\left\{T, \frac{Q_T^{\mathrm{cap}} + D_T^{\mathrm{srv}} + 1}{p_{\min}}\right\}. \tag{21}$$

When $D_T^{\mathrm{srv}} = 0$, this is $O(\min\{K_T T, K_T^2 \eta^{-2} \log(K_T/\delta)/p_{\min}\})$.

Here busy means that the current W_t is nonempty after arrivals and before observation. If every current arm has reached its checkpoint, the gap intervals are narrow enough that W_t is empty. Each successful service therefore advances a finite remaining checkpoint, and phase accounting bounds the number of busy rounds. This proves the theorem without requiring a cost-optimal plan. Classification has $p_{\min} = 1$ and no refusals, hence the worst-area order matches the general lower bound in its common parameter range. Finite-budget Balanced must retain the denial term; it has no unconditional prescribed-area guarantee. An arbitrary predictable λ_t preserves label reliability, but completion speed needs the displayed positive lower bound.

Corollary 3.5 (Finite-arm completion)**.** *Suppose $\beta_C > 0$ is fixed, busy rounds have a uniform positive service rate, and arrivals eventually stop with finitely many arms. On $\mathcal{E}$, every arm outside the boundary band eventually receives a definite label. This conclusion holds regardless of continuing reward-kernel exploration and its separate ledger consumption.*

A conservative finite-prefix bound, the compatibility proof, and a one-round obstruction to exact all-environment dominance appear in Supplement S4. Completion on a fixed-confidence event concerns the boundary band; it is not unconditional almost-sure consistency for the exact oracle set.

4 Fundamental costs of continual maintenance

4.1 One uncertain arrival can obstruct many reports

The first lower bound retains one reward environment throughout the time axis. It therefore measures a cumulative burden without adding the worst cases of incompatible experiments. Fix $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, and $0 < \eta < \varepsilon/2$. An initial reference has Bernoulli mean $1/4 + \varepsilon/2$. Candidates arrive on any prescribed schedule. In environment zero each has mean $p = 1/4$; environment j changes only candidate j to mean $p + 2\varepsilon$. All candidates are good in environment zero, with boundary distance $\varepsilon/2$. After candidate j arrives in environment j, every other candidate is bad.

Let M_t count arrived candidates and $a_t = M_t - M_{t-1}$. Define

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \qquad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$
$$\mathcal{B}_T = \sum_{t \leq T} (M_t - 1)_+ \mathbf{1}\{Q_t > 0\}. \tag{22}$$

This analytical queue is an information-capacity certificate: arrivals add q_- units per candidate, while only one observation is possible per round. It is distinct from F-CCP's phased service rule and from the fixed-checkpoint witness's FIFO queue. It imposes no pathwise sampling quota on every candidate. In the stated range $q_- \geq 2$.

Theorem 4.1 (Broadcast uncertainty)**.** *For every policy reliable on environment zero and all its one-candidate alternatives,*

$$\mathbb{E}_0 \mathcal{U}_T(\eta) \geq (3/4 - \delta)\mathcal{B}_T.$$

If $\mathcal{B}_T > 0$, then

$$\mathbb{P}_0\{\mathcal{U}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4.$$

These conclusions still hold when the policy knows the full arrival schedule, the horizon, and the reference distribution.

At a busy time, some deterministic candidate label has expected sample count below q_-. Changing that candidate alone has too little history KL to permit positive certification of another candidate with probability at least $1/4$. Reliability also bounds incorrect negative reports. The remaining probability must be unresolved. The same reference environment supports this argument at every time; the difficult label is selected from expected counts rather than from a favorable realized history. The supplementary proof includes the bounded-variable step needed for the probability statement.

If m candidates have arrived by time s, the queue obeys the deterministic inequality

$$\sum_{t \leq s+qm-1} (M_t - 1)_+ \mathbf{1}\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1), \qquad q \geq 2. \tag{23}$$

The factor m^2 reflects the number of affected reports as well as the work needed to exclude high-quality alternatives. The subtraction of one unit in $q - 1$ accounts for the last observation of each busy period: reports are issued after service.

4.2 Worst-schedule maintenance area

Let $\mathfrak{A}_{T,K}$ be the smallest a attainable by a full-model reliable policy with

$$\sup_{\boldsymbol{\tau}: K_T \leq K} \sup_{\boldsymbol{r}} \mathbb{P}_{\boldsymbol{r}}\{E_{\text{lab}}^c \text{ or } \mathcal{U}_T(\eta) > a\} \leq \delta.$$

This optimization imposes no regret budget. Policies may know T, K and the schedule for the lower bound. The upper bound uses neither T nor K as input.

Theorem 4.2 (Worst-schedule area and joint attainability)**.** *For integers $K \geq 2$ and $T \geq 1$, $0 < \varepsilon \leq 1/8$, $0 < \eta \leq \varepsilon/32$, and $0 < \delta \leq 1/16$,*

$$2^{-24} \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\} \leq \mathfrak{A}_{T,K} \leq 256 \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\}. \tag{24}$$

Checkpoint-UCB with $\rho = \eta/4$ attains the upper order.

The same Checkpoint-UCB witness also satisfies

$$\mathbb{E}R_T \le \min\left\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\right\}. \tag{25}$$

For the upper bound, completed checkpoints imply endpoint errors at most $\eta/4$, sufficient to classify all arms outside the boundary band. The unfinished-work queue is dominated by adding q_i at each arrival and serving at even rounds. On the simultaneous coverage event,

$$\mathcal{U}_T(\eta) \le \min\left\{K_T T, 2K_T \sum_{i \le K_T} q_i\right\}. \tag{26}$$

Additional UCB observations reduce the work; they are not charged again as new checkpoint samples. The lower bound uses $K-1$ candidates with independent means $1/2 \pm 2\eta$ and a reference of mean $1/2 + \varepsilon$, all available at the first pull. Simultaneous partial recovery and a horizon truncation give the probability statement required by the definition of $\mathfrak{A}_{T,K}$. The statistical $\log K$ factor has fixed-arm multiple-testing precedents; this worst-schedule lower construction itself requires no repeated arrivals.

The classification-priority mode of F-CCP, with $\lambda = 1$ and no certification guard, attains the same worst-area order by Theorem 3.4. Its constants and reward bound are stated separately. The explicit constants in (24) and (25) continue to refer to the fixed-checkpoint witness in Supplementary Section S5.

4.3 Growth rates and the limits of total-count criteria

Corollary 4.3 (Arrival-growth boundary)**.** *Fix the task parameters of Theorem 4.1 and one infinite arrival schedule. If a policy is reliable for every reward environment and has a deterministic envelope $a_T = o(T)$ satisfying (2) in every environment, then $K_t = o(\sqrt{t})$. Conversely, $K_T^2 \log(eK_T) = o(T)$ suffices for one horizon-free policy to have both $\mathbb{E}R_T = o(T)$ and such a sublinear area envelope. In particular, for $K_t \asymp t^\alpha$, $0 < \alpha < 1$, simultaneous attainment is possible if and only if $\alpha < 1/2$.*

The necessity follows from (23). For sufficiency, the general checkpoint completion bound in Supplementary Lemma S5.2 applies throughout $0 < \eta < \varepsilon/2$, together with Supplementary Proposition S5.1. This argument does not require the narrower range $\eta \le \varepsilon/32$ in Theorem 4.2. The exponent statement leaves a logarithmic gap for general critical schedules.

Within the parameter range of Theorem 4.2, (24) also gives a uniform terminal-count statement: over all schedules with $K_T \le k(T)$, sublinear expected regret and sublinear service envelopes are jointly attainable if and only if $k(T)^2 \log k(T) = o(T)$. Here the worst schedule can change with T. This statement concerns a different schedule class from the single infinite schedule in Corollary 4.3.

The necessary condition $K_t = o(\sqrt{t})$ is not sufficient on every fixed schedule. For example, batches of size

$$m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor, \qquad s_n = 2^{2^n},$$

give $K_t = o(\sqrt{t})$ but preclude any $o(T)$ area envelope uniform over reward environments. Near s_n, simultaneous classification of that burst requires accumulated unresolved area of order $m_n^2 \log m_n$, which exceeds s_n by a growing factor. The supplementary argument selects a single infinite schedule; the difficult reward environment is allowed to vary with the endpoint, as required by the stated uniform guarantee.

5 Temporary tasks and limits of the regret–maintenance trade-off

5.1 A repeated-update family

We keep the reward environment, arrival schedule, endpoint, and reporting parameters fixed when comparing two area budgets. Choose integers $J \geq 1$, $M \geq \max\{2, J\}$ and

$$0 < \varepsilon \leq 1/8, \quad 0 < \delta \leq 1/16, \quad 0 < v, \quad Jv \leq \varepsilon/4, \quad 0 < d \leq v/32, \quad \eta = d/2. \tag{27}$$

At $a_e = 1 + eH$, $e = 0, \ldots, J$, a reference with mean $b_e = 1/2 + ev$ arrives. For each $e < J$, the same batch contains M candidates with means

$$r_{e,k} = b_e - \varepsilon + \sigma_{e,k} d, \qquad \sigma_{e,k} \in \{-1, +1\}. \tag{28}$$

All rewards are Bernoulli and all arms remain available. The last batch contains only its reference, and $T = (J + 1)H$. The family Θ includes every sign configuration and every within-batch permutation. The policy is not told which arm is the reference. Lower bounds may disclose those identities and the schedule.

In its birth window a candidate is good exactly when its sign is positive. Its reward gap is $\varepsilon - \sigma_{e,k} d$, while its distance to the membership boundary is d. At the next update its gap becomes $\varepsilon + v - \sigma_{e,k} d$, so both signs are bad, with boundary distance at least $v - d$. Every reference remains good: the final good set consists of all references. The tasks that distinguish the candidate signs expire even though their arms remain available. The restriction $Jv \leq \varepsilon/4$ limits the number of updates at any fixed positive improvement size.

5.2 Low area forces enough temporary tasks to finish

The lower-bound ingredient permits arbitrary interleaving and stopping. For m independent uniform signs with observation laws P_i^+, P_i^-, put

$$\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty).$$

If a bounded stopping time has at least half the signs determined with prior probability at least $1/2$, and all configurations have familywise report error at most $\delta \leq 1/16$, then

$$\overline{\mathbb{E}} \sum_{i=1}^{m} N_i \geq \frac{m}{1024\kappa} \log \frac{m}{\delta}. \tag{29}$$

Other observations must not encode these signs. The proof factors the conditional posterior after the adaptive sampling and bounded stopping kernels cancel, and combines simultaneous recovery with coordinatewise change of measure (Kaufmann et al., 2016). It does not assume independence of the random sample counts. The product prior is a proof device; the final lower bound concerns a fixed environment.

A window that never reaches half-completion contributes at least $MH/2$ area. Thus, on the event $\mathcal{U}_T(\eta) \le V/8$, where $V = JMH$, at least $3J/4$ windows half-complete. A common product prior over all windows turns this area requirement into enough instances of (29). Each candidate observation costs at least $\varepsilon - d$ regret, and observations from distinct batches are charged once. This argument does not impose a deadline on every individual batch; the policy can choose where to defer. The heterogeneous completion probabilities and their information-allocation relaxation are given in Supplementary Section S8. That relaxation is a necessary constraint, with no general causal attainability claim.

5.3 Shared evidence and the two endpoint witnesses

A saved upper endpoint for an old candidate can be compared with the lower endpoint of a later reference. In the separated-window construction,

$$u_{e,k} < \ell_{e+1,\mathrm{ref}} - \varepsilon$$

certifies exclusion after the new reference matures. No new observation of that old candidate is needed for this report, although the reward layer remains free to select it. Reference identities in this explanation are analytical roles, not information supplied to the policy. Supplementary Proposition S9.1 gives the complete heterogeneous conditions, unresolved-area bound, and unconditional regret bound for Checkpoint-UCB. Each quota is charged once over the arm's lifetime, and the birth-window condition is essential. These guarantees do not apply to arbitrary F-CCP configurations by renaming the method.

5.4 Rapid and deferred maintenance at the same endpoint

For the homogeneous family (27)–(28), set

$$K = J(M+1)+1, \quad L_K = \log \frac{4K(K+1)}{\delta}, \quad \rho_f = d/4, \quad \rho_c = v/8,$$
$$q_f = \lceil 32 L_K/d^2 \rceil, \quad q_c = \lceil 128 L_K/v^2 \rceil, \quad H = 128(M+J+1)q_f, \quad V = JMH, \quad T = (J+1)H. \tag{30}$$

The quantities q_f, q_c are uniform analysis bounds on the actual armwise quotas. Neither final arm count nor future times are policy inputs. Fine precision can be set from the task input as $\rho_f = \eta/2$; the coarse precision is a separate preselected configuration. Automatic adaptation to an unknown improvement scale is not asserted.

Theorem 5.1 (Rapid and deferred maintenance at the same endpoint)**.** *On the preceding family and horizon,*

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}. \tag{31}$$

The coarse-precision Checkpoint-UCB policy has joint label-and-area success probability at least $1-\delta$ *with area budget* $9V/8$*, and*

$$\sup_{\vartheta\in\Theta}\mathbb{E}_\vartheta R_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon}+\frac{J^2}{v}\right)\log(2T)+1. \tag{32}$$

Consequently, at fixed legal $J, M, \varepsilon, v, \delta$ *and* $d\downarrow 0$*, with* $\eta = d/2$ *and the horizon in* (30)*,* $\mathfrak{R}_\Theta(V/8) = \Theta(d^{-2})$*, whereas* $\mathfrak{R}_\Theta(9V/8) = O(\log(1/d))$*.*

For the lower bound, low area forces enough windows to satisfy (29). For the upper bound, (S18) gives the two area budgets. At the specified horizon the deterministic bound $R_T \le 2\varepsilon T$ suffices for the rapid order; it would not justify the same conclusion at arbitrarily longer windows. The coarse bound instead counts checkpoint observations and excess ordinary UCB pulls separately, as in (S19).

The two budgets concern the same $\Theta, \varepsilon, \eta, \delta, T$, but allow different service levels. Their ratio does not describe an individual delay for every arm. The theorem does not claim that the coarse policy meets the rapid budget, that its regret upper bound is matched, or that all intermediate budgets are characterized on this family and horizon. A preselected mixture of fine and coarse batches gives additional attainable points in the supplement. The separation concerns this specified repeated-update family, with η and T changing together as $d\downarrow 0$.

5.5 A probability lower bound under the same area requirement

An expectation lower bound alone could reflect unusual sample paths. The following result gives a stronger conclusion without replacing cumulative area by individual deadlines.

Theorem 5.2 (A nonvanishing probability of the rapid-maintenance cost)**.** *In the repeated-update family, suppose a reliable policy satisfies*

$$\mathbb{P}_\sigma\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le V/8\} \ge 1-\delta$$

for every sign configuration. Then one fixed configuration satisfies

$$\mathbb{P}_\sigma\left\{R_T \ge 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \ge \frac{11}{80}. \tag{33}$$

The proof stops each window at half-completion, its endpoint, or its $(q+1)$st candidate observation, where q is a fixed sample budget. Stopping at the first observation beyond q retains reports that become determinate after a reference observation while the candidate count is still q. Applying the partial-recovery lemma to this bounded experiment makes the probability of a cheap completion less than $1/2$. Low total area requires many completions; counting cheap ones

then leaves enough expensive windows with probability at least $1/5 - \delta \geq 11/80$. Their actual observations give (33). No independence between window-completion events is needed. Averaging under the common prior then selects one fixed environment.

Since $11/80 > \delta$, the rapid endpoint in (31) also has matching order when regret and area must be bounded on one event of probability at least $1 - \delta$. The exact joint-quantile definition and corollary are in Supplementary Section S10. This consequence concerns the rapid endpoint at the prescribed horizon and gives no matching joint-quantile result for deferred maintenance.

5.6 Information limits and intermediate service levels

A policy designed separately for each known schedule has more information than one causal policy required to work across an unknown schedule class. Both still lack the true reward means. Supplementary Theorem S12.1 constructs early and late improvements with the same horizon, area budget, final arm count, and common initial history. Timely service on the late schedule forces an inverse-square cost on the early schedule, while an early-schedule comparator has a logarithmic upper bound. This rules out a universal constant-factor comparison with every separately optimized schedule frontier. It does not rule out an optimal policy for a common causal minimax problem.

There is also a positive intermediate-budget result on a more strongly separated repeated-update family. Write $m = JM$, $V = mH$, and $k(a) = \lceil (V - a)/H \rceil$. The activation-budgeted maintenance witness (ABM) uses coarse lifetime checkpoints, then at most s fine candidates and corresponding references. Fine evidence is collected from new observation blocks after predictable activation, with confidence budgets assigned by activation rank. Under the separate window conditions in Supplementary Theorem S13.4, for $A_{\min} \leq a < V$,

$$\frac{\varepsilon k(a)}{3072 d^2} \log \frac{k(a)}{\delta} \leq \mathfrak{R}_{\Theta_H}(a) \leq R_0(T) + 1024 \frac{\varepsilon k(a)}{d^2} \log \frac{k(a)}{\delta}.$$

The supplement defines $A_{\min}$ and the retained additive reward cost $R_0(T)$, and gives the full construction and proof. This family uses a different H from (30). Given s, the witness does not need future arrivals; calibrating s from an area target uses the performance family's m, H and other analysis quantities. The interval result belongs to ABM, not to the preference parameter of F-CCP, and leaves smaller budgets and general overlapping arrivals unresolved.

6 Numerical studies

6.1 Protocol and comparisons

We study reward performance and the cost of reliable reporting separately. The frozen main suite contains 12 scenarios, 24 configurations and 96 paired replications per scenario and configuration, giving 27,648 records. A subsequent correction of one ALL comparator contributes 1,152 additional records. The original and corrected comparator records are both retained. The primary analysis replaces the original adaptive ALL configuration by its corrected version, leaving 24 configurations and 27,648 records.

The scenarios have 2–20 arms and horizons of 12,000–16,000. Eleven use Bernoulli rewards; one uses bounded two-point rewards. The reporting parameters ε and η vary by scenario, while $\delta = .05$ throughout. Supplementary Section S15 gives every arrival time, mean and configuration. Within a replication, all methods share the same potential reward stream for each arm, and the mean labels of arms arriving together are jointly permuted. The action routines receive only revealed arms, past observations and their own randomization; they do not receive the means, future arrivals or evaluation horizon. All failed-report paths remain in every reported mean, area and quantile.

The five reward comparators are UCB1, Growing-MOSS, MC-MOSS, Thompson sampling and KL-UCB (Garivier and Cappé, 2011). They make no designated certification pulls and carry the same valid reporter as F-CCP. Thompson sampling uses Beta$(1, 1)$ updates for Bernoulli rewards; its fractional updates for the bounded two-point scenario are an action heuristic, not a correctly specified Bernoulli posterior. F-R uses temperature .5 and reward protection $\beta_R = 16$ with service disabled. The compatibility configuration F-R-TS-exact instead uses temperature one and no reward guard; it reproduces the specified Thompson rule rather than serving as a second independent comparator.

For joint performance we compare the prespecified F configurations with CCP-original, ALL-area and ALL-auto-costmatched. CCP-original is the earlier CCP configuration in this test suite, not the checkpoint method in Supplementary Section S17. The ALL methods are natural retained-history adaptations, rather than implementations of the original FAREAST or $(ST)^2$ procedures. The corrected adaptive ALL method uses the same projected-mean sampling price as F-CCP, the same reward kernel and the same two budgets. Its initial implementation returned zero as that price; the correction changes that comparator input alone, with the scenarios, seeds and all F-CCP parameters fixed. The superseded output remains a diagnostic in Supplementary Section S16.

Development and testing use separate environments and seeds. The reward configuration was selected by the prespecified equal-weight development mean regret, and the moderate-service configuration was fixed before the main test. No preference, temperature or confidence parameter is selected from the test results. The parameter values in these experiments are not generally in the narrower regime of the minimax lower bound; these comparisons do not validate its asymptotic rate.

6.2 Reward performance with service disabled

F-R has a lower mean regret than all five reward comparators in 11 of the 12 scenarios, and a strictly lower mean in 59 of the 60 scenario–comparator comparisons. Its mean is 19.5% below Thompson sampling when the 12 scenario means are averaged with equal weights. This ratio compares two descriptive suite averages; it is not an average of scenario-specific percentage reductions or a risk under one common environment distribution. The gain in this experiment concerns the reward-score configuration and its protection, since the certificate scheduler is disabled.

The two-close-arms scenario is an exception. F-R minus Thompson has mean 0.70 and a paired, two-sided 95% approximate t interval $[-5.28, 6.68]$. Across the full family of 60 prespecified comparisons, 52 one-sided Bonferroni-adjusted upper endpoints are below zero. Under the separate, more permissive margin of 5% of the comparator mean plus two regret units, 54 are below the

Table 2: Mean cumulative dynamic regret in the frozen test suite. Each entry uses 96 paired replications; Monte Carlo standard errors are in Supplementary Table S3. All six methods devote no pulls to certification.

Scenario	UCB1	G-MOSS	MC-MOSS	TS	KL-UCB	F-R
Two close arms	75.52	52.16	52.16	51.37	62.63	52.07
Few arrivals	32.60	10.71	11.02	7.47	12.52	5.63
Stable benchmark	393.14	98.11	92.20	66.47	181.01	41.90
Successive improvements	313.99	105.44	86.29	57.61	123.14	53.59
Burst	532.82	140.79	131.95	80.01	177.09	68.56
Membership boundary	482.45	115.74	113.09	83.32	225.66	53.57
Many close means	341.20	188.60	190.01	202.27	267.46	169.80
Late improvement	364.87	126.45	101.93	83.89	196.49	70.56
Rare rewards	146.50	102.35	81.43	60.16	78.06	50.29
Ceiling	366.96	126.06	102.01	28.42	59.24	26.46
Larger pool	702.32	189.70	179.02	121.25	254.50	98.74
Bounded low variance	365.81	111.25	88.96	62.89	171.08	37.61

Table 3: Descriptive equal-weight averages of the 12 scenario means. The first six rows keep $\beta_C = 1$ wherever certification is active; F-area uses no certification cap. Area is in 10^3 arm-time units. These operating points are not an optimal frontier.

Method / preference	β_C	Regret	Area / 10^3
F-R	inactive	60.73	62.22
F, $\lambda = 0.1$	1	114.68	48.29
F, $\lambda = 0.3$	1	213.55	38.46
F, $\lambda = 0.6$	1	392.99	31.70
F, $\lambda = 0.9$	1	425.03	30.84
F, $\lambda = 1$	1	421.52	31.00
F-area	∞	1210.02	27.12
CCP-original	1	440.32	30.07
ALL-auto-costmatched	1	435.68	36.72
ALL-area	∞	1581.72	36.81

corresponding threshold. The latter criterion is not zero-margin noninferiority. Neither the low means nor a failure to detect a difference proves uniform dominance over the baselines.

6.3 Preference and certification cost

Increasing service preference reduces unresolved arm-time over much of the tested range, at a substantial reward cost. The points at $\lambda = .9$ and 1 under the fixed certification budget are not monotone: the latter has slightly lower regret and slightly larger area. We retain that reversal. F-area removes the certification cap; it is a different budget configuration and is not a continuation of the same-budget curve.

Relative to ALL-auto-costmatched, F-auto-0.6 has lower equal-weight mean regret (392.99 versus 435.68) and area (31.70 versus 36.72 thousand arm-time units). The comparison with CCP-original is a trade-off: F-auto-0.6 has lower mean regret but higher mean area. F-area has lower suite mean regret and area than ALL-area. These are finite-suite averages, rather than per-scenario or pathwise

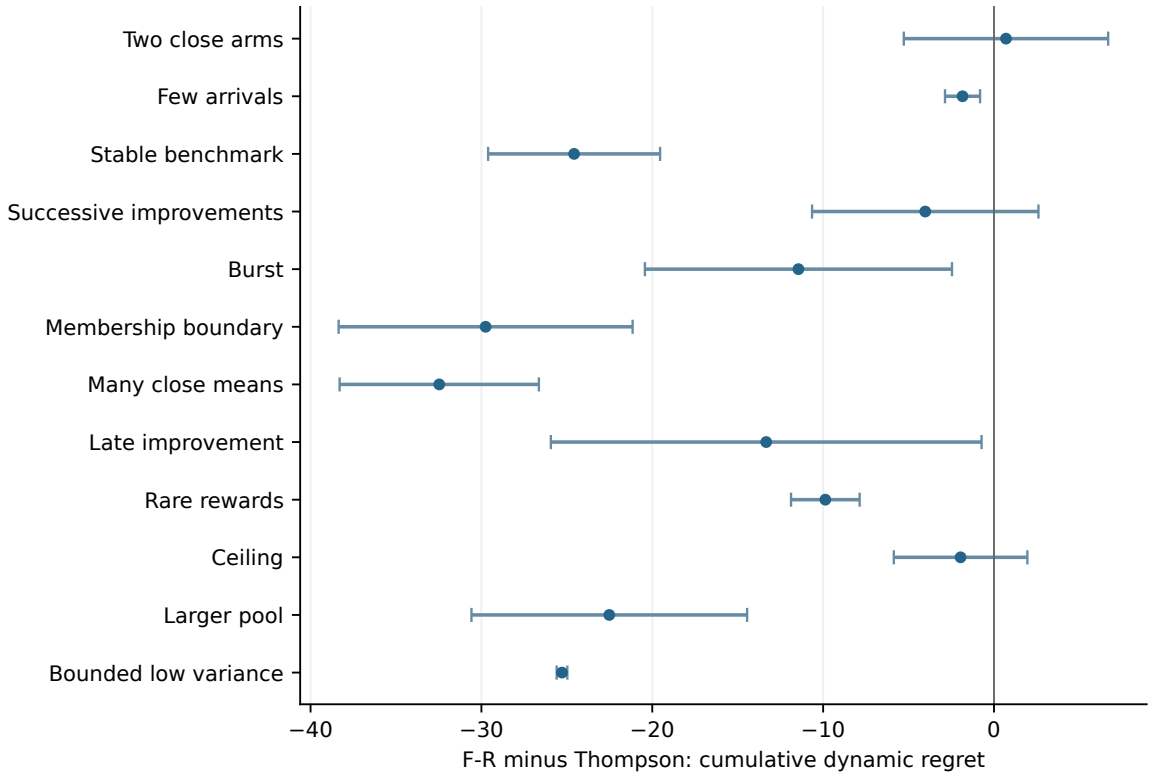


Figure 1: F-R minus Thompson cumulative dynamic regret in each frozen scenario. Points are means of 96 within-replication differences; error bars are individual two-sided 95% approximate t intervals. Negative differences favor F-R. These 12 intervals do not by themselves control the 60-comparison family; the adjusted counts are reported in the text.

Pareto guarantees. Supplementary Tables S4 and S6 give the scenario decomposition and paired uncertainty.

The controller's proposed rate differs from the realized service fraction because the phase must accumulate and the certification guard can reject a proposal. Supplementary Table S7 reports proposed busy-round rates, successful pulls, denials and the final unresolved count. Fixed-rate, uncapped-service, temperature-one and unprotected-reward variants are retained as prespecified ablations. We use those comparisons only for the mechanisms they actually change.

6.4 Tail behavior and report failures

The lower mean regret of F-R does not extend to all tail summaries. Its empirical 95th percentile exceeds Thompson's in the burst, ceiling and larger-pool scenarios in Table 4. Each percentile uses only 96 replications; it should not be read as a precise population tail estimate.

F-auto-0.6 produces an incorrect determined label on one of its 1,152 runs; F-R and F-area produce none. Coverage failures and empty intersections also occur: the respective counts are $(6, 1)$ for F-R, $(7, 1)$ for F-auto-0.6, and $(8, 0)$ for F-area. An empty intersection triggers persistent all-unresolved reporting, not successful completion. All these runs enter the cost calculations. Since

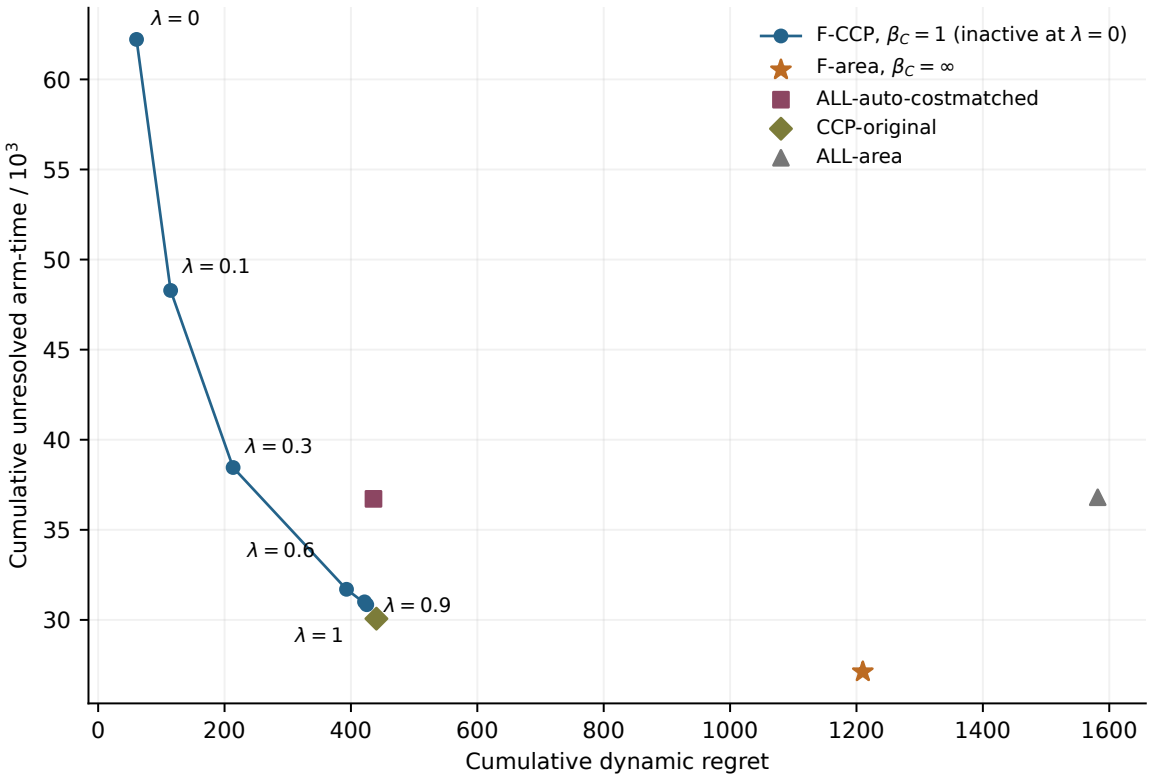


Figure 2: Descriptive averages of terminal costs over the 12 frozen scenarios. Connected circles follow the prespecified order $\lambda = 0, .1, .3, .6, .9, 1$, with $\beta_R = 16$, temperature .5, and $\beta_C = 1$ whenever service is active. The star is F-area with $\beta_C = \infty$; the other markers are named comparators. The connecting segments identify configurations and do not estimate a continuous or optimal frontier. Area is divided by 10^3.

methods share potential streams, their failures are dependent and cannot be pooled as independent reliability trials. These empirical counts neither replace the reliability theorem nor justify an all-times claim from 96 repetitions. Full failure counts, empirical maxima and implementation qualifications appear in Supplementary Sections S15–S16.

7 Discussion

Reliable maintenance has costs that reward allocation alone does not measure. An uncertain arrival can obstruct many positive reports, while a later improvement can end a fine membership task. The general lower bounds quantify these effects. The worst-area lower bound answers a different question and is already attained by an initial static batch; the fixed-schedule and terminal-count growth consequences retain different quantifiers.

F-CCP adds an implementable choice of emphasis to this task. Its evidence layer uses every observation, its certificate planner proposes a single next observation, and separate accounts control reward and certification costs. Finite guards protect the spectral regret order but can reject

Table 4: Empirical 95th percentiles of cumulative dynamic regret from 96 replications per method and scenario. These are finite-sample tail summaries, not high-probability regret bounds.

Scenario	F-R	Thompson
Burst	208.887	123.069
Ceiling	90.940	48.971
Larger pool	237.301	184.175

service; the area bound retains that rejection cost. Classification priority with no certification guard achieves the worst-area order, without being fastest on every instance. A compatibility endpoint recovers a specified baseline from the start under identical state and randomization. The default cold-temperature reward mode has a different contract.

The frozen experiments support lower mean reward costs for F-R in most of the specified comparisons and useful finite-budget trade-offs for the service modes. They also show an exception in the two-arm close-mean case, heavier empirical regret tails than Thompson sampling in three environments, and a nonzero label-error record for the balanced mode. Pure-reward improvements cannot be attributed to certificate scheduling that was switched off. The corrected matched-cost ALL comparison is distinct from the retained superseded controller result. These findings concern the frozen suite, not uniform dominance or a general frontier.

The activation-budgeted witness characterizes an intermediate interval on a sufficiently separated family, with an additive reward term and its own information requirements. General causal frontiers under unknown and overlapping arrivals remain open. The schedule-information separation does not make every causal comparison impossible. Likewise, the one-step zero-remainder counterexample leaves room for comparisons with an explicit adaptation cost or a restricted performance family.

Automatic precision, reward kernels with better tail behavior, a faithful adaptation of the original ALL-ε algorithms, and large-pool planning costs need further study. The supplement's eventual exact-set consistency theorem belongs to a separate revisable multiresolution procedure under finite final arrivals and strict boundary separation. F-CCP's coverage-event completion outside a boundary band is a different statement.

Data and code availability

The accompanying materials contain simulated data, frozen configurations, code, and the scripts used to generate the article's tables and figures. The principal test has 27,648 recorded runs; a subsequent 1,152-run matched-cost ALL correction is retained separately, together with the superseded comparator. Replacing that comparator for the principal analysis leaves 24 configurations and 27,648 records. Earlier checkpoint comparisons contain 5,760 records and are reported separately in the supplement; the preceding CCP and frontier research suites retain their own protocols and archives. All reported failures remain in the analysis. The available implementation distinguishes the frozen dual-account production engine from the interactive interface and the development branches. These materials are provided locally with the project; no public repository or permanent archive identifier has been verified for this version.

Supplementary material

The Cost of Reliable Near-Optimal Set Maintenance
with Arriving Arms

Notation and scope

The observation model has independent arm streams, each consisting of independent, identically distributed rewards in $[0,1]$, deterministic permanent arrivals $\mathcal{A}_t$, and one observation per physical time step. Each finite $\mathcal{A}_t$ is nonempty, and arbitrarily large finite batches are allowed. Arrivals occur before the action, and reports follow its observation. Let $r^*(t) = \max_{i\in\mathcal{A}_t} r_i$, $\Delta_i(t) = r^*(t) - r_i$, and $\mathcal{G}_t^\varepsilon = \{i \in \mathcal{A}_t : \Delta_i(t) \le \varepsilon\}$. Reports partition $\mathcal{A}_t$ as $(\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t)$. Write

$$E_{\mathrm{lab}} = \{\forall t : \mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}, \qquad \mathcal{U}_T(\eta) = \sum_{t\le T}\sum_{i\in\mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, |\Delta_i(t) - \varepsilon| > \eta\},$$

and $R_T = \sum_{t\le T} \Delta_{I_t}(t)$. Reliability means $P(E_{\mathrm{lab}}) \ge 1-\delta$ for every allowed environment. $\mathfrak{R}_\Theta(a)$ minimizes worst-case ER_T over full-model reliable policies with $P_\vartheta(E_{\mathrm{lab}}, \mathcal{U}_T \le a) \ge 1-\delta$ for all $\vartheta \in \Theta$. A bar denotes averaging under the independent uniform prior on all candidate signs. These priors are proof devices and impose no Bayesian requirement on the policy.

The probability and sample-budget arguments are given below. Classical change of measure and multiple-recovery scales are credited to the existing literature (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018; Mason et al., 2020).

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S1 Probability tools and risk conventions

For the reporting constructions, index arms in arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. Every arm has an independent potential reward stream $(Y_{i,n})_{n\geq 1}$ in $[0, 1]$. The policy observes its nth entry on its nth pull. Arrival times are deterministic, and reports are issued after the current observation. Write $N_i(t)$ for lifetime observations through round t and $\hat{r}_i(n) = n^{-1} \sum_{s\leq n} Y_{i,s}$. The observed reward in physical-time notation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. The conclusions below also hold conditionally on an independent exogenous arrival schedule.

For independent $[0, 1]$ observations, Hoeffding's inequality gives

$$P\{|\hat{r}_i(n) - r_i| > x\} \leq 2e^{-2nx^2}.$$

For completeness, the centered log moment generating function has value and derivative zero at the origin; its second derivative is a variance under an exponentially tilted distribution and is at most $1/4$. Integrating twice gives $Ee^{\lambda(X-EX)} \leq e^{\lambda^2/8}$. Multiplication over independent observations and exponential Markov inequality prove the display.

For two Bernoulli mean vectors $\boldsymbol{r} = (r_i)_i$ and $\boldsymbol{r}' = (r'_i)_i$ and a bounded stopping time τ, action kernels, policy randomization, and stopping kernels cancel in the history likelihood ratio. Padding after stopping by null symbols and taking iterated expectations yields

$$D(P_{\boldsymbol{r}}^{H_\tau} \| P_{\boldsymbol{r}'}^{H_\tau}) = \sum_i E_{\boldsymbol{r}} N_i(\tau) \operatorname{kl}(r_i, r'_i). \tag{S1}$$

The same proof applies to arbitrary observation laws with finite two-way KL. For an event F measurable at stopping, splitting the likelihood integral over F, F^c and applying Jensen gives

$$D(P\|Q) \geq \operatorname{kl}(P(F), Q(F)) \geq P(F) \log \frac{1}{Q(F)} - \log 2.$$

The second inequality uses the binary entropy bound $h(p) \leq \log 2$. We repeatedly use

$$2(p-q)^2 \leq \operatorname{kl}(p, q) \leq \frac{(p-q)^2}{q(1-q)}.$$

The upper bound follows from $\log x \leq x - 1$; for the lower bound the second derivative in p is $1/\{p(1-p)\} \geq 4$ and both value and derivative vanish at $p = q$.

The primary risk combines an unconditional expectation for R_T with a joint high-probability requirement for labels and unresolved area at each fixed deterministic endpoint. A common coverage event can yield upper bounds simultaneously in all endpoints; it does not change the quantifiers in the lower-bound risk. The fixed-checkpoint, all-prefix, and mixed-evidence reporters below have separate failure conventions. In particular, the historical all-prefix convention in Section S17.6 is not the persistent-failure convention of F-CCP.

S2 Sequential evidence and certificate planning

This section establishes the evidence statements used by F-CCP. The bounds hold under the bounded iid model and do not require Bernoulli rewards. All random counts refer to the original potential streams; no distributional assertion is made conditional on a selected count.

S2.1 The bounded-reward mixture

Fix a stream of mean $m \in (0,1)$ and let its first n rewards be $Y_1, \dots, Y_n \in [0,1]$. For $\vartheta \in (0,1)$ put

$$f_{m,\vartheta}(x) = (\vartheta/m)^x \{(1-\vartheta)/(1-m)\}^{1-x}.$$

This is the exponential of an affine function of x, hence convex. The chord inequality on $[0,1]$, followed by conditional expectation, gives

$$\mathbb{E}\{f_{m,\vartheta}(Y_n) \mid Y_1, \dots, Y_{n-1}\} \le (1-m)\frac{1-\vartheta}{1-m} + m\frac{\vartheta}{m} = 1.$$

Consequently $\prod_{r \le n} f_{m,\vartheta}(Y_r)$ is a nonnegative supermartingale. Integrating against the $\mathrm{Beta}(1/2, 1/2)$ density and using conditional Tonelli gives another nonnegative supermartingale, initially one, whose value is

$$\mathsf{M}_n(m) = \frac{\mathrm{B}(\Sigma_n + 1/2, n - \Sigma_n + 1/2)}{\mathrm{B}(1/2,1/2) m^{\Sigma_n} (1-m)^{n-\Sigma_n}}.$$

For a nonnegative supermartingale initially one, stop at the first crossing of $1/\alpha$ before a deterministic integer L. The stopped expectation is at most one, so the crossing probability is at most α. Letting L increase proves Ville's inequality

$$\mathbb{P}\{\exists n \ge 1 : \mathsf{M}_n(m) > 1/\alpha\} \le \alpha.$$

The possible distinction between strict and non-strict crossings is resolved by taking limits. At $m = 0$ or 1, boundedness forces the stream to be identically that endpoint almost surely; the formula and inversion use their limits. In particular, the true endpoint cannot be lost.

For a fixed positive count, the log of $\mathsf{M}_n(m)$ is a constant minus $\Sigma_n \log m + (n - \Sigma_n)\log(1-m)$. It is convex in m, so its sublevel set is an interval, possibly empty. Intersecting these sets over observed prefixes retains simultaneous coverage. The union bound over arrival ranks gives total mixture failure probability at most $\sum_i \alpha_i = \delta/2$. Adaptive actions reveal prefixes of these streams; they cannot invalidate an event defined simultaneously for every potential prefix. Independent policy randomization can be included in the physical filtration without altering this argument.

At the deterministic per-arm prefix q_i, Hoeffding's inequality gives

$$\mathbb{P}\{|\hat{r}_i(q_i) - r_i| > \rho/2\} \le 2e^{-q_i \rho^2/2} \le \frac{\delta}{2i(i+1)}.$$

This prefix is defined even when the actual experiment never reaches it. A second union bound costs at most $\delta/2$. The two types of evidence need not be independent. On their common event $\mathcal{E}$,

of probability at least $1-\delta$, every intersection contains its true mean at every physical time, and any arm that has reached q_i has interval width at most ρ forever afterward. The failure flag is never set on $\mathcal{E}$.

S2.2 Exact rectangular gap bounds and paused work

Fix a nonempty rectangle $\mathcal{R} = \prod_{i\in\mathcal{A}}[\ell_i, u_i]$. For $K > 1$, minimizing $\max_j r_j - r_i$ over $\mathcal{R}$ sets $r_i = u_i$ and every other coordinate to its lower endpoint. Maximizing it sets $r_i = \ell_i$ and every other coordinate to its upper endpoint. The resulting extrema are exactly

$$g_i^- = \max\{0, \max_{j\neq i} \ell_j - u_i\}, \qquad g_i^+ = \max\{0, \max_{j\neq i} u_j - \ell_i\}.$$

For $K = 1$ the gap is identically zero, regardless of interval width. This explains why using the same arm's upper and lower endpoints as independent competitors would be unnecessarily conservative. The extrema are attained. If $g_i^- \le \varepsilon < g_i^+$, the rectangle contains both membership possibilities, so no definite label follows from this rectangle alone. This is a logical property of rectangular information, not optimality among all sequential tests or all joint confidence regions.

On $\mathcal{E}$ the true mean vector lies in $\mathcal{R}$, which proves Proposition 3.1. If $[g_i^-, g_i^+] \subseteq [\varepsilon-\eta, \varepsilon+\eta]$, the true gap satisfies $|\Delta_i - \varepsilon| \le \eta$. Such an arm remains in $\mathcal{V}$ but contributes no charged area. At a fixed physical time after arrivals, the subsequent observation only shrinks the rectangle. Thus its attainable gap intervals only shrink. An arm already given a definite label cannot become unresolved during this update, and an arm in the paused set Z cannot return to W during it. Every post-action arm contributing area therefore belonged to the pre-action W. New arrivals can change this conclusion across rounds, which is why reports and paused work are always recomputed.

If every available interval has width at most ρ, then

$$g_i^+ - g_i^- \le (u_i - \ell_i) + \max_{j\neq i}(u_j - \ell_j) \le 2\rho = \eta/2.$$

For a still-unresolved arm the gap interval crosses ε. It is consequently contained in the boundary band, and the arm belongs to Z. Hence $W = \varnothing$ when all current checkpoints are complete. Equivalently, on $\mathcal{E}$, nonempty W implies that an incomplete checkpoint exists and the service fallback can find one.

S2.3 Solution of the two-endpoint planning problem

Let $n_k^0 = \max(1, n_k)$ for $k = i, j$, let $a_k = v_k^{\text{plan}}\sqrt{n_k^0}$, and fix positive prices c_k and $h > 0$. If $a_i/\sqrt{n_i^0} + a_j/\sqrt{n_j^0} \le h$, the unique minimum of (10) has $x_k = n_k^0$ and zero proxy cost. Otherwise, with both $a_k > 0$, the active constraint has Lagrange conditions

$$c_k = \frac{\omega a_k}{2x_k^{3/2}}, \qquad k = i, j,$$

for every coordinate above its lower bound. If both coordinates are interior, substitution gives

$$D = \frac{a_i^{2/3} c_i^{1/3} + a_j^{2/3} c_j^{1/3}}{h}, \qquad x_k = D^2 (a_k/c_k)^{2/3}, \quad k = i, j. \tag{S2}$$

The feasible set is closed, convex, and nonempty; positive prices make its objective sublevel sets compact. If the interior candidate is below n_i^0, fix $x_i = n_i^0$ and take

$$x_j = \max\{n_j^0, (a_j/(h - v_i^{\text{plan}}))^2\}.$$

The denominator is positive in this boundary case. Exchange i, j for the other boundary case. When work is needed, both unconstrained coordinates cannot lie below their lower bounds. Equivalently, the solution is characterized by

$$x_k(\omega) = \max\{n_k^0, (\omega a_k/(2c_k))^{2/3}\}, \qquad \sum_k a_k/\sqrt{x_k(\omega)} = h.$$

The radius sum decreases strictly after leaving its initial plateau, which proves uniqueness. A zero a_k leaves that coordinate at its lower bound and reduces the problem to the other coordinate. With two zero radii no work is needed.

In the service routine, a legal endpoint additionally requires $n_k < q_k$ and $x_k > n_k$, using the actual count in this comparison. Among its two legal endpoints, choose the larger $v_k^{\text{plan}}/(\max(1, n_k)c_k)$; a tie favors the current candidate i. Scan candidates in arrival order and retain the first pair of minimum cost. Empirical-best and largest-upper reference ties also use first arrival. If no legal pair is found, scan incomplete checkpoints for the first interval of largest width. These conventions specify the deterministic pair-plan routine used by the main algorithm. The convention $\max(1, n_k)$ neither increments n_k nor shrinks the report interval. The proxy's frozen centers and assumed radius decay are not assertions about future random observations.

The numerical implementation replaces zero planning radii by 10^{-12}, uses a 10^{-8} tolerance for positive planned increments, and retains the original actual counts for all statistical updates. Its fixed-prefix evidence is not altered by these numerical conventions.

S3 Virtual indices and the optimism-gap bound

We give the arbitrary-action bridge used in Theorem 3.3. The potential-prefix and arrival-clock arguments originate in the companion manuscript Temporal Geometry. Their use here must allow actions that do not maximize the virtual index. No confidence-success conditioning is used anywhere in this section.

S3.1 Potential-prefix bounds

For one original potential stream of mean r, write $S_n = \sum_{v=1}^{n}(r - Y_v)$ in this subsection only, and define

$$Z(v) = \sup_{n\geq 1}\left[r - \hat{r}(n) - \sqrt{[\log(v/n)]_+/n}\right]_+, \qquad v > 0.$$

The symbol S_n here is a centered partial sum, not the reward sum Σ_i or the arrival spectrum $\mathcal{S}_t$. For bounded rewards Hoeffding's moment bound gives $\mathbb{E}e^{\gamma(r-Y_v)} \leq e^{\gamma^2/8}$. Applying the nonnegative-supermartingale crossing inequality to $e^{\gamma S_n - n\gamma^2/8}$ yields

$$\mathbb{P}\{\exists n > v : S_n/n \geq x\} \leq e^{-2vx^2}, \qquad v, x > 0, \tag{S3}$$

$$\mathbb{P}\{\max_{n\leq L} S_n \geq y\} \leq e^{-2y^2/L}, \qquad L \in \mathbb{N},\ y > 0. \tag{S4}$$

For the first inequality use $\gamma = 4x$; for the second use $\gamma = 4y/L$. The first event is an infinite union of countable prefixes, obtained by increasing finite stopping horizons.

Lemma S3.1 (Potential underestimation)**.** *For every $v > 0$, $\mathbb{E}Z(v) \leq 5/\sqrt{v}$.*

Proof. Use (S3) for $n > v$. For $n \leq v$, let $m_k = v2^{-(k+1)}$ and partition the positive integer counts into $m_k < n \leq 2m_k$, $k \geq 0$. If a prefix in this interval makes $Z(v) > x$, then

$$S_n > m_k x + \sqrt{m_k k \log 2}.$$

For a nonempty integer interval, (S4) with $L = \lfloor 2m_k \rfloor$ bounds its probability by $2^{-k}e^{-vx^2/2^{k+1}}$. Therefore

$$\mathbb{P}\{Z(v) > x\} \leq e^{-2vx^2} + \sum_{k\geq 0} 2^{-k}e^{-vx^2/2^{k+1}}.$$

Integrating the nonnegative terms gives

$$\mathbb{E}Z(v) \leq \frac{\sqrt{\pi}}{2\sqrt{2v}} + \frac{\sqrt{\pi}}{\sqrt{2v}}\sum_{k\geq 0} 2^{-k/2} < \frac{5}{\sqrt{v}}.$$

This statement is about the original stream simultaneously over its counts, so it also bounds a count selected using its observed past. □

Lemma S3.2 (Distinct arm–count pairs)**.** *Select at most h pairs (i, n) from k fixed potential streams, where $n \geq 1$ and no pair is used more than once. The selection may depend on the entire adaptive experiment. Then*

$$\mathbb{E} \sum_{\text{selected } (i,n)} (\hat{r}_i(n) - r_i)_+ \leq \sqrt{2kh}.$$

For every $v > 0$,

$$\sum_{n \geq 1} \sqrt{[\log(v/n)]_+ / n} \leq \sqrt{2\pi v}.$$

Proof. For every $x > 0$ the first sum is bounded pathwise by

$$xh + \sum_{i=1}^{k} \sum_{n \geq 1} (\hat{r}_i(n) - r_i - x)_+.$$

The expectation of the double sum for one arm is at most

$$\int_x^\infty \sum_{n \geq 1} e^{-2ny^2} \, dy = \int_x^\infty \frac{dy}{e^{2y^2} - 1} \leq \frac{1}{2x}.$$

Minimize $xh + k/(2x)$ at $x = \sqrt{k/(2h)}$; the empty-selection case is immediate. This proof does not condition on any chosen empirical mean. For the second statement the summand decreases on $(0, v)$, so its sum is at most

$$\int_0^v \sqrt{\log(v/x)/x} \, dx = \sqrt{v} \int_0^\infty y^{1/2} e^{-y/2} \, dy = \sqrt{2\pi v}.$$

If $v < 1$ the sum is zero, which also satisfies the bound. □

S3.2 A segment with arbitrary actions

Fix the deterministic arrival schedule, reward laws, and a virtual clock segment of capacity $d = d_{\text{clock}}$ and prefix length h. The start and end of this segment depend only on arrivals. At most $k \leq d$ arms are available during the segment: its starting competitor count is c, $d = 2c$, and any later count $K_t - 1 \geq 2c$ would start a new segment before the action. Retain all samples from before the segment.

At every time choose a true optimal label i_t^* by a fixed rule using only the fixed means and schedule. For $n_{i_t^*} > 0$, clipping at one cannot move its index below $r^*(t) - Z_{i_t^*}(s/d)$, since $r^*(t) \leq 1$. For $n_{i_t^*} = 0$, its index is one and the same lower bound holds. For an observed selected arm, the identity $J_{I_t} = \max_j J_j - \zeta_t(I_t)$ and the upper bound $J_{I_t} \leq \hat{r}_{I_t}(n) + \sqrt{[\log\{s/(dn)\}]_+ / n}$ give

$$\Delta_{I_t}(t) \leq \zeta_t(I_t) + Z_{i_t^*}(s/d) + (\hat{r}_{I_t}(n) - r_{I_t})_+ + \sqrt{[\log\{s/(dn)\}]_+ / n}, \quad n = N_{I_t}(t-1) > 0. \tag{S5}$$

This comparison does not assume that the chosen arm maximizes J. In particular, an unobserved optimal arm creates no exceptional conditioning issue. The genuinely first pulls of selected arms are charged separately at cost at most one each.

If $h \geq d$, first pulls cost at most $d \leq \sqrt{dh}$. Lemma S3.1, applied to the deterministic label at each local time, gives

$$\sum_{s=1}^{h} \mathbb{E} Z_{i_t^*}(s/d) \leq 5 \sum_{s=1}^{h} \sqrt{d/s} \leq 10\sqrt{dh}.$$

An arm's lifetime count increases at every actual pull, whether it is called a reward pull or a service pull. Hence its selected pre-pull count never repeats. Lemma S3.2 bounds the empirical excess by $\sqrt{2dh}$. For bonuses replace s by h, extend each arm's counts to all positive integers, and obtain $k\sqrt{2\pi h/d} \leq \sqrt{2\pi dh}$. Summing proves

$$\mathbb{E} R_{\text{segment}} \leq 15\sqrt{dh} + \mathbb{E} \sum_{t \in \text{segment}} \zeta_t(I_t). \tag{S6}$$

The coefficients sum to $1 + 10 + \sqrt{2} + \sqrt{2\pi} < 15$. If $h < d$, the trivial regret bound $h \leq \sqrt{dh}$ gives (S6) as well. These are unconditional expectations under the original experiment, so old observations and arbitrary predictable score kernels are allowed.

S3.3 Deterministic charging of arrival clocks

For $b > 0$ put $f_b(k) = \min\{b, \sqrt{kb}\}$. It is subadditive: $f_b(\sum_r k_r) \leq \sum_r f_b(k_r)$, since a saturated term settles the assertion and otherwise the square root is subadditive. We use two deterministic spectral facts from Temporal Geometry, recording their proofs to fix the interface.

Deleting competitors and delaying the remaining competitors, without changing the endpoint, produces a spectrum at most twice the original one. Indeed, arms originally in dyadic group m can only move to groups $\ell \leq m$. If $k_{m,\ell}$ counts these arms, their new total cost is bounded both by

$$\sum_{\ell \leq m} 2^\ell < 2 \cdot 2^m, \qquad \sum_{\ell \leq m} \sqrt{k_{m,\ell} 2^\ell} \leq \sqrt{2k_m 2^m}.$$

It is therefore at most $2f_{2^m}(k_m)$. Subadditivity separates the original groups; summing proves the comparison. Also, if m competitors have lifetime at least H, their spectral contribution is at least $f_H(m)$: each containing group has upper endpoint at least H, and subadditivity combines the groups.

Before the first competitor arrives, regret is zero. Index later segments by $j = 0, \ldots, J$, with starts σ_j, lengths h_j, starting competitor counts $c_j = K_{\sigma_j} - 1$, capacities $d_j = 2c_j$, and remaining lifetimes $H_j = T - \sigma_j + 1$. The counts satisfy $c_{j+1} \geq 2c_j$. If j_* is the first index for which $c_j \geq H_j$, charge that segment and the entire later tail by $H_{j_*} \leq \mathcal{S}_T$, using the preceding lifetime fact. If no such index exists, all segments belong to the following sum.

For $j < j_*$ put $c_{-1} = 0$ and retain the competitors arriving between consecutive segment starts, including initial competitors for $j = 0$. There are $m_j = c_j - c_{j-1} \geq c_j/2$ such arms. Delete all other competitors and delay these m_j arms to σ_j; this produces $\mathcal{S}'_T \leq 2\mathcal{S}_T$. Group the retained starts by the dyadic scale of H_j. For a nonempty group of upper endpoint $b_\ell = 2^\ell$, with last segment j_{last},

geometric growth gives

$$\sum_{j:H_j\in(b_\ell/2,b_\ell]}\sqrt{c_jH_j}\le\frac{\sqrt{c_{j_{\mathrm{last}}}b_\ell}}{1-2^{-1/2}}.$$

The number k'_ℓ of delayed competitors in this group satisfies

$$c_{j_{\mathrm{last}}}/2\le k'_\ell\le c_{j_{\mathrm{last}}}<H_{j_{\mathrm{last}}}\le b_\ell.$$

Its spectral cost is unsaturated. Therefore, using $h_j\le H_j$,

$$\sum_{j<j_*}\sqrt{c_jh_j}\le\frac{2\sqrt{2}}{1-2^{-1/2}}\mathcal{S}_T.$$

Together with (S6), the capacities $d_j=2c_j$, and the tail bound, this yields

$$\mathbb{E}R_T\le\left(1+\frac{60}{1-2^{-1/2}}\right)\mathcal{S}_T+\mathbb{E}\sum_{t\le T}\zeta_t(I_t)\le 300\mathcal{S}_T+\mathbb{E}\sum_{t\le T}\zeta_t(I_t). \tag{S7}$$

The nonnegative index-gap terms may be extended from the retained segments to the whole horizon. Since $R_T\le T$, this proves (18). The endpoint-dependent grouping is used only in the proof; none of it is an algorithm input.

Finally, for any $u\le T$, remove competitors arriving after u and delay every remaining competitor by $T-u$. Their ages at T are exactly their ages at u, so the same deletion–delay comparison gives $\mathcal{S}_u\le 2\mathcal{S}_T$ and thus

$$\overline{\mathcal{S}}_T\le 2\mathcal{S}_T. \tag{S8}$$

The counting anchor is never removed or delayed. Neither this bound nor the arbitrary-action inequality reestablishes the companion minimax lower theorem; that theorem is a separately attributed input when interpreting the regret order.

S4 Guarantees for Flexible CCP

The reward and service results have different hypotheses. Reliability allows any legal predictable preference, including a preference tending to zero. The regret bound uses fixed budget coefficients; an area rate requires a positive lower bound on service while work remains. These conditions are not supplied by the word "adaptive."

S4.1 Compatibility and state preservation

To prove Proposition 3.2, couple the potential reward streams, arrivals, full initial kernel state, and backend random-number source. At the first round the service rate is zero, the reward guard is disabled, and the same kernel receives the same observed state. Its scores and selected action agree. Both procedures reveal the same next potential reward from that arm, so the kernel states still agree. Repeat inductively. The evidence updates do not affect decisions in this mode and consume

no action random numbers. Any auxiliary kernel state, including its clock and random-number state, must be included in the coupling. This proves equality of actions, lifetime counts, and regret simultaneously at every physical round and endpoint.

For standard Beta$(1,1)$ Thompson sampling, this endpoint uses $\theta = 1$, $\lambda = 0$, and $\beta_R = \infty$. It does not include F-R, which has $\theta = .5$ and a finite guard. The batch backend and interactive backend use different random-number implementations; matching their numerical seeds is not this coupling. Likewise, a preference switch after earlier service preserves the actual history, not the history of a baseline independently run from the beginning. The interface changes preferences only between a completed observation and the next action and resets no count, interval, phase, or ledger.

S4.2 The two-ledger regret bound

For a fixed finite budget coefficient the corresponding budget $\beta\overline{\mathcal{S}}_t$ is nondecreasing. Initially each ledger is zero. On a service round its guard explicitly verifies $C_C(t-1) + \zeta_t(I_t) \le \beta_C\overline{\mathcal{S}}_t$; on other rounds this ledger is unchanged. Induction gives

$$C_C(T) \le \beta_C\overline{\mathcal{S}}_T.$$

On a reward round the maximization is restricted to $\mathcal{F}_R(t)$. This set is nonempty: a largest-index arm has $\zeta_t = 0$, and the previous reward ledger remains within the nondecreasing budget. Thus

$$C_R(T) \le \beta_R\overline{\mathcal{S}}_T.$$

This argument also covers $\beta_R = 0$ and $\beta_C = 0$; only zero-gap actions can then consume the corresponding type of round. Disabling a guard means bypassing its comparison, not evaluating $\infty \cdot 0$. All these decisions occur before the sole reward observation at that physical time.

Every actual round is exactly one of the two types, and therefore

$$\sum_{t\le T} \zeta_t(I_t) = C_R(T) + C_C(T) \le (\beta_R + \beta_C)\overline{\mathcal{S}}_T \le 2(\beta_R + \beta_C)\mathcal{S}_T.$$

Apply (18). Its expectation includes failed reporting histories and arbitrary reward scores, so this substitution does not incur a missing δT term. Truncation by T proves (19). When service is disabled throughout the run its ledger is identically zero, which removes β_C.

Even without the certification guard, every actual service selects an arm of actual count below q_i and increments that count once. Lifetime counts are shared with reward allocation and never reset, so the total number of services through T is at most $Q_T^{\mathrm{cap}} = \sum_{i\in\mathcal{A}_T} q_i$ on every path. The persistent failure rule can only reduce this number. Since $0 \le \zeta_t \le 1$, $C_C(T) \le Q_T^{\mathrm{cap}}$. Combining it with the reward guard and (S8) proves (20).

The resulting preset bounds are

$$\begin{aligned}
&\text{F– R}: && \mathbb{E}R_T \le \min\{T, 332\mathcal{S}_T\},\\
&\text{Balanced}: && \mathbb{E}R_T \le \min\{T, 334\mathcal{S}_T\},\\
&\text{Classification}: && \mathbb{E}R_T \le \min\{T, 332\mathcal{S}_T + Q_T^{\mathrm{cap}}\}.
\end{aligned}$$

The compatibility endpoint has no finite reward-guard coefficient, so these bounds are not inherited automatically for an arbitrary kernel. The two ledgers bound observable optimism gaps, not the realized regret difference from an independently run kernel. The established companion minimax lower bound supplies the matching spectral order for finite fixed coefficients, with no assertion of sharp constants or optimality under a prescribed area constraint.

S4.3 Phase accounting and the area guarantee

Let A_T^{att} be the number of designated service attempts and B_T^{busy} the number of pre-action rounds with nonempty W_t. In exact arithmetic $0 \le \phi_t < 1$, because $0 \le p_t \le 1$ and each crossing causes exactly one subtraction. The telescoping identity is

$$\phi_T = \sum_{t \le T} p_t - A_T^{\mathrm{att}}, \qquad A_T^{\mathrm{att}} = \left\lfloor \sum_{t \le T} p_t \right\rfloor. \tag{S9}$$

A service denial does not undo this subtraction and is not retried within the same physical round.

On $\mathcal{E}$, Supplement S2 shows that busy work implies an incomplete checkpoint and hence a legal proposal. Every attempt is then either an executed service or a certification budget denial. If D_T^{srv} counts these denials and S_T^{srv} executed services, then

$$A_T^{\mathrm{att}} = S_T^{\mathrm{srv}} + D_T^{\mathrm{srv}}, \qquad S_T^{\mathrm{srv}} \le Q_T^{\mathrm{cap}}.$$

When $p_t \ge p_{\min} > 0$ on busy rounds, the rate is zero on all other rounds and (S9) gives

$$p_{\min} B_T^{\mathrm{busy}} \le \sum_{t \le T} p_t < Q_T^{\mathrm{cap}} + D_T^{\mathrm{srv}} + 1.$$

Every post-action arm that contributes area was in the pre-action W_t, so $\mathcal{U}_T(\eta) \le K_T B_T^{\mathrm{busy}}$. Together with the deterministic $K_T T$ bound, this proves Theorem 3.4 on the single event $\mathcal{E}$, simultaneously for all endpoints. It is a high-probability area statement, not an expected-area replacement for the model's risk.

For adaptive control $p_0 \le w_t \le 1$. The function $\lambda w/(1-\lambda+\lambda w)$ increases in w, so a fixed $\lambda > 0$ gives (17). Fixed-rate control also obeys this lower bound. At $\lambda = 1$ the rate is one on busy rounds. When the certification guard is disabled, $D_T^{\mathrm{srv}} = 0$, and

$$Q_T^{\mathrm{cap}} = O\{K_T \eta^{-2} \log(K_T/\delta)\}.$$

Consequently Classification has area $O(\min\{K_T T, K_T^2 \eta^{-2} \log(K_T/\delta)\})$. The comparison with the

retained worst-area lower bound uses its common range $K_T \geq 2$, $\varepsilon \leq 1/8$, $\eta \leq \varepsilon/32$, and $\delta \leq 1/16$. This is an order comparison over the worst environments and schedules; it does not identify the fastest classification rule on each instance. Finite certification budgets must retain the denial count. If the preferences can approach zero without a common positive lower bound, (S9) still holds but gives no bound on the number of unweighted busy rounds by itself.

S4.4 A finite-arm nonstarvation bound

Fix an endpoint T, write $Q = Q_T^{\mathrm{cap}}$, and suppose $\beta_C > 0$ with a positive busy-round lower rate. On $\mathcal{E}$,

$$\mathcal{U}_T(\eta) \leq K_T \min\left\{T, \left\lceil (Q/\beta_C)^2 \right\rceil + 1 + \frac{Q+1}{p_{\min}}\right\}. \tag{S10}$$

For a disabled certification guard use the no-denial area bound instead. To prove (S10), let τ_c be the first time a second arm is available. A singleton is certified positive without observations, so it contributes no unresolved area. The first competitor remains available forever. Its current age gives

$$\mathcal{S}_t \geq \sqrt{t - \tau_c + 1}, \qquad t \geq \tau_c.$$

Thus after at most $\lceil (Q/\beta_C)^2 \rceil + 1$ rounds from τ_c, the certification budget is at least Q. Before that point charge all busy rounds by this length.

If a service proposal is still legal, fewer than Q services have occurred: otherwise all quotas of the prefix would already have been reached, even without counting reward pulls. The ledger plus the next service gap is at most the previous service count plus one, and therefore at most Q. Once the budget reaches Q, no remaining service can be rejected. Over any subsequent interval the inherited phase can only help; phase accounting bounds its remaining busy rounds by $(Q+1)/p_{\min}$. Combining the two parts proves (S10).

If arrivals eventually stop, use the final finite value of Q. The budget eventually exceeds Q and the number of further busy rounds is finite. After the final arrival the confidence rectangles only shrink, so a persistently unresolved arm outside the boundary band would require infinitely many busy rounds, a contradiction. This proves Corollary 3.5. The reward ledger does not enter the argument; continuing reward exploration cannot consume the separately reserved certification budget. The bound is conservative and is not a worst-area matching result for finite budgets. It applies on $\mathcal{E}$ and leaves the boundary band unrestricted, so it is not unconditional almost-sure exact-set consistency. When $\beta_C = 0$ this proof provides no eventual-service guarantee.

S4.5 Why compatibility does not imply universal dominance

Consider two initially available arms, one physical round, and two specified baselines. Baseline A is MC-MOSS with the stated fixed-label initialization and chooses arm 1 first. Baseline B is symmetric Beta$(1,1)$ Thompson sampling and chooses either arm with probability $1/2$. Let an arbitrary unknown-mean policy choose arm 1 with probability p. In the bounded environment $(r_1, r_2) = (3/4, 1/4)$, its expected regret exceeds that of the best specified baseline by $(1-p)/2$. In

environment $(1/4, 3/4)$, the excess is $p/2 - 1/4$. The first decision has the same information in the two environments, and

$$\min_{p\in[0,1]} \max\{(1-p)/2,\, p/2 - 1/4\} = 1/8,$$

attained at $p = 3/4$. Thus no unknown-mean procedure is exactly no worse than both baselines in every environment and at every endpoint with zero remainder. This example concerns these initializations and that literal requirement. It does not preclude comparisons with an adaptation cost, asymptotic noninferiority, spectral minimax rates, or finite-suite numerical improvements.

S4.6 Implementation arithmetic and branch identity

The mathematical contract uses exact interval inversion, phase arithmetic, and budget comparisons. The frozen implementation uses 35 bisection steps with outward mixture endpoints and a 10^{-12} budget-comparison tolerance. Planning regularization was described in Supplement S2. At a cumulative phase close to an integer, floating addition can delay an attempt by one round. These choices belong to the numerical implementation; the displayed exact phase identity is not asserted as a bitwise identity for every floating trajectory.

The production batch rule has two ledgers. If a proposed service fails its certification check, it draws the chosen reward kernel's scores once and maximizes them over the reward-feasible set. It does not necessarily choose a largest-MC-index arm. No unexecuted proposal reveals a reward. Each completed choice is followed by exactly one physical observation and one update to the selected arm's actual count and sum. The earlier single-ledger development branch is not this production rule. The interactive interface implements the same dual-budget decision contract; its corrected ALL-controller wrapper changes only that comparator's observed proposal-cost input and does not alter the F-CCP branch.

S5 Checkpoint-UCB as an attainability witness

The original explicit upper bounds use Checkpoint-UCB as a fixed-precision attainability witness. It is distinct from F-CCP in its evidence, service schedule, and reward rule. Its reporting intervals are formed once, while its reward estimates use every observation. Index arms by arrival order $i = 1, 2, \ldots$, with a fixed order within each batch. Choose $0 < \rho < 1$ before observing rewards and set

$$q_i(\rho) = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil. \tag{S11}$$

Before this lifetime count is reached, arm i has reporting interval $[0, 1]$. At its q_ith observation the interval becomes permanently

$$[\ell_i, u_i] = [\hat{r}_i(q_i) - \rho/2, \hat{r}_i(q_i) + \rho/2] \cap [0, 1]. \tag{S12}$$

Although its completion time is adaptive, the interval uses a predetermined potential prefix. Different precisions ρ_i may also be prescribed before the corresponding arms are observed; data-dependent

checkpoint selection is not covered.

At even times the policy serves the first incomplete checkpoint in arrival order, if one remains. At every other time it maximizes, over all $i \in \mathcal{A}_t$,

$$J_i(t) = \hat{r}_i(N_i(t-1)) + \sqrt{\frac{3\log(2t)}{N_i(t-1)}}. \tag{S13}$$

A zero count gives index $+\infty$; ties are broken by the smallest arrival-order label. Both kinds of pull update the same count and reward sum. After observing the selected reward, define $\underline{r}^*(t) = \max_{i\in\mathcal{A}_t} \ell_i$ and $\overline{r}^*(t) = \max_{i\in\mathcal{A}_t} u_i$, and report

$$\mathcal{P}_t = \{i : \ell_i \geq \overline{r}^*(t) - \varepsilon\}, \qquad \mathcal{N}_t = \{i : u_i < \underline{r}^*(t) - \varepsilon\}, \qquad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t). \tag{S14}$$

Algorithm S1 observes exactly one reward per physical round. Its variables n_i and s_i are the lifetime count and reward sum. An ordinary UCB pull can complete any checkpoint, including one behind the queue's front. The queue discards such completed entries when they reach its front. An interval is computed only when $n_i = q_i$, and is not recomputed when $n_i > q_i$. Reaching q_i forms fixed-precision evidence; it need not produce a determinate report if other arms are still uncertain or the arm lies closer to the boundary than that evidence can resolve.

All arrived arms remain eligible for (S13), including negatively reported arms. The intervals are nonempty on every path, so $\mathcal{P}_t$ and $\mathcal{N}_t$ are disjoint. A count, sum, and interval are stored for every arm; a direct index or report update costs $O(K_t)$ operations. The policy uses no terminal horizon, future arrivals, true means, or reference identities.

Proposition S5.1 (Simultaneous reporting and unconditional reward cost)**.** *On the full bounded model, Checkpoint-UCB satisfies $\mathbb{P}(E_{\mathrm{lab}}) \geq 1 - \delta/2$. If $\mathcal{I}_T$ is its set of designated service time steps, then at every deterministic T,*

$$\mathbb{E}R_T \leq \min\left\{T, K_T + 4\sqrt{3K_T T\log(2T)} + 1 + \mathbb{E}\sum_{t\in\mathcal{I}_T} \Delta_{I_t}(t)\right\}. \tag{S15}$$

Pathwise, $|\mathcal{I}_T| \leq \sum_{i\leq K_T} q_i$.

S5.1 Reporting proof and fixed-prefix evidence

The sum of coverage failures is at most $\sum_i \delta/\{2i(i+1)\} = \delta/2$. On coverage, each endpoint differs from the true mean by at most ρ_i. These intervals are nonempty on every path and only contract once. If $\underline{r}^*(t) = \max_i \ell_i(t)$ and $\overline{r}^*(t) = \max_i u_i(t)$, use

$$\mathcal{P}_t = \{i : \ell_i(t) \geq \overline{r}^*(t) - \varepsilon\}, \quad \mathcal{N}_t = \{i : u_i(t) < \underline{r}^*(t) - \varepsilon\}, \quad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t).$$

The sets $\mathcal{P}_t, \mathcal{N}_t$ are disjoint because $\underline{r}^*(t) \leq \overline{r}^*(t)$ and $\ell_i \leq u_i$. Coverage implies $\mathcal{P}_t \subseteq \mathcal{G}_t^\varepsilon \subseteq \mathcal{A}_t \setminus \mathcal{N}_t$. Moreover, $\mathcal{N}_t$ is increasing as observations and arrivals accumulate: $\underline{r}^*(t)$ cannot decrease and an existing u_i cannot increase.

Algorithm S1 Checkpoint-UCB

Require: $\varepsilon \in (0,1)$, $\delta \in (0,1/2)$, predetermined $\rho \in (0,1)$

$\mathcal{A}^{\mathrm{cur}} \leftarrow \varnothing$, FIFO queue $\mathcal{Q}^{\mathrm{srv}} \leftarrow \varnothing$, $k \leftarrow 0$
for physical time $t = 1, 2, \ldots$ **do**
  for each arm arriving at t, in a fixed within-batch order **do**
    $k \leftarrow k+1$; assign label $i \leftarrow k$; $\mathcal{A}^{\mathrm{cur}} \leftarrow \mathcal{A}^{\mathrm{cur}} \cup \{i\}$
    $n_i \leftarrow 0$, $s_i \leftarrow 0$, $[\ell_i, u_i] \leftarrow [0,1]$
    $q_i \leftarrow \lceil (2/\rho^2) \log\{4i(i+1)/\delta\} \rceil$; append i to $\mathcal{Q}^{\mathrm{srv}}$
  end for
  while $\mathcal{Q}^{\mathrm{srv}} \neq \varnothing$ and $n_{\mathrm{front}(\mathcal{Q}^{\mathrm{srv}})} \geq q_{\mathrm{front}(\mathcal{Q}^{\mathrm{srv}})}$ **do**
    Remove the front of $\mathcal{Q}^{\mathrm{srv}}$
  end while
  if t is even and $\mathcal{Q}^{\mathrm{srv}} \neq \varnothing$ **then**
    $I \leftarrow \mathrm{front}(\mathcal{Q}^{\mathrm{srv}})$ ▷ designated service pull
  else
    $J_i \leftarrow +\infty$ if $n_i = 0$, and $J_i \leftarrow s_i/n_i + \sqrt{3\log(2t)/n_i}$ otherwise, for $i \in \mathcal{A}^{\mathrm{cur}}$
    $I \leftarrow \min \arg\max_{i \in \mathcal{A}^{\mathrm{cur}}} J_i$
  end if
  Pull I once and observe x; $n_I \leftarrow n_I + 1$, $s_I \leftarrow s_I + x$
  if $n_I = q_I$ **then**
    $m \leftarrow s_I/q_I$; $\ell_I \leftarrow \max\{0, m - \rho/2\}$, $u_I \leftarrow \min\{1, m + \rho/2\}$
  end if
  $L \leftarrow \max_{i \in \mathcal{A}^{\mathrm{cur}}} \ell_i$, $U \leftarrow \max_{i \in \mathcal{A}^{\mathrm{cur}}} u_i$
  $\mathcal{P}_t \leftarrow \{i \in \mathcal{A}^{\mathrm{cur}} : \ell_i \geq U - \varepsilon\}$, $\mathcal{N}_t \leftarrow \{i \in \mathcal{A}^{\mathrm{cur}} : u_i < L - \varepsilon\}$
  $\mathcal{V}_t \leftarrow \mathcal{A}^{\mathrm{cur}} \setminus (\mathcal{P}_t \cup \mathcal{N}_t)$; report $(\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t)$
end for

S5.2 Unconditional reward and completion guarantees

A reward bound that does not condition on report coverage. At physical time t, order arms by their first observed rewards. At most t streams have been exposed, each at at most t potential prefixes. A previously unobserved stream is independent of the history selecting it. Applying Hoeffding to each newly exposed stream and union bounding gives

$$P\left\{\exists \text{ observed arm and prefix } n \le t : |\hat{r}_i(n) - r_i| > \sqrt{3\log(2t)/n}\right\} \le 2t^2(2t)^{-6} = \frac{1}{32t^4}.$$

The exposure order is used only for this union bound, not to assert independence among observed counts. An ordinary UCB pull with finite index, on this event's complement, has $\Delta_{I_t}(t) \le 2\sqrt{3\log(2t)/N_{I_t}(t-1)}$. Each arm–pre-pull-count pair is charged at most once; interleaved service observations skip counts. Therefore the ordinary, nonfirst-pull regret is at most

$$4\sqrt{3\log(2T)}\sum_i \sqrt{N_i(T)} \le 4\sqrt{3K_T T\log(2T)}$$

on good rounds. First pulls cost at most K_T, and the sum of expected bad-round losses is less than one. Adding the actual regret of service rounds proves

$$ER_T \le \min\left\{T, K_T + 4\sqrt{3K_T T\log(2T)} + 1 + E\sum_{t\in\mathcal{I}_T}\Delta_{I_t}(t)\right\}.$$

There is no δT term: the reward policy never permanently discards an arm because of a reporting error.

Lemma S5.2 (Completion under arbitrary arrivals)**.** *Fix $0 < \eta < 1$ and use Checkpoint-UCB with the predetermined precision $\rho = \eta/4$. Let $\overline{Q}_t$ be the post-service workload of the virtual FIFO queue that inserts q_i units at arm i's arrival and serves one unit at every even physical round when nonempty. On the simultaneous checkpoint-coverage event, for each fixed deterministic endpoint T,*

$$\mathcal{U}_T(\eta) \le \sum_{t\le T} K_t 1\{\overline{Q}_t > 0\} \le \min\{K_T T, 2K_T\sum_{i\le K_T} q_i\}.$$

Proof. If all current checkpoints are complete, all endpoint errors are at most $\eta/4$ and every arm whose boundary distance exceeds η has a determinate report. A virtual FIFO queue adding q_i work at each arrival and serving one unit at every even round dominates unfinished checkpoints; additional UCB observations only reduce actual work. Its busy periods, including alignment to the first even round, have total length at most $2\sum_i q_i$. Thus on simultaneous coverage the unresolved area is bounded by the queue-weighted sum in the statement, which is at most $K_T T$ and at most $2K_T\sum_{i\le K_T} q_i$. □

S6 Broadcast uncertainty and queue area

Precise claim. Let $0 < \varepsilon \le 1/16$, $0 < \delta \le 1/16$, $0 < \eta < \varepsilon/2$. The reference arm has mean $1/4 + \varepsilon/2$. All candidates have mean $p = 1/4$ in environment zero; environment j raises only candidate j to $p + 2\varepsilon$. Let M_t be the candidate count, a_t the new-candidate count, and

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \quad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$

$$\mathcal{B}_T = \sum_{t \le T} (M_t - 1)_+ 1\{Q_t > 0\}.$$

Every policy reliable on these environments obeys

$$E_0 \mathcal{U}_T(\eta) \ge (3/4 - \delta)\mathcal{B}_T, \qquad P_0\{\mathcal{U}_T(\eta) \ge \mathcal{B}_T/4\} \ge 1/4$$

when $\mathcal{B}_T > 0$.

Proof. The denominator defining q_- is at most $\mathrm{kl}(1/4, 3/8) \le 1/15$, whereas its numerator is at least $\frac{1}{4}\log 4 + \frac{3}{4}\log(4/5) \ge 7/48 > 2/15$. Thus $q_- \ge 2$. The recursion gives, by induction,

$$Q_t = \max_{0 \le u \le t} \left\{ q_- \sum_{r=u+1}^{t} a_r - (t - u) \right\}.$$

If $Q_t > 0$, there is a suffix in which total candidate demand exceeds all physical observation slots. Its arms had no observations before that suffix. Consequently at least one deterministic candidate label j satisfies $E_0 N_j(t) < q_-$. The choice is made from expected counts, not from a realized history. For any other candidate i, membership of i in $\mathcal{P}_t$ is erroneous under environment j. Its probability there is at most δ. The history KL is strictly below $\mathrm{kl}(1/4, \delta)$, so data processing and monotonicity of binary KL force $P_0(i \in \mathcal{P}_t) < 1/4$. Since all candidates are genuinely good under environment zero, $P_0(i \in \mathcal{N}_t) \le \delta$. It follows that $P_0(i \in \mathcal{V}_t) \ge 3/4 - \delta$. These candidates all have boundary distance $\varepsilon/2 > \eta$. Summing at deterministic busy rounds proves the expectation bound, entirely under the one environment zero.

Let Y count candidate unresolved area only at busy rounds with $M_t \ge 2$. Then $Y \le \mathcal{U}_T$, $0 \le Y \le 2\mathcal{B}_T$, and $E_0 Y \ge 11\mathcal{B}_T/16$. If $p = P_0(Y \ge \mathcal{B}_T/4)$, then $11\mathcal{B}_T/16 \le \mathcal{B}_T/4 + (7\mathcal{B}_T/4)p$, so $p \ge 1/4$. □

Queue area. For an integer $q \ge 2$, suppose m candidates arrive by time s and put $D = s + qm - 1$. Then

$$\sum_{t \le D} (M_t - 1)_+ 1\{Q_t(q) > 0\} \ge \frac{q-1}{2} m(m-1).$$

To prove this, remove all subsequent arrivals, which can only decrease both the queue and its weights. The qm service units finish by D. At the nth actual service, at least $\lceil n/q \rceil$ candidates have arrived, so service rounds weighted by $M_t - 1$ sum to at least $\sum_{n=1}^{qm} (\lceil n/q \rceil - 1) = qm(m-1)/2$. The

post-service queue is zero at the last service round of each busy period. Their cumulative arrival counts are distinct integers in $\{1, \dots, m\}$, so removing their weights costs at most $m(m-1)/2$. This proves the claim.

In particular, if along a fixed infinite schedule a policy reliable in every environment has a deterministic area envelope $a_T = o(T)$, then $K_t = o(\sqrt{t})$. Otherwise along s_n there are at least $c\sqrt{s_n}$ candidates. Choose the first $m_n = \lfloor c\sqrt{s_n} \rfloor$; eventually $q_- m_n \le s_n$. The queue bound and the preceding probability statement at $2s_n$ give $P_0(\mathcal{U}_{2s_n} \ge c' s_n) \ge 1/4$, contradicting $\delta \le 1/16$ and the proposed envelope.

S7 Partial recovery, worst area, and growth

Precise claim. Let $m \ge 1$ independent uniform signs specify independent observation laws P_i^+, P_i^-. All nuisance parameters and observations are independent of these signs. Sampling may be adaptive and interleaved, and τ is bounded. At every sign configuration, the probability of any erroneous determinate report is at most $\delta \le 1/16$. Let J_τ be the number of determinate reports and $\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty)$. If $\overline{P}(J_\tau \ge m/2) \ge 1/2$, then

$$\overline{E} \sum_i N_i(\tau) \ge \frac{m}{1024\kappa} \log \frac{m}{\delta}.$$

Proof. Let L_i be the observed positive-versus-negative log likelihood ratio for coordinate i. After conditioning on the full stopped history, the action and stopping kernels cancel. The likelihood factors over coordinates and so does the posterior, even though the sample counts need not be independent. The smallest coordinatewise posterior error is $e_i = (1 + e^{|L_i|})^{-1}$. If S is the reported subset, its conditional probability of complete correctness is at most $\prod_{i \in S}(1 - e_i)$; choosing a non-MAP label cannot improve it. Thus

$$\overline{E}\left[1 - \prod_{i \in S}(1 - e_i)\right] \le \delta.$$

With probability at least $1 - 2\delta$ this product is at least $1/2$ and hence $\sum_{i \in S} e_i \le \log 2$. Intersecting with $J_\tau \ge m/2$ costs at most 2δ and leaves probability at least $3/8$. For $m \ge 4096$, Jensen gives on this intersection

$$\sum_i |L_i| \ge \sum_{i \in S} \{\log(1/e_i) - \log 2\} \ge \frac{m}{2} \log \frac{m}{4 \log 2}.$$

Successive conditional expectations of signed likelihood increments give

$$\overline{E} \sum_i \sigma_i L_i \le \kappa \overline{E} \sum_i N_i(\tau).$$

Finite two-way KL makes the stopped increments integrable. Given the history, $E(\sigma_i L_i \mid H_\tau) =$

$|L_i|(1-2e_i)$, and $2|L_i|e_i \le 1$ for every real L_i. Therefore

$$\kappa \overline{E} \sum_i N_i \ge \overline{E} \sum_i |L_i| - m \ge m \left\{ \frac{3}{16} \log \frac{m}{4 \log 2} - 1 \right\} \ge \frac{m}{256} \log m,$$

where the last inequality holds for $m \ge 4096$.

For the confidence term, fix a sign vector and flip coordinate i only. The event reporting the original sign on coordinate i has original probability at least its determinate-report probability minus δ, and flipped probability at most δ. Average the data-processing inequalities over vectors and coordinates and use joint convexity of binary KL. Since $\overline{E} J_\tau \ge m/4$, this yields

$$\kappa \overline{E} \sum_i N_i \ge m \, \mathrm{kl}(1/4 - \delta, \delta) \ge \frac{m}{256} \log(1/\delta).$$

To check the last scalar inequality, write $L = \log(1/\delta)$. For $L \le 8$, Pinsker gives a lower bound $2(1/4 - 2\delta)^2 \ge 1/32 \ge L/256$. For $L \ge 8$, binary entropy at most $\log 2$ gives $\mathrm{kl}(1/4 - \delta, \delta) \ge 3L/16 - \log 2 \ge L/16$. When $m \ge 4096$, the maximum of the two preceding bounds is at least half their sum. When $m < 4096$, $\log m \le 3 \log(1/\delta)$, and the confidence bound alone suffices. Both cases imply the stated constant $1/1024$. □

Single-window truncation. If such a batch has H rounds with qualifying boundary distance larger than η, set

$$h_* = \min \left\{ H, \left\lceil \frac{m \log(m/\delta)}{1024 \kappa} \right\rceil - 1 \right\}.$$

At every $r \le h_*$ there are at most r batch observations. The recovery lemma's contrapositive makes the probability of half-completion less than $1/2$, so the expected unresolved count is at least $m/4$. Thus batch area has expectation at least $m h_*/4$. Its first-h_*-round area Y lies in $[0, m h_*]$; consequently $\overline{P}(Y \ge m h_*/8) \ge 1/7$. One fixed sign configuration attains at least this probability. If $h_* = 0$, only the trivial zero lower bound is asserted.

S7.1 Worst-schedule area and growth consequences

Suppose $0 < \varepsilon \le 1/8$, $0 < \eta \le \varepsilon/32$, $0 < \delta \le 1/16$, and $K \ge 2$. Let $\mathfrak{A}_{T,K}$ be the smallest area threshold attainable with joint label-and-area probability at least $1 - \delta$, uniformly over reward environments and schedules with $K_T \le K$. The policy must be reliable on the full model.

Upper bound. Use fixed precision $\rho = \eta/4$. Then

$$q_i = \left\lceil 32 \eta^{-2} \log \frac{4i(i+1)}{\delta} \right\rceil \le 99 \eta^{-2} \log(K/\delta), \qquad i \le K.$$

Lemma S5.2 proves, on coverage,

$$\mathcal{U}_T \le \min\{KT, 2K \sum_{i \le K} q_i\} \le 256 \min\{KT, K^2 \eta^{-2} \log(K/\delta)\}.$$

The unconditional reward bound in Section S5.2 and at most $\sum_i q_i$ service pulls also give

$$ER_T \le \min\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\}.$$

Lower bound. At round one release a reference of mean $1/2 + \varepsilon$ and $m = K - 1$ candidates of means $1/2 \pm 2\eta$. Their signs are independent and specify their relative-goodness labels. The reference is always optimal; its known distribution gives no sign information. Boundary distances equal $2\eta > \eta$, and two-way KL is at most $128\eta^2$. Set

$$q_* = \frac{m\log(m/\delta)}{131072\eta^2}, \qquad h = \min\{T, \lceil q_* \rceil - 1\}.$$

Since $\eta \le 1/256$, $q_* \ge m\log(m/\delta)/2 > 1$ and $h \ge \frac{1}{2}\min\{T, q_*\}$. Single-window truncation gives a fixed environment with

$$P\{\mathcal{U}_T \ge mh/8\} \ge 1/7 > \delta.$$

Therefore no valid uniform area quantile can be smaller than $m\min\{T, q_*\}/16$. The comparisons $m \ge K/2$ and $\log(m/\delta) \ge \frac{1}{2}\log(K/\delta)$ yield

$$\mathfrak{A}_{T,K} \ge 2^{-24}\min\{KT, K^2\eta^{-2}\log(K/\delta)\}.$$

The lower bound uses a static first-round burst; it does not attribute its statistical logarithm uniquely to dynamic arrivals.

At fixed task parameters, uniformly over schedules satisfying a terminal count limit $K_T \le k(T)$, sublinear expected regret and sublinear area envelopes are simultaneously attainable if and only if $k(T)^2\log k(T) = o(T)$. Necessity is the last lower bound; sufficiency follows from the displayed upper bounds. This terminal-count-class conclusion allows the worst schedule to depend on T.

Growth on a fixed schedule. At fixed η, δ, the checkpoints are $O(\log(ei))$, so $K_T^2\log(eK_T) = o(T)$ suffices for both sublinear area and expected regret. Combined with the broadcast consequence in Section S6, a path with $K_t \asymp t^\alpha$, $0 < \alpha < 1$, admits both if and only if $\alpha < 1/2$. This is an exponent statement and does not remove the critical logarithm.

A fixed sparse schedule that still defeats a uniform area envelope. Set

$$s_n = 2^{2^n}, \qquad m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor.$$

Release one reference at time one and m_n candidates at each s_n. Earlier batch sizes sum to $o(m_n)$, so $K_{s_n}/\sqrt{s_n} \to 0$; between batches the ratio decreases. Thus $K_t = o(\sqrt{t})$ on this one schedule. For each selected batch n, give the reference mean $1/2 + \varepsilon$, that batch means $1/2 \pm 2\eta$, and other

candidates mean 1/2. Let

$$h_n = \left\lceil \frac{m_n \log(m_n/\delta)}{131072\eta^2} \right\rceil - 1, \qquad T_n = s_n + h_n - 1.$$

For large n, $h_n = o(s_n)$ and the next batch has not arrived. The old history is independent of the new signs. Single-window truncation supplies one fixed configuration for this endpoint with $P(\mathcal{U}_{T_n} \geq m_n h_n/8) \geq 1/7$. But

$$\frac{m_n h_n}{T_n} \asymp_{\eta,\delta} \sqrt{\log s_n} \longrightarrow \infty.$$

No deterministic envelope $a_T = o(T)$ can consequently hold uniformly over reward environments for this schedule. The reward environment may vary with the chosen endpoint; this is exactly the uniform-in-environment statement and is not a claim of simultaneous failure at all endpoints in one fixed reward environment.

S8 Area-constrained temporary tasks

In window e, suppose M_e candidates have gaps $\varepsilon - \sigma_{e,k} d_e$, and a new reference makes their signs irrelevant after that window. Put $w_e = M_e H_e$, $V = \sum_e w_e$, and let p_e be the product-prior probability that at least half of batch e becomes determinate simultaneously before its window ends. Define

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1-\delta-u, \delta), & u < 1-2\delta, \\ 0, & u \geq 1-2\delta, \end{cases}$$

$$g_{m,\delta}(p) = \max\left\{0, \frac{(p-2\delta)_+}{2}\left[\log\frac{m}{4\log 2}\right]_+ - 1, \Phi_\delta(1-p/2)\right\}.$$

Each displayed component is convex and nondecreasing in p.

Proof of the general area-to-regret lower bound. If window e never half-completes, at every round more than half its candidates remain unresolved. Its area is at least $w_e/2$. On area success, $\sum_e w_e 1_{S_e^c} \leq 2a$; outside that event this sum is at most V. Consequently

$$\sum_e w_e p_e \geq (1-\delta)V - 2a.$$

Stop batch e at first half-completion or at its endpoint. The likelihood argument above now intersects the product-posterior event with completion probability p_e, giving intersection probability at least $(p_e - 2\delta)_+$ and expected determinate count at least $M_e p_e/2$. The absolute-likelihood and coordinate-flip derivations therefore give

$$\kappa_e \overline{E} N_e \geq M_e g_{M_e,\delta}(p_e).$$

For the absolute-likelihood term, when the bracketed logarithm is positive its Jensen expression is increasing for subset sizes at least $M_e/2$; when it is nonpositive the stated term is at most zero.

Each observation counted before stopping has regret at least $\varepsilon - d_e$. The signs of all batches belong to one product prior, and samples of different batches are disjoint. Hence

$$\sup_\sigma E_\sigma R_T \geq \inf_{\substack{0\leq p_e\leq 1 \\ \sum_e w_e p_e \geq (1-\delta)V - 2a}} \sum_e \frac{(\varepsilon - d_e)M_e}{\kappa_e} g_{M_e,\delta}(p_e). \tag{S16}$$

This is an information lower relaxation, not a claim that every feasible allocation is achieved by a causal policy. □

Homogeneous consequence. Take $M_e = M$, $H_e = H$, $d_e = d$, and $a = JMH/8$. On success at least $3J/4$ windows half-complete; thus $\sum_e p_e \geq 45J/64$. At least $3J/8$ windows have $p_e \geq 1/2$: otherwise the sum is at most $3J/8 + (5J/8)/2 = 11J/16 < 45J/64$. The Bernoulli means lie in $[1/4, 3/4]$, giving $\kappa_e \leq 32d^2$. The partial-recovery lemma and $\varepsilon - d \geq \varepsilon/2$ yield

$$\sup_\sigma E_\sigma R_T \geq \frac{3J}{8} \frac{\varepsilon M}{2 \cdot 1024 \cdot 32 d^2} \log \frac{M}{\delta} \geq 2^{-20} \frac{\varepsilon JM}{d^2} \log \frac{M}{\delta}.$$

S9 Shared evidence and two endpoint witnesses

S9.1 Shared evidence for area and reward guarantees

Checkpoint-UCB may use a predetermined precision ρ_e for each arriving batch. Consider the heterogeneous repeated-update family: batch F_e has reference mean b_e and M_e candidate means $b_e - \varepsilon \pm d_e$, all in the allowed reward range. The windows partition physical times $1, \ldots, T$; write a_e for the arrival time of batch e, H_e for its window length, and $S_e = \sum_{i \in F_e} q_i(\rho_e)$ for its total checkpoint work; the final batch contains only a reference. Set $M_{-1} = M_J = 0$. Suppose

$$H_e \geq 2S_e, \qquad b_J - b_0 \leq \varepsilon/4, \qquad d_e + \rho_e + \rho_{e+1} < b_{e+1} - b_e, \qquad 2\max_e \rho_e \leq \varepsilon/2. \tag{S17}$$

The neighboring-batch condition is imposed for $e < J$; the window and precision bounds include the final batch. These are conditions on the performance family, not future information supplied to the policy. Put $\mathcal{F} = \{e < J : 2\rho_e < d_e\}$. For any arm that has a positive gap at some time through T, define

$$\gamma_i = \min_{t \leq T : i \in \mathcal{A}_t,\, \Delta_i(t) > 0} \Delta_i(t), \qquad D_i^T = \max_{t \leq T : i \in \mathcal{A}_t} \Delta_i(t).$$

Only such arms are included in the sum containing γ_i below.

Proposition S9.1 (Shared update from lifetime evidence)**.** *Under* (S17)*, on the checkpoint coverage event,*

$$\mathcal{U}_T(\eta) \leq \sum_{e \notin \mathcal{F}} M_e H_e + 2 \sum_{e \in \mathcal{F}} M_e S_e + 2 \sum_{e=0}^{J} (e + 1 + M_{e-1}) S_e. \tag{S18}$$

When the next batch completes its checkpoints, every preceding-batch candidate can be excluded using its saved interval and the new reference lower bound, without another observation of that old

candidate for certification. For the same policy, the unconditional expected regret satisfies

$$\mathbb{E}R_T \le \sum_{e<J}\sum_{k=1}^{M_e}(\varepsilon + d_e)q_{e,k} + \sum_{i:\gamma_i \text{ exists}} D_i^T \left[\left\lceil \frac{12\log(2T)}{\gamma_i^2}\right\rceil - q_i + 1\right]_+ + 1. \tag{S19}$$

The last area term charges temporarily affected references and the preceding candidate cohort, rather than all historical candidates at every update. For $d_e > \eta$, batches in $\mathcal{F}$ complete their current fine membership task; a batch with $d_e \le \eta$ lies inside the evaluation band in its birth window, and its corresponding area term is conservative. An old candidate's upper endpoint and the new reference's lower endpoint differ by more than ε, giving the later negative report without new observations of that candidate. This does not prevent the reward layer from selecting it again.

Every q_i is a lifetime quota. The condition $H_e \ge 2S_e$ ensures that all observations up to that quota occur in the arm's birth window. The first term of (S19) charges candidate observations in that window, while reference observations then have zero regret. The second term counts excess ordinary UCB pulls using distinct pre-pull counts. The expectation is not conditioned on checkpoint coverage. The birth-window condition is essential to this charging and does not cover arbitrary overlapping arrivals.

Proof. The algorithm is not told which arm is the reference or when the next window starts. By induction every batch completes its checkpoints within its first $2S_e$ rounds, before the next batch appears. On coverage, successive reference levels satisfy $b_e + \rho_e < b_{e+1} - \rho_{e+1}$. Once batch e completes, the maximum upper endpoint of all available arms is at most $b_e + \rho_e$. Its candidates are therefore completely classified when $2\rho_e < d_e$.

An old candidate in batch $e-1$ has upper endpoint at most $b_{e-1} - \varepsilon + d_{e-1} + \rho_{e-1}$, whereas the new reference has lower endpoint at least $b_e - \rho_e$. Their difference exceeds ε, excluding the entire preceding candidate batch without any further old-arm observations. Older exclusions persist. Every reference has gap at most $\varepsilon/4$ and total endpoint error at most $\varepsilon/2$, so all references become positively certified. If $F = \{e < J : 2\rho_e < d_e\}$, the area is consequently at most

$$\sum_{e\notin F} M_e H_e + 2\sum_{e\in F} M_e S_e + 2\sum_{e=0}^{J}(e + 1 + M_{e-1})S_e.$$

The terms cover current nonfine candidates, fine candidates, and temporarily unresolved references plus the preceding candidate batch. All earlier candidate batches have already been excluded.

For expected regret, every observation with pre-pull count below q_i occurs in its arm's birth window. Such reference observations have zero regret, and candidate observations cost at most $\varepsilon + d_e$ each. All observations with pre-pull count at least q_i are ordinary UCB pulls. For each arm that has a positive gap by T, let $\gamma_i = \min_{t\le T: i\in\mathcal{A}_t,\, \Delta_i(t)>0} \Delta_i(t)$ and $D_i^T = \max_{t\le T: i\in\mathcal{A}_t} \Delta_i(t)$. Use the good-round concentration bound from Section S5.2. At a good UCB round, any positive-gap pull has pre-pull count at most $12\log(2T)/\gamma_i^2$. Since lifetime counts do not repeat, this gives the

unconditional bound

$$ER_T \le \sum_{e<J}\sum_{k=1}^{M_e}(\varepsilon+d_e)q_{e,k} + \sum_{i:\gamma_i \text{ exists}} D_i^T\left[\left\lceil\frac{12\log(2T)}{\gamma_i^2}\right\rceil - q_i + 1\right]_+ + 1. \tag{S20}$$

The completion-before-next-batch assumption is essential to the first charging term. It is not an assertion about arbitrary overlapping arrivals.

□

S9.2 Rapid and deferred endpoints on one family

Here $J \ge 1$, $M \ge \max\{2, J\}$, $0 < \varepsilon \le 1/8$, $0 < \delta \le 1/16$, $Jv \le \varepsilon/4$, $0 < d \le v/32$, and $\eta = d/2$. The reference means are $b_e = 1/2 + ev$ and the M candidates born in window $e < J$ have means $b_e - \varepsilon + \sigma_{e,k}d$. Each window has length H, and the final window contains only a new reference. Set

$$K = J(M+1)+1, \quad L_K = \log\frac{4K(K+1)}{\delta}, \quad q_f = \lceil 32L_K/d^2\rceil, \quad q_c = \lceil 128L_K/v^2\rceil,$$

$$H = 128(M+J+1)q_f, \qquad T = (J+1)H, \qquad V = JMH.$$

These are analysis quantities: the policy uses the selected precision $\rho_f = d/4$ or $\rho_c = v/8$, not K, H, T or the best-arm identities. Both choices satisfy the separated-window conditions.

The preceding area bound is at most $B_{J,M}q_f$ for the fine policy and $V + B_{J,M}q_c$ for the coarse policy, where

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M+J+1).$$

For $2 \le M$ and $J \le M$, $B_{J,M} \le 12JM^2$. Indeed, $M+1 \le 3M/2$, $2M+1 \le 5M/2$, and $J-1 \le M$ bound the main term by $9JM^2$, and the final term by $3JM^2$. The definition of H gives $B_{J,M}q_f \le V/8$, while $q_c \le q_f$ gives coarse area at most $9V/8$. Both statements hold on the same fixed-checkpoint coverage event of probability at least $1 - \delta/2$.

For rapid maintenance the area-to-regret lower bound applies. For its upper bound, every available arm in this family has gap at most 2ε. Hence, on every path, $R_T \le 2\varepsilon T$. Since $(J+1)(M+J+1)/(JM) \le 5$,

$$2\varepsilon T \le 1280\varepsilon JMq_f.$$

Furthermore $K \le 2M^2$, $K+1 \le 3M^2$ imply $L_K \le 5\log(M/\delta)$, and $q_f \le 165d^{-2}\log(M/\delta)$. Thus

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

This upper bound uses the stated horizon and must not be extended unchanged to arbitrarily long windows.

For the coarse policy, within-checkpoint candidate regret is at most $2\varepsilon JMq_c$. Candidates have $\gamma_i \ge \varepsilon - d \ge 3\varepsilon/4$, $D_i^T \le 2\varepsilon$; old references have $\gamma_i \ge v$, $D_i^T \le Jv$. Dropping the favorable quota

subtraction in (S20) and absorbing integer terms gives

$$\sup_{\Theta} ER_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

At fixed $J, M, \varepsilon, v, \delta$, $d \downarrow 0$ makes $T = \Theta(d^{-2})$. Consequently the rapid endpoint is $\Theta(d^{-2})$ and the deferred endpoint is at most $O(\log(1/d))$. No matching lower bound for the latter endpoint is asserted.

S9.3 Earlier intermediate configurations and disclosed-reference diagnostic

In the homogeneous repeated-update family, define

$$B_{J,M} = 2(M+1)\{J(2M+1) + J(J-1)/2\} + 2(M+J+1).$$

If a predetermined set of s of the J candidate batches uses precision $\rho_f = d/4$ and the rest uses $\rho_c = v/8$, the shared-evidence bound gives

$$\mathcal{U}_T(\eta) \le (J-s)MH + B_{J,M}q_f$$

on checkpoint coverage. The unconditional reward bound gives

$$ER_T \le 2\varepsilon M\{sq_f + (J-s)q_c\} + 64\left(\frac{JM}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

Every adjacent precision pair satisfies $d + \rho_e + \rho_{e+1} < v$, and every window completes its checkpoints, so the proof follows by applying the same batchwise accounting. The choice of which batches receive fine service must be made without their reward data. These are attainable points, not a proof of optimality at every area budget. They are distinct from the activation-ranked construction in Section S13, which uses a different window contract.

An additional diagnostic gives each batch's reference identity to the policy and discloses that its role is best within its own window. Its numerical mean remains unknown. Sample the reference to its checkpoint, then its candidates to theirs, and otherwise sample the latest reference. Keep the same reporting comparisons. For windows long enough to complete this work, the regret in the specified family is pathwise

$$R_T = \sum_{e,k} (\varepsilon - \sigma_{e,k} d) q_{e,k}.$$

Reference sampling and idle exploitation have zero regret, whereas each candidate checkpoint occurs in its birth window and pays its displayed gap. The coarse version has $R_T \le 2\varepsilon JMq_c$ with no extra $\log T$ reward term. Comparing with the rapid lower bound isolates a squared v/d cost ratio. This additional role information makes the allocation problem easier; this policy is not the main unknown-reference algorithm. Its interval-based label reliability still holds on the full bounded model, while its reward statement only holds on the disclosed-role performance family.

S10 Regret quantiles under the same maintenance protocol

The following result strengthens the expectation lower bound. Its statement uses the same reporting event and the same cumulative unresolved area; it does not introduce a separate deadline requirement. Write $\overline{P}$ for the product prior on all candidate signs in the repeated-update family.

Lemma S10.1 (Completion with a bounded candidate-sampling budget)**.** *Consider one batch of m candidates in its validity window, under the assumptions of the partial-recovery lemma. Let τ be the first round at which at least $m/2$ candidates have determinate reports, or the end of the window if this does not occur. Let S denote that completion occurs, and let $N(\tau)$ count only samples of this batch. Put*

$$L = \frac{m \log(m/\delta)}{1024\kappa}, \qquad q = \lfloor L/2 \rfloor.$$

If $L \geq 4$, then

$$\overline{P}\{S,\ N(\tau) \leq q\} < \frac{1}{2}.$$

Proof. Stop at the earliest of τ, the window endpoint, and the first round with $q+1$ candidate samples. At this bounded stopping time ζ, the number of candidate samples is at most $q+1 < L$. On $\{S, N(\tau) \leq q\}$, the budget stopping condition has not fired before τ, so $\zeta = \tau$ and at least $m/2$ reports are determinate. If the displayed event had probability at least $1/2$, the partial-recovery lemma at ζ would give $\overline{E}N(\zeta) \geq L$, contradicting the deterministic bound $N(\zeta) \leq q+1 < L$.

The extra sample in the stopping rule is needed. Stopping exactly at the qth candidate sample would miss a completion that uses q candidate samples and later observations of the reference arm. □

Theorem S10.2 (Regret quantiles required by rapid maintenance)**.** *In the repeated-update family with J validity windows, M candidates per window, $M \geq \max\{2, J\}$, $Jv \leq \varepsilon/4$, $d \leq v/32$, $\eta = d/2$, and $0 < \delta \leq 1/16$, suppose a policy is δ-reliable on the family and, for every sign configuration σ,*

$$P_\sigma\{E_{\mathrm{lab}},\ \mathcal{U}_T(\eta) \leq JMH/8\} \geq 1 - \delta.$$

Then there is one fixed configuration σ for which

$$P_\sigma\left\{R_T \geq 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \geq \frac{11}{80}.$$

This conclusion permits the policy to know the reference identities, the arrival schedule, and the horizon.

Proof. Let S_e denote completion of at least half of the current candidates before the next reference arrives, and let N_e be that batch's sample count at its first such completion, or at the window endpoint when completion fails. Set

$$L_0 = \frac{M \log(M/\delta)}{32768 d^2}, \qquad q_0 = \lfloor L_0/2 \rfloor.$$

The two directional Bernoulli divergences are at most $32d^2$, so the partial-recovery lower bound is at least L_0. The assumptions imply $d \le 1/1024$ and $M \ge 2$, hence $L_0 > 4$. The preceding budget argument, with $q_0 + 1 < L_0$, gives

$$\overline{P}(F_e) < \frac{1}{2}, \qquad F_e := \{S_e, N_e \le q_0\}.$$

The argument is valid with all previous-window information included in the history. Those observations and later reference distributions carry no information about this batch's independent signs.

On the area-success event, any window without S_e contributes at least $MH/2$ area. Consequently $\sum_e 1_{S_e} \ge 3J/4$ on an event of prior probability at least $1 - \delta$. If $F = \sum_e 1_{F_e}$, then $\overline{E}F \le J/2$ and Markov's inequality gives

$$\overline{P}\{F \le 5J/8\} \ge 1/5.$$

No independence across windows is asserted or needed. Intersecting these two events has probability at least $1/5 - \delta \ge 11/80$. On their intersection at least $J/8$ windows complete with $N_e > q_0$, and therefore with $N_e > L_0/2$. Each candidate observation in its birth window has gap at least $\varepsilon - d \ge \varepsilon/2$. These are observations of distinct batches in disjoint windows. Pathwise on that intersection,

$$R_T \ge \frac{\varepsilon}{2}\frac{J}{8}\frac{L_0}{2} = 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

A mixture probability at least $11/80$ implies that one fixed configuration has at least that probability. Thus no different worst environment is selected for different windows. □

Corollary S10.3 (Rapid maintenance with a joint regret quantile)**.** *Let $\mathfrak{R}_\Theta^{\mathrm{joint}}(a)$ be the infimum of r over policies that are reliable on the full bounded model and satisfy*

$$\inf_{\vartheta\in\Theta} P_\vartheta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le a, R_T \le r\} \ge 1 - \delta.$$

For the repeated-update horizon $H = 128(M + J + 1)\lceil 32d^{-2}\log\{4K(K+1)/\delta\}\rceil$, where $K = J(M+1)+1$, universal positive constants c, C satisfy

$$c\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta^{\mathrm{joint}}(JMH/8) \le C\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}.$$

Proof. For the lower bound, every policy feasible for the joint criterion satisfies the area premise of Theorem S10.2. Since $11/80 > \delta$, a joint regret bound strictly below its threshold is impossible. Taking an infimum gives the claimed lower bound. For the upper bound, the fine-checkpoint policy has the required area bound on its simultaneous interval-coverage event, whose probability is at least $1 - \delta/2$. Independently of coverage, every available arm in the family has gap at most 2ε, so $R_T \le 2\varepsilon T$ on every sample path. The specified horizon and $J \le M$ yield the displayed upper order, with the same universal constant as the rapid-maintenance upper bound. □

S11 General lower-bound extensions

S11.1 Compatible-window truncated area

Proposition S11.1 (Additive truncated area in one compatible experiment)**.** *Consider finitely many batches indexed by e. Batch e contains m_e independent signs and has a deterministic validity window W_e of length H_e that begins at that batch's arrival round. During that window each sign is its arm's relative-goodness label and its boundary distance exceeds η. The signs across all batches are independent, other distributions do not encode them, and no information about batch e is available before its arrival. Samples remain interleaved and adaptive. Suppose every sign configuration has familywise reporting error at most $0 < \delta \le 1/16$. Let $0 < \kappa_e < \infty$ bound the two directional divergences in batch e, and define*

$$h_e = \min\left\{H_e, \left\lceil \frac{m_e \log(m_e/\delta)}{1024\kappa_e} \right\rceil - 1\right\}, \qquad S = \sum_e m_e h_e.$$

When $S > 0$, one fixed full sign configuration satisfies

$$P\{\mathcal{U}_T(\eta) \ge S/8\} \ge 1/7,$$

provided every selected window lies before T and every candidate–time pair is counted only once. The windows may overlap when they concern distinct arms and their stated qualification structure is jointly realizable.

Proof. At the rth round after batch e arrives, $r \le h_e$, the entire experiment has supplied at most r samples of that batch. The contrapositive of the partial-recovery lemma shows that at least half the batch is determinate with probability less than $1/2$. Its expected unresolved count is therefore at least $m_e/4$. Let Y sum unresolved candidate–time pairs over the first h_e rounds of all selected windows. All these pairs have boundary distance larger than η, so $Y \le \mathcal{U}_T(\eta)$, while $0 \le Y \le S$ and $\overline{E}Y \ge S/4$. If $p = \overline{P}(Y \ge S/8)$, then $S/4 \le S/8 + (7S/8)p$, giving $p \ge 1/7$. Select a fixed configuration from the common product prior. Compatibility of the windows is essential: this proof cannot sum bounds from mutually incompatible reference environments. □

S11.2 General information-allocation relaxation

This result is useful beyond the separated family, but does not assert attainability. Fix a reference environment $\boldsymbol{r}$, endpoint T, and a deterministic query set

$$\mathcal{Q} \subseteq \{(i,t) : i \in \mathcal{A}_t, |\Delta_i(t) - \varepsilon| > \eta\}, \qquad W = |\mathcal{Q}|.$$

For each query choose any allowed alternative $\boldsymbol{r}'$ that reverses arm i's membership at time t. Put $u_{i,t} = P_{\boldsymbol{r}}(i \in \mathcal{V}_t)$ and

$$\Phi_\delta(u) = \begin{cases} \mathrm{kl}(1-\delta-u, \delta), & 0 \le u < 1-2\delta, \\ 0, & 1-2\delta \le u \le 1. \end{cases}$$

This function is convex and nonincreasing: the positive branch is an affine composition of binary KL, and its value and derivative vanish at the join.

Proposition S11.2 (One protocol for area and information)**.** *For any reliable policy,*

$$\sum_j E_{\boldsymbol{r}} N_j(t)\,\mathrm{kl}(r_j, r'_j) \geq \Phi_\delta(u_{i,t}).$$

If its joint area budget at $\boldsymbol{r}$ is a, then $\sum_{(i,t)\in\mathcal{Q}} u_{i,t} \leq a + \delta W$. Consequently $E_{\boldsymbol{r}} R_T$ is at least the value of

$$\inf_{x,u} \sum_{s\leq T} \sum_{j\in\mathcal{A}_s} \Delta_j(s) x_{s,j},$$

subject to $x_{s,j} \geq 0$, $\sum_{j\in\mathcal{A}_s} x_{s,j} = 1$, no prearrival allocations, the preceding unresolved-probability constraint, and for every query and chosen alternative,

$$\sum_{s\leq t} \sum_j x_{s,j}\,\mathrm{kl}(r_j, r'_j) \geq \Phi_\delta(u_{i,t}).$$

For non-Bernoulli laws replace binary arm divergences by their distributional KL.

Proof. Let F be the event of issuing, at (i,t), the determinate label correct in the reference environment. Then $P_{\boldsymbol{r}}(F) \geq 1 - u_{i,t} - \delta$, whereas $P_{\boldsymbol{r}'}(F) \leq \delta$. The history-KL identity and data processing give the claimed constraint whenever $1 - u_{i,t} - \delta > \delta$; otherwise nonnegativity suffices. This is the precise point where the truncated KL is needed. On area success the queried unresolved sum is at most a; outside success it is at most W. Taking expectations gives $a + \delta W$. The actual occupation probabilities $x_{s,j} = P_{\boldsymbol{r}}(I_s = j)$ satisfy every constraint, with objective exactly $E_{\boldsymbol{r}} R_T$. □

One observation can enter many information constraints but is charged once in the objective. Neither convex feasibility nor an oracle allocation proves that a policy with unknown means and future arrivals attains this relaxation. Dropping δW would also change the stated risk protocol.

S12 Schedule information and causal comparison

Two comparison classes use different schedule information. For a fixed schedule τ, write $\mathfrak{R}^{\mathrm{sch}}_{T,\tau}(a;\Theta)$ for the infimum of worst-case expected regret when the policy can be designed knowing T and τ. It must remain label-reliable on the full bounded model, and its area requirement is imposed on Θ. In $\mathfrak{R}^{\mathrm{causal}}_{T,\mathcal{T}}(a;\Theta)$, one policy sees only arrivals that have already occurred and must satisfy the area requirement on every schedule in $\mathcal{T}$. Neither class is told the realized reward environment. The following separation concerns simultaneous comparison with the first class, rather than nonexistence of an optimum in the second class.

Theorem S12.1 (A schedule-information separation)**.** *Fix $\varepsilon = 1/8$, $v = 1/32$, $0 < \delta \leq 1/16$, and take $0 < d \leq 1/1024$, with $\eta = d/2$. There are three Bernoulli arms with means*

$$r_0 = 1/2, \qquad r_1 = 1/2 - \varepsilon + \sigma d, \qquad r_2 = 1/2 + v, \qquad \sigma \in \{-1, +1\}.$$

Let Θ be the two sign environments specified above. Arms 0 and 1 arrive at time 1. Define

$$Q_f = \left\lceil 32d^{-2}\log(24/\delta)\right\rceil, \quad h = 16Q_f, \quad L_d = \frac{\mathrm{kl}(1-\delta,\delta)}{\mathrm{kl}(1/2-\varepsilon-d,1/2-\varepsilon+d)},$$

$$w = \lfloor L_d/2\rfloor, \qquad a = h+w, \qquad T = 2(a+1).$$

The early schedule has $\tau_2 = h+1$, and the late schedule has $\tau_2 = T$. They have the same endpoint, area budget, final arm count, and task parameters. Every one causal policy satisfying the joint label–area requirement on both signs of the late schedule obeys

$$E_{-,\mathrm{early}} R_T \geq \varepsilon(L_d/2-1) = \Omega_\delta(d^{-2}).$$

For all sufficiently small d, both schedules have feasible area requirements, while

$$\mathfrak{R}^{\mathrm{sch}}_{T,\tau^{\mathrm{early}}}(a;\Theta) = O_{\varepsilon,v,\delta}(\log(1/d)).$$

Consequently no universal constant gives one unknown-future policy a simultaneous constant-factor comparison with the separately optimized schedule-informed values. This remains so if the policy knows T, a, the two possible schedules, the final count, and the performance family.

Proof. On the late schedule, arm 1's membership through $t_0 = a+1 < T$ equals its sign and has boundary distance $d > \eta$. If it has never received a determinate report by t_0, it alone contributes $a+1$ unresolved area. Joint label–area success therefore forces a correct report by t_0. Let F be the event that arm 1 has been reported non-near-optimal by t_0. Then $P_-F \geq 1-\delta$ and $P_+F \leq \delta$. Applying the history-KL identity in (S1) gives

$$E_{-,\mathrm{late}} N_1(t_0) \geq L_d.$$

Only $w+1$ observations can occur from $h+1$ through t_0, whence

$$E_{-,\mathrm{late}} N_1(h) \geq L_d - w - 1 \geq L_d/2 - 1.$$

The arrival and reward histories through h have identical laws under the early and late schedules. The same causal policy therefore has the same law of $N_1(h)$ on both. Every pull of arm 1 in this common prefix costs at least ε, proving the lower bound. The Bernoulli parameters remain in the interior of $[0,1]$, so $L_d \asymp_\delta d^{-2}$.

It remains to verify feasibility and the early upper bound. On the late schedule, use Checkpoint-UCB with $\rho_f = d/4$. Each of the first two quotas is at most Q_f, and even-round service completes them within $4Q_f$ physical rounds. On checkpoint coverage the first two arms are then classified, so their initial area is at most $8Q_f$. The last-round arrival contributes at most 3 additional arm–time pairs. Thus $\mathcal{U}_T \leq 8Q_f + 3 < a$ for small d.

On the early schedule, use instead $\rho_c = v/8$ and set

$$Q_c = \left\lceil 128 v^{-2} \log(48/\delta) \right\rceil .$$

All three quotas are at most Q_c, independent of d. The initial coarse evidence is complete within $4Q_c$ rounds. It certifies the old reference positively; arm 1 may remain unresolved. The new reference completes its evidence within $2Q_c$ rounds of arrival. Since $d + 2\rho_c < v$, its lower endpoint and arm 1's saved upper endpoint exclude arm 1, while both reference arms are positively certified. Consequently

$$\mathcal{U}_T \le h + 10Q_c \le h + w = a$$

for all sufficiently small d. Both feasibility statements hold on the fixed-checkpoint coverage event, whose probability is at least $1 - \delta/2$.

The early coarse policy uses at most $3Q_c$ designated observations. Every positive gap in this family is at least v. The unconditional UCB counting argument of Section S5.2 therefore bounds its expected regret by $3Q_c + O_v(\log T)$, and $T \asymp_\delta d^{-2}$. Both policies report through valid full-model confidence intervals; they do not label arms by inserting the performance family's means. The lower bound already holds in a fixed negative-sign environment; no endpoint-dependent selection from a prior is required here. □

The result does not rule out causal minimax optimization over a common schedule class, nor optimization on a known schedule. It also gives no claim that F-CCP attains the causal joint frontier.

Finite spectrum protection and very rapid maintenance. In the original repeated-update family, fix $J, M, \varepsilon, v, \delta$ and let $d \downarrow 0$ under its prescribed endpoint. Then $T \asymp d^{-2}$ and K is fixed, while the rapid area budget requires $\Omega(d^{-2})$ expected regret. Any strategy with a fixed-constant bound $ER_T \le C\mathcal{S}_T = O(\sqrt{T}) = O(d^{-1})$ cannot satisfy that rapid budget. Thus a fixed finite protection coefficient cannot cover this regime; the allowed coefficient must grow at least on the d^{-1} scale or the budget mechanism must change. This is a compatibility consequence of the existing upper and lower bounds, not a simultaneous attainability statement.

S13 Activation-budgeted intermediate maintenance

This section gives a separate theoretical construction, Activation-Budgeted Maintenance (ABM). Its confidence allocation is by fine-task activation rank. ABM is neither F-CCP nor the shared-certificate scheduler evaluated during development. Its window contract below is different from the fixed horizon used for the earlier two endpoints.

S13.1 Recovery of an arbitrary number of coordinates

Theorem S13.1 (Arbitrary-subset simultaneous recovery)**.** *Let $m \geq 1$ independent uniform signs specify independent laws P_i^+, P_i^-. Other observations carry no information about the signs. Sampling may be adaptive and interleaved, with independent policy randomization and a bounded stopping time τ. Suppose*

$$0 < \kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} < \infty.$$

At stopping, a data-dependent subset S receives sign reports, with probability of any reported error at most $0 < \delta \leq 1/16$ under every sign configuration. If $1 \leq s \leq m$ is an integer and $\overline{P}(|S| \geq s) \geq 1/2$, then

$$\overline{E} \sum_i N_i(\tau) \geq \frac{s}{48\kappa} \log \frac{s}{\delta}.$$

The bar uses one independent-sign product prior, solely as a proof device.

Proof. Let L_i be coordinate i's observed positive-versus-negative log likelihood ratio. Conditional on the full stopped history, action, stopping, and reporting kernels cancel. The likelihood and posterior factor over signs even though the random sample counts are dependent. The minimum posterior error for sign i is

$$e_i = (1 + e^{|L_i|})^{-1} \in (0, 1/2].$$

Any actual report has conditional probability of complete correctness at most $\prod_{i \in S}(1 - e_i)$, hence

$$\overline{E}\left[1 - \prod_{i \in S}(1 - e_i)\right] \leq \delta.$$

With probability at least $3/4$, this product is at least $1 - 4\delta$. On that event,

$$\sum_{i \in S} e_i \leq v_\delta := -\log(1 - 4\delta) \leq 16\delta/3.$$

Its intersection with $|S| \geq s$ has probability at least $1/4$. Define

$$f(e) = (1 - 2e)\log((1 - e)/e), \qquad 0 < e \leq 1/2.$$

The function is nonnegative, decreasing, and convex, with $f''(e) = 1/\{e^2(1 - e)^2\}$. Conditional on the stopped history, $E(\sigma_i L_i \mid H_\tau) = f(e_i)$. Taking conditional expectations of the successive signed likelihood increments gives

$$\overline{E} \sum_i f(e_i) = \overline{E} \sum_i \sigma_i L_i \leq \kappa \overline{E} \sum_i N_i(\tau).$$

Bounded stopping and finite two-way KL make these increments integrable. On the intersection

event, write $k = |S| \geq s$. Convexity and monotonicity give

$$\sum_i f(e_i) \geq k f(v_\delta/k) \geq s f(v_\delta/s).$$

Here $v_\delta/s \leq 1/3$, so

$$f(v_\delta/s) \geq \frac{1}{3} \log \frac{s}{8\delta} \geq \frac{1}{12} \log \frac{s}{\delta}.$$

The last step uses $s/\delta \geq 16$. Multiply by the intersection probability $1/4$ and use the information-drift bound. In particular, no subtraction proportional to all m coordinates is needed when $s \ll m$. Zero posterior errors can be handled by limits; finite stopping and the integrability just noted exclude uncontrolled infinite expectations. □

The half-recovery lemma in Section S7 remains the dependency for the older constants and probability bounds. The new constant is used only for the arbitrary-recovery extension below.

S13.2 Area-to-cost lower bound with a free window length

Let

$$J \geq 1, \quad M \geq \max\{2, J\}, \quad 0 < \varepsilon \leq 1/8, \quad 0 < \delta \leq 1/16, \quad Jv \leq \varepsilon/4, \quad 0 < d \leq v/32, \quad \eta = d/2.$$

Set $m = JM$ and $K = J(M+1)+1$. At time $1 + eH$, for $e = 0, \ldots, J-1$, release a reference of mean $b_e = 1/2 + ev$ and M candidates of means

$$r_{e,k} = b_e - \varepsilon + \sigma_{e,k} d.$$

At time $1 + JH$ release only the final reference b_J. Include all sign configurations and within-batch permutations in the performance family, denoted Θ_H. Set $T = (J+1)H$ and $V = mH$. Reference identities are not given to the upper-bound policy. Every candidate has regret gap at least $\varepsilon - d \geq \varepsilon/2$ at every time after its arrival.

Theorem S13.2 (Necessary cost at any area below V)**.** *For $0 \leq a < V$ and $k(a) = \lceil (V-a)/H \rceil$,*

$$\mathfrak{R}_{\Theta_H}(a) \geq \frac{\varepsilon k(a)}{3072 d^2} \log \frac{k(a)}{\delta}.$$

The assertion includes the case of an infeasible area budget, when the left side is infinite.

Proof. For each candidate retain its first determinate label within its own birth window, if there is one. Only that label estimates its sign; a negative report after the next reference arrives is not an answer to the expired sign question. A candidate never labeled in its window contributes H area, because its boundary distance there is $d > \eta$. Thus joint label–area success forces at least $k(a)$ retained labels, all correct.

Use one product prior on all m signs and apply Theorem S13.1 at deterministic stopping time T. The recovery-count event has probability at least $1 - \delta > 1/2$. Interior Bernoulli parameters

give two-way KL at most $32d^2$, so

$$\overline{E}\sum_{e,k} N_{e,k}(T) \geq \frac{k(a)}{1536d^2}\log\frac{k(a)}{\delta}.$$

This count includes observations after each validity window. Such observations can still encode the old sign and must remain in the KL accounting; their regret is still at least $\varepsilon/2$ each. The expected regret under the common prior therefore has the stated lower bound. At least one fixed sign configuration attains the prior average, and taking the infimum over feasible policies completes the proof. □

S13.3 ABM and predictable fresh-block evidence

ABM takes $\varepsilon, \delta, \rho_c, \rho_f$ and an integer fine-candidate budget $s \geq 0$. Its operational inputs contain no horizon, window length, future arrivals, means, or reference identities. It records arrival batches and keeps all observations in lifetime reward estimates. For arrival-order arm i, its coarse quota is

$$q_i^c = \left\lceil 2\rho_c^{-2}\log\{4i(i+1)/\delta\}\right\rceil.$$

When this lifetime count is first attained, its first q_i^c observations give a saved coarse interval with radius $\rho_c/2$, clipped to $[0,1]$. These coarse errors total at most $\delta/2$.

After all coarse checkpoints of a batch complete, choose its largest saved coarse mean, breaking ties by arrival-order label. From the remaining arms choose, in label order, at most the remaining fine-candidate budget. Whenever at least one candidate is selected, also activate a fine task for that batch's empirical best arm. There are at most s selected candidates and at most s associated reference tasks. Rank these fine activations by $r = 1, 2, \ldots$, placing the empirical best first within its batch. Activation rank r is assigned the quota

$$q_r^f = \left\lceil 2\rho_f^{-2}\log\{4r(r+1)/\delta\}\right\rceil.$$

Its count starts at zero immediately after activation. Only subsequently observed rewards of that arm enter this block, including ordinary UCB observations. The observation which completed the coarse batch and triggered activation is excluded from its new fine block. On reaching q_r^f fresh observations, save a fine interval of radius $\rho_f/2$. Old data remain in the lifetime reward estimates and coarse evidence; they are not recycled as an activation-selected fixed prefix.

At even physical rounds, coarse FIFO tasks have priority, followed by fine tasks in activation order. At other rounds, or when both queues are empty, use the full-history UCB rule (S13). Both kinds of observation update the same lifetime statistics, and an active fine block receives an observation only if it was active before that observation. Reports use the intersection of available coarse and fine intervals and the comparisons (S14); negative reporting does not remove reward eligibility. An empty intersection triggers all-unresolved reports thereafter, while the prescribed reward and queue operations remain defined.

Algorithm S2 Activation-Budgeted Maintenance (ABM)

Require: $\varepsilon, \delta, \rho_c, \rho_f \in (0,1)$; integer $s \geq 0$

Initialize empty coarse and fine FIFO queues, remaining budget s, fine rank $r = 0$
for physical time $t = 1, 2, \ldots$ **do**
    Receive the new batch; assign arrival-order labels and initialize lifetime counts and sums
    Set each new arm's coarse quota q_i^c; append it to the coarse queue
    Remove completed tasks from the fronts of both queues
    if t is even and the coarse queue is nonempty **then**
        Select its front arm
    else if t is even and the fine queue is nonempty **then**
        Select its front arm
    else
        Select a largest full-history UCB index, using smallest-label ties
    end if
    Observe one reward and update the selected arm's lifetime statistics
    Update its fine block if active before this observation and not yet complete
    At a quota crossing, form its corresponding coarse or fine interval
    if this observation newly completes all coarse checkpoints in its batch **then**
        Let j be the batch's largest saved coarse mean, with smallest-label ties
        Select other batch arms in label order, up to the remaining candidate budget
        if at least one candidate was selected **then**
            Activate j first, then the selected candidates, each with the next rank r
            Give rank r quota q_r^f, zero fresh count and sum, and append its task
            Decrease the remaining budget by the number of selected candidates
        end if
    end if
    Report from saved-interval intersections; an empty intersection gives persistent all-unresolved reports
end for

The fresh-block Hoeffding bound remains valid conditional on activation history. The next unrevealed suffix of the selected iid arm stream retains its iid law, even when its identity and activation time depend on past data. Apply the conditional bound first at bounded activation times and then take expectations; a task never activated has no failure event. Fine-rank errors total at most $\sum_{r\geq 1}\delta/\{2r(r+1)\} = \delta/2$. Together with coarse coverage, this gives full-model all-time label reliability at least $1-\delta$. This argument requires fresh post-activation observations and does not validate reassigning a smaller error budget to a previously selected old prefix.

S13.4 A sufficiently separated window contract

On Θ_H, choose $\rho_c = v/8$ and $\rho_f = d/4 = \eta/2$. The coarse precision is a selected service parameter, not an estimate that automatically learns v. Define the analysis quantities

$$Q_c = \left\lceil 128v^{-2}\log\frac{4K(K+1)}{\delta}\right\rceil, \quad Q_f(r) = \left\lceil 32d^{-2}\log\frac{4r(r+1)}{\delta}\right\rceil,$$

$$L_c = 2(M+1)Q_c, \quad L = L_c + 2(M+1)Q_f(2m),$$

$$B_{\mathrm{rec}} = L_c\left\{\frac{(J+1)(J+2)}{2} + JM\right\}, \qquad H \geq 4(L + B_{\mathrm{rec}}).$$

Here B_{rec} is a recovery-area constant; it is not a best mean, reward sum, or F-CCP budget. The displayed condition defines this performance family and is not future information given to ABM.

Theorem S13.3 (ABM area and unconditional reward cost)**.** *For $0 \leq s \leq m$, ABM on this family satisfies*

$$P\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \leq (m-s)H + sL + B_{\mathrm{rec}}\} \geq 1-\delta.$$

For $s \geq 1$,

$$\sup_{\Theta_H} ER_T \leq R_0(T) + 4\varepsilon s Q_f(2s),$$

where

$$R_0(T) = 2\varepsilon K Q_c + 128\left(\frac{m}{\varepsilon} + \frac{J^2}{v}\right)\log(2T) + 1.$$

For $s = 0$, the upper bound is $R_0(T)$ without a fine-layer term.

Proof. For every reward path, a batch requires at most $(M+1)Q_c$ coarse observations and $(M+1)Q_f(2m)$ fine fresh observations. Even-round service, helped by ordinary observations, completes them within L rounds of birth. Since $H \geq 4(L + B_{\mathrm{rec}})$, induction over batches leaves no queue backlog at the next arrival. This workload statement does not rely on choosing the correct empirical reference.

On simultaneous coverage, each candidate is below its batch reference by at least $\varepsilon - d$, so the saved coarse means select the true reference. There are exactly s selected candidates. Upon fine completion, the selected candidate and reference endpoints have errors at most ρ_f; $2\rho_f < d$ resolves their birth-window labels. No old coarse upper endpoint can obstruct this fine comparison: successive reference spacing satisfies $v > \rho_c + \rho_f$, and all candidates are farther below the current

reference. Thus these s candidates contribute at most sL in their birth windows, while the others contribute at most $(m-s)H$.

The new batch's coarse evidence completes within L_c rounds. Because $d + 2\rho_c < v$, its reference lower endpoint and the preceding batch's saved candidate upper endpoints exclude that older cohort without further old-candidate observations. Older exclusions persist. Every reference has gap at most $\varepsilon/4$ and combined coarse endpoint error at most $2\rho_c < \varepsilon/2$; all references are therefore positively certified after the current coarse layer completes. In window $e = 0, \dots, J$, this recovery can temporarily affect at most $e+1$ references and the preceding candidate batch. Its total area is bounded by

$$L_c \sum_{e=0}^{J} (e + 1 + M_{e-1}) = B_{\mathrm{rec}}, \qquad M_{-1} = M_J = 0.$$

This proves the joint area bound.

For expected regret, do not condition on interval coverage. Every observation below its coarse quota is charged among at most KQ_c observations, each with gap at most 2ε. All designated fine observations beyond these coarse counts are bounded by $2sQ_f(2s)$, and each again costs at most 2ε. The remaining observations are ordinary full-history UCB pulls. Candidate arms have minimum positive gap at least $3\varepsilon/4$ and maximum gap at most 2ε; older references have minimum positive gap at least v and maximum gap at most Jv. The distinct-count argument in Section S5.2, with its summable bad-round loss below 1, is bounded by $128(m/\varepsilon + J^2/v)\log(2T) + 1$. First observations were already charged to the coarse quotas. These counts remain valid on failure paths, including paths on which a wrong empirical reference is chosen. This gives the unconditional reward bound. □

S13.5 A matching interval with an explicit additive cost

Theorem S13.4 (Intermediate area budgets on Θ_H)**.** *Set $A_{\min} = mL + B_{\mathrm{rec}}$. For $A_{\min} \le a < V = mH$, let*

$$s(a) = \left\lceil \frac{V + B_{\mathrm{rec}} - a}{H - L} \right\rceil.$$

Then $1 \le s(a) \le m$, ABM with candidate budget $s(a)$ attains area budget a, and

$$\frac{\varepsilon k(a)}{3072 d^2} \log \frac{k(a)}{\delta} \le \mathfrak{R}_{\Theta_H}(a) \le R_0(T) + 1024 \frac{\varepsilon k(a)}{d^2} \log \frac{k(a)}{\delta}.$$

Proof. The premise $a \ge mL + B_{\mathrm{rec}}$ gives $s(a) \le m$, and its definition guarantees $(m-s)H + sL + B_{\mathrm{rec}} \le a$. If $k = k(a)$, then $V - a \le kH$. The separation condition gives

$$\frac{V + B_{\mathrm{rec}} - a}{H - L} \le \frac{kH + B_{\mathrm{rec}}}{H - L} \le 2k.$$

Since $2k$ is an integer, $s(a) \le 2k$. For $s \ge 1$,

$$\log \frac{4(2s)(2s+1)}{\delta} \le 3 \log(s/\delta), \quad Q_f(2s) \le 97 d^{-2} \log(s/\delta).$$

Also $\log(2k/\delta) \le (5/4)\log(k/\delta)$. The fine term in Theorem S13.3 is therefore bounded by

$$388\varepsilon s d^{-2}\log(s/\delta) \le 970\varepsilon k d^{-2}\log(k/\delta),$$

which is at most the stated expression with constant 1024. The lower bound is Theorem S13.2. □

For the explicit choice $H = 16(L + B_{\mathrm{rec}})$,

$$\frac{A_{\min}}{V} \le \frac{1 + 1/m}{16} \le \frac{3}{32} < \frac{1}{8}.$$

Thus this separately calibrated family includes $V/8$ in its matching interval and extends through intermediate $a < V$. At fixed $J, M, \varepsilon, v, \delta$ and $d \downarrow 0$, it has $H, T = \Theta(d^{-2})$ and $R_0(T) = O(\log(1/d))$; the fine term is at least of order d^{-2}. The additive $R_0(T)$ remains part of the finite-sample statement and need not be negligible in joint growth regimes.

The result does not characterize $a < A_{\min}$, the interval $[V, V + B_{\mathrm{rec}})$, arbitrary overlapping arrivals, or the full unknown-schedule frontier. For $a \ge V + B_{\mathrm{rec}}$, the coarse strategy supplies an upper bound without a matching lower bound here. ABM given s uses no future arrivals; calibrating $s(a)$ from a uses the performance-family values m, H. It is therefore not an automatic future-free area-budget controller. Neither ABM's ordinary UCB guarantee nor this intermediate interval is asserted for F-CCP.

S14 A separate exact-consistency extension

This extension uses a separate, revisable multiresolution policy, rather than the fixed-checkpoint policy or F-CCP. Its reports are repeatedly recomputed, so it has no persistent reporting-failure state. This almost-sure conclusion is not a property of F-CCP. Let epochs be $E_e = \{16^e, \dots, 16^{e+1} - 1\}$ and use cumulative targets $q_e = \lceil 4^e(e+1)^2 \rceil$. At even rounds serve, by FIFO, any arrived arm below the current target, including arms formerly reported excluded. Other rounds use full-history UCB. In epoch e report from the raw intervals with radius

$$b_{i,e}(n) = \sqrt{\frac{\log\{16 i^2 (e+1)^2 n^2/\delta\}}{2n}},$$

clipped to $[0,1]$ and equal to $[0,1]$ at zero count. Intervals are recomputed rather than permanently intersected across epochs.

Hoeffding and a union over i, e, n bound the probability of any coverage failure by $\delta\zeta(2)^3/8 < \delta$. This proves all-time reporting reliability. Suppose arrivals eventually stop with $K < \infty$. For all large e, the current epoch's remaining checkpoint work is at most Kq_e, and its $2Kq_e$ physical service time is smaller than $15 \cdot 16^e$. Therefore each arm has at least q_{e-1} observations at the beginning of every sufficiently late epoch. Uniformly for $t \in E_e$ and actual $n \in [q_{e-1}, t]$, the radius is $O(\sqrt{\{e + \log i + \log(1/\delta)\}/q_{e-1}}) = o(1)$.

For every positive rational z, Hoeffding makes $\sum_n P(|\hat{r}_i(n) - r_i| > z) < \infty$. Tail union bounds

imply only finitely many deviations of size z almost surely. Taking a countable intersection over z and the finitely many final arms gives simultaneous empirical-mean convergence. If $\min_i |\max_j r_j - r_i - \varepsilon| > 0$, then all interval comparisons are eventually correct and complete, almost surely. This conclusion is not conditioned on early coverage, and early reporting errors can be corrected.

Through $T \in E_e$, service pulls number at most Kq_e, because targets are cumulative and lifetime counts never decrease. This is $O(K\sqrt{T}(1+\log T)^2)$. Apply the unconditional UCB counting bound in Section S5.2 to the remaining pulls to obtain

$$ER_T = O\{K\sqrt{T}(1+\log T)^2 + \sqrt{KT\log(2T)}\} = o(T)$$

for each eventually finite instance. Exact boundary cases are not included in the complete-classification claim.

S15 Main numerical protocol and full results

S15.1 Frozen scenarios and randomization

The original test has 27,648 records ($12 \times 24 \times 96$); the corrected ALL comparator adds 1,152. All 28,800 records are retained. Replacing, rather than pooling, the superseded adaptive ALL method yields the 24-method primary roster. A replication is a shared-stream paired experiment, not 24 independent environments. The reward-stream seed is 117110100 and the policy seed is 117120100, with the recorded scenario and replication offsets. Within-batch permutations use a separate generator with offset 50,000,000. Complete settings and exact offsets are stored in the two frozen JSON protocols.

Two close arms. Arrivals and corresponding means, grouped by physical arrival time, are
$t = 1$: $(0.493, 0.51)$.

Few arrivals. Arrivals and corresponding means, grouped by physical arrival time, are
$t = 1$: (0.565); $t = 4013$: $(0.785, 0.295)$.

Stable benchmark. Arrivals and corresponding means, grouped by physical arrival time, are
$t = 1$: $(0.735, 0.625)$; $t = 2809$: $(0.18, 0.35, 0.515)$; $t = 6803$: $(0.66, 0.26, 0.43)$.

Successive improvements. Arrivals and corresponding means, grouped by physical arrival time, are
$t = 1$: $(0.425, 0.19)$; $t = 3301$: $(0.645, 0.385, 0.51)$; $t = 7013$: $(0.825, 0.615, 0.48)$; $t = 11003$: $(0.955, 0.715, 0.88)$.

Burst. Arrivals and corresponding means, grouped by physical arrival time, are
$t = 1$: $(0.635, 0.465)$; $t = 4219$: $(0.915, 0.885, 0.81, 0.76, 0.675, 0.61, 0.57, 0.49, 0.41, 0.335, 0.225, 0.13)$.

Table S1: Frozen main-test scenarios; $\delta = .05$ in every row. The arrays below specify arrivals and arm means before within-batch random permutation.

Scenario	K_T	T	ε	η	Reward
two_close	2	12,000	0.027	0.006	Bernoulli
few_arrivals	3	12,000	0.18	0.045	Bernoulli
stable	8	12,000	0.2	0.05	Bernoulli
successive_improvements	11	14,000	0.2	0.05	Bernoulli
burst	14	14,000	0.2	0.05	Bernoulli
membership_boundary	9	14,000	0.2	0.035	Bernoulli
many_close_means	13	12,000	0.115	0.03	Bernoulli
late_improvement	8	14,000	0.2	0.05	Bernoulli
rare_rewards	9	12,000	0.018	0.005	Bernoulli
ceiling	9	12,000	0.08	0.02	Bernoulli
larger_pool	20	16,000	0.2	0.05	Bernoulli
bounded_low_variance	10	12,000	0.2	0.05	Two-point

Membership boundary. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.765, 0.545, 0.585, 0.66, 0.41, 0.19)$; $t = 5701$: $(0.735, 0.57, 0.52)$.

Many close means. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.615, 0.605, 0.59, 0.57, 0.55)$; $t = 4507$: $(0.635, 0.6, 0.565, 0.54, 0.51)$; $t = 8903$: $(0.65, 0.62, 0.585)$.

Late improvement. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.54, 0.445, 0.285)$; $t = 4201$: $(0.35, 0.51)$; $t = 11701$: $(0.935, 0.71, 0.875)$.

Rare rewards. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.035, 0.027, 0.018, 0.008)$; $t = 3713$: $(0.042, 0.024, 0.009)$; $t = 8819$: $(0.053, 0.031)$.

Ceiling. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.977, 0.954, 0.9, 0.79)$; $t = 4211$: $(0.991, 0.934, 0.825)$; $t = 9001$: $(0.965, 0.865)$.

Larger pool. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.81, 0.73, 0.615, 0.4)$; $t = 4019$: $(0.15, 0.22, 0.29, 0.36, 0.43, 0.5, 0.57, 0.64, 0.71, 0.78, 0.845, 0.89)$; $t = 10501$: $(0.94, 0.91, 0.76, 0.665)$.

Bounded low variance. Arrivals and corresponding means, grouped by physical arrival time, are

$t = 1$: $(0.705, 0.535, 0.395, 0.225)$; $t = 4703$: $(0.845, 0.625, 0.775)$; $t = 9173$: $(0.815, 0.555, 0.345)$.

For Bernoulli arms, each latent observation is an indicator with the listed mean. In the two-point scenario an arm with mean r_i takes values $r_i \pm h_i$ with equal probability, where $h_i = \min\{.045, r_i, 1 - r_i\}$. All observations lie in $[0, 1]$. Arm labels within a simultaneous batch are permuted jointly for all methods in a replication. An evaluator holds the future streams and true

means; the selection routines do not receive them. Every stored run has total actual pulls equal to its stated horizon.

S15.2 Method identity and recorded metrics

Table S2: Method identifiers and frozen controls. M22 is retained solely as a superseded controller diagnostic; M25 is its corrected primary comparator. An inactive service budget is never tested because service is disabled.

Code	Frozen method ID	λ	θ	β_R	β_C	Rate
M01	`UCB1`	0	1	∞	inactive	fixed
M02	`UCB-conservative`	0	1	∞	inactive	fixed
M03	`Growing-MOSS`	0	1	∞	inactive	fixed
M04	`MC-MOSS`	0	1	∞	inactive	fixed
M05	`MC-MOSS-clipped`	0	1	∞	inactive	fixed
M06	`Thompson`	0	1	∞	inactive	fixed
M07	`KL-UCB`	0	1	∞	inactive	fixed
M08	`F-R`	0	0.5	16	inactive	fixed
M09	`F-R-TS-exact`	0	1	∞	inactive	fixed
M10	`Cold-TS-unprotected`	0	0.5	∞	inactive	fixed
M11	`Warm-TS-guard`	0	0.5	16	inactive	fixed
M12	`CCP-original`	0.5	1	0	1	fixed
M13	`F-auto-0.1`	0.1	0.5	16	1	auto
M14	`F-auto-0.3`	0.3	0.5	16	1	auto
M15	`F-auto-0.6`	0.6	0.5	16	1	auto
M16	`F-auto-0.9`	0.9	0.5	16	1	auto
M17	`F-area`	1	0.5	16	∞	fixed
M18	`F-all-service-budgeted`	1	0.5	16	1	fixed
M19	`F-fixed-0.6`	0.6	0.5	16	1	fixed
M20	`F-auto-0.6-no-service-cap`	0.6	0.5	16	∞	auto
M21	`F-auto-0.6-TS`	0.6	1	16	1	auto
M22	`ALL-auto-0.6`	0.6	0.5	16	1	auto
M23	`ALL-area`	1	0.5	16	∞	fixed
M24	`ALL-original-budget`	0.5	1	0	1	fixed
M25	`ALL-auto-costmatched`	0.6	0.5	16	1	auto

All methods use prior parameter one for Beta action scores and the same report evidence. The default reward guard coefficient is 16 where enabled; the adaptive controller uses price .1, floor .02 and backlog scale four. The scheduler is the new certificate-cost planner for the F configurations, the retained earlier scheduler for CCP-original, and the ALL adaptation for the ALL configurations. Pure reward controls have no designated certification. Frozen null service caps in F-R and F-R-TS-exact are inactive because $\lambda = 0$; the interactive presets express the same behavior with service cap zero.

The CSV field `regret` stores the sum of current mean gaps, called cumulative dynamic regret in the manuscript. `area` is unresolved arm-time outside the scenario's η boundary band. `final_unresolved` counts only these boundary-separated unresolved arms, whereas `final_all_unresolved` includes unresolved arms inside the band. The latter is not substituted for the former. `service_pulls` counts actual certification observations; `service_attempts` also

includes rejected proposals. `service_mass` sums proposed rates over busy rounds. The two `tax` fields accumulate virtual optimism gaps for actual reward and service choices; they are distinct from `service_regret`, which the evaluator computes from true means. `label_error`, `coverage_error` and `empty_intersection` are separate run-level indicators.

S15.3 Complete costs and Monte Carlo uncertainty

Table S3: Cumulative dynamic regret, mean (Monte Carlo standard error), from 96 paired replications per cell. G-MOSS denotes Growing-MOSS and TS denotes Thompson sampling.

Scenario	UCB1	G-MOSS	MC-MOSS	TS	KL-UCB	F-R
Two close arms	75.52 (1.44)	52.16 (2.82)	52.16 (2.82)	51.37 (4.71)	62.63 (2.09)	52.07 (5.77)
Few arrivals	32.60 (0.73)	10.71 (0.40)	11.02 (0.47)	7.47 (0.36)	12.52 (0.48)	5.63 (0.69)
Stable benchmark	393.14 (3.63)	98.11 (1.82)	92.20 (1.71)	66.47 (1.89)	181.01 (2.89)	41.90 (2.76)
Successive improvements	313.99 (2.57)	105.44 (1.69)	86.29 (2.56)	57.61 (1.54)	123.14 (1.82)	53.59 (4.00)
Burst	532.82 (3.15)	140.79 (1.77)	131.95 (6.16)	80.01 (3.14)	177.09 (2.35)	68.56 (6.93)
Membership boundary	482.45 (3.95)	115.74 (2.10)	113.09 (2.11)	83.32 (2.20)	225.66 (3.15)	53.57 (5.09)
Many close means	341.20 (2.40)	188.60 (3.68)	190.01 (4.97)	202.27 (4.22)	267.46 (3.28)	169.80 (5.71)
Late improvement	364.87 (3.92)	126.45 (2.35)	101.93 (2.19)	83.89 (2.94)	196.49 (3.53)	70.56 (7.70)
Rare rewards	146.50 (0.40)	102.35 (0.96)	81.43 (1.33)	60.16 (1.50)	78.06 (0.99)	50.29 (1.77)
Ceiling	366.96 (1.70)	126.06 (1.32)	102.01 (1.53)	28.42 (1.14)	59.24 (1.24)	26.46 (2.44)
Larger pool	702.32 (3.76)	189.70 (2.39)	179.02 (6.66)	121.25 (4.10)	254.50 (3.18)	98.74 (6.89)
Bounded low variance	365.81 (0.25)	111.25 (0.17)	88.96 (0.13)	62.89 (0.22)	171.08 (0.24)	37.61 (0.17)

Table S4: All 25 stored method IDs in every scenario, including the superseded M22 diagnostic. Each row uses all 96 replications. Regret, area, service pulls and final boundary-separated unresolved counts are means (Monte Carlo standard errors); area is divided by 10^3. The two regret tail columns give the empirical 95th percentile and observed maximum. Method codes are in Table S2.

Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	max R	Service (SE)	$V_{\text{sep}}(T)$ (SE)
Two close arms						
M01	75.52 (1.44)	23.81 (0.10)	95.21	117.30	0.0 (0.0)	1.95 (0.02)
M02	80.42 (1.17)	23.78 (0.11)	95.82	115.09	0.0 (0.0)	1.91 (0.03)
M03	52.16 (2.82)	23.95 (0.05)	101.38	119.39	0.0 (0.0)	1.99 (0.01)
M04	52.16 (2.82)	23.95 (0.05)	101.38	119.39	0.0 (0.0)	1.99 (0.01)
M05	52.16 (2.82)	23.95 (0.05)	101.38	119.39	0.0 (0.0)	1.99 (0.01)
M06	51.37 (4.71)	24.00 (0.00)	147.02	200.97	0.0 (0.0)	2.00 (0.00)
M07	62.63 (2.09)	23.88 (0.08)	93.93	119.14	0.0 (0.0)	1.96 (0.02)
M08	52.07 (5.77)	24.00 (0.00)	189.44	202.21	0.0 (0.0)	2.00 (0.00)
M09	51.37 (4.71)	24.00 (0.00)	147.02	200.97	0.0 (0.0)	2.00 (0.00)
M10	53.36 (6.44)	24.00 (0.00)	200.92	203.90	0.0 (0.0)	2.00 (0.00)
M11	50.78 (5.86)	24.00 (0.00)	188.41	203.03	0.0 (0.0)	2.00 (0.00)
M12	94.21 (0.72)	23.65 (0.12)	101.49	104.14	5867.8 (40.4)	1.86 (0.04)
M13	50.91 (5.19)	23.93 (0.05)	169.66	200.40	259.9 (7.9)	1.98 (0.01)
M14	53.60 (4.23)	23.91 (0.08)	146.07	192.20	991.2 (23.9)	1.98 (0.01)
M15	67.93 (2.24)	23.75 (0.13)	105.46	124.18	2936.2 (48.6)	1.95 (0.02)
M16	96.50 (0.88)	23.66 (0.12)	103.86	104.35	7539.3 (134.5)	1.86 (0.04)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\text{sep}}(T)$ (SE)
M17	101.44 (0.29)	23.59 (0.13)	104.71	105.30	12000.0 (0.0)	1.83 (0.04)
M18	96.13 (1.01)	23.65 (0.12)	104.25	104.91	10149.4 (248.1)	1.88 (0.03)
M19	95.98 (0.82)	23.65 (0.12)	102.82	104.45	6827.1 (76.5)	1.85 (0.04)
M20	67.93 (2.24)	23.75 (0.13)	105.46	124.18	2936.2 (48.6)	1.95 (0.02)
M21	72.19 (2.07)	23.74 (0.13)	104.18	114.43	2946.5 (48.2)	1.95 (0.02)
M22	80.80 (1.21)	23.71 (0.12)	108.17	111.98	4076.7 (12.2)	1.88 (0.03)
M23	99.90 (1.27)	23.59 (0.13)	116.22	138.60	12000.0 (0.0)	1.83 (0.04)
M24	90.98 (1.13)	23.69 (0.12)	104.60	110.65	5749.1 (53.9)	1.86 (0.04)
M25	67.28 (2.23)	23.75 (0.13)	110.52	122.47	2947.5 (49.9)	1.95 (0.02)
Few arrivals						
M01	32.60 (0.73)	2.74 (0.23)	45.20	59.29	0.0 (0.0)	0.06 (0.02)
M02	49.80 (0.92)	1.13 (0.09)	64.31	77.91	0.0 (0.0)	0.00 (0.00)
M03	10.71 (0.40)	7.76 (0.14)	17.64	23.03	0.0 (0.0)	0.93 (0.03)
M04	11.02 (0.47)	8.10 (0.06)	19.36	30.13	0.0 (0.0)	0.96 (0.02)
M05	11.02 (0.47)	8.10 (0.06)	19.36	30.13	0.0 (0.0)	0.96 (0.02)
M06	7.47 (0.36)	8.44 (0.02)	13.25	23.55	0.0 (0.0)	1.00 (0.00)
M07	12.52 (0.48)	7.70 (0.15)	22.17	27.44	0.0 (0.0)	0.90 (0.03)
M08	5.63 (0.69)	10.21 (0.19)	11.35	55.42	0.0 (0.0)	1.04 (0.02)
M09	7.47 (0.36)	8.44 (0.02)	13.25	23.55	0.0 (0.0)	1.00 (0.00)
M10	5.63 (0.69)	10.21 (0.19)	11.35	55.42	0.0 (0.0)	1.04 (0.02)
M11	7.15 (0.61)	8.48 (0.03)	11.89	59.90	0.0 (0.0)	1.00 (0.00)
M12	41.56 (1.63)	0.54 (0.03)	67.68	102.41	3736.2 (61.0)	0.00 (0.00)
M13	26.60 (1.30)	3.88 (0.18)	49.05	66.61	139.9 (2.9)	0.02 (0.01)
M14	37.83 (1.41)	1.32 (0.05)	60.11	86.42	573.3 (9.6)	0.00 (0.00)
M15	97.64 (2.71)	0.60 (0.02)	134.96	165.31	1619.2 (23.8)	0.00 (0.00)
M16	128.07 (2.60)	0.55 (0.03)	149.71	164.18	3130.3 (40.0)	0.00 (0.00)
M17	490.97 (13.58)	0.36 (0.01)	592.15	609.59	7530.7 (129.5)	0.00 (0.00)
M18	128.40 (2.31)	0.55 (0.03)	149.44	161.24	4227.2 (53.6)	0.00 (0.00)
M19	124.89 (3.16)	0.55 (0.03)	150.53	165.33	3846.6 (59.2)	0.00 (0.00)
M20	97.73 (2.72)	0.60 (0.02)	134.96	165.80	1620.1 (23.8)	0.00 (0.00)
M21	97.89 (2.71)	0.60 (0.02)	134.13	165.53	1620.1 (23.8)	0.00 (0.00)
M22	125.70 (3.04)	0.56 (0.03)	148.99	163.64	1215.5 (25.3)	0.00 (0.00)
M23	547.41 (20.43)	0.37 (0.01)	745.36	824.03	7488.6 (142.2)	0.00 (0.00)
M24	124.70 (2.97)	0.56 (0.03)	148.28	162.17	2049.4 (52.9)	0.00 (0.00)
M25	129.61 (3.39)	0.60 (0.02)	157.66	172.13	1224.4 (23.7)	0.00 (0.00)
Stable benchmark						
M01	393.14 (3.63)	28.21 (0.53)	453.64	491.34	0.0 (0.0)	2.89 (0.08)
M02	553.06 (3.70)	20.89 (0.54)	618.61	642.23	0.0 (0.0)	1.75 (0.07)
M03	98.11 (1.82)	44.74 (0.29)	130.20	140.52	0.0 (0.0)	5.70 (0.05)
M04	92.20 (1.71)	45.77 (0.18)	123.23	133.81	0.0 (0.0)	5.78 (0.04)
M05	92.20 (1.71)	45.77 (0.18)	123.23	133.81	0.0 (0.0)	5.78 (0.04)
M06	66.47 (1.89)	47.19 (0.06)	98.67	132.36	0.0 (0.0)	5.98 (0.01)
M07	181.01 (2.89)	41.05 (0.43)	232.68	271.85	0.0 (0.0)	4.95 (0.07)
M08	41.90 (2.76)	54.90 (0.18)	71.87	244.59	0.0 (0.0)	6.75 (0.04)
M09	66.47 (1.89)	47.19 (0.06)	98.67	132.36	0.0 (0.0)	5.98 (0.01)
M10	42.33 (3.11)	54.91 (0.18)	71.87	285.80	0.0 (0.0)	6.75 (0.04)
M11	52.35 (2.23)	47.36 (0.05)	80.61	217.26	0.0 (0.0)	6.00 (0.00)
M12	568.11 (4.03)	13.19 (0.40)	627.96	674.18	2933.2 (26.4)	0.54 (0.05)
M13	113.49 (1.71)	39.57 (0.30)	142.58	167.49	250.1 (2.2)	4.10 (0.07)
M14	231.20 (2.90)	27.71 (0.36)	274.10	320.66	1026.9 (9.6)	2.18 (0.07)
M15	500.30 (3.93)	16.50 (0.32)	559.82	583.11	2525.0 (21.3)	0.66 (0.05)
M16	545.02 (4.44)	14.16 (0.41)	608.29	650.28	2889.6 (27.5)	0.62 (0.06)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\text{sep}}(T)$ (SE)
M17	1649.94 (13.22)	8.75 (0.21)	1866.31	2148.39	11659.3 (57.0)	0.27 (0.05)
M18	537.14 (4.41)	14.09 (0.42)	598.61	625.81	2995.7 (32.2)	0.67 (0.07)
M19	553.14 (4.44)	13.92 (0.40)	616.72	661.78	2990.2 (26.4)	0.57 (0.05)
M20	504.49 (3.74)	16.39 (0.30)	562.38	583.11	2541.5 (22.2)	0.65 (0.05)
M21	506.47 (3.93)	16.21 (0.32)	569.02	585.40	2505.6 (21.3)	0.65 (0.05)
M22	551.41 (3.21)	20.30 (0.51)	605.92	646.04	2593.9 (18.6)	1.26 (0.09)
M23	1888.56 (14.98)	11.72 (0.37)	2165.96	2327.43	11673.2 (56.7)	0.10 (0.03)
M24	559.47 (3.44)	19.35 (0.50)	615.65	653.08	2549.2 (18.3)	1.25 (0.09)
M25	552.25 (3.66)	22.00 (0.40)	610.54	619.76	2513.3 (15.0)	1.30 (0.08)
Successive improvements						
M01	313.99 (2.57)	23.99 (0.21)	358.08	376.68	0.0 (0.0)	0.89 (0.04)
M02	425.34 (2.82)	22.31 (0.21)	465.88	497.09	0.0 (0.0)	0.51 (0.05)
M03	105.44 (1.69)	33.00 (0.20)	131.07	172.34	0.0 (0.0)	2.41 (0.05)
M04	86.29 (2.56)	35.76 (0.20)	125.34	201.99	0.0 (0.0)	2.88 (0.05)
M05	86.29 (2.56)	35.76 (0.20)	125.34	201.99	0.0 (0.0)	2.88 (0.05)
M06	57.61 (1.54)	38.86 (0.21)	86.09	101.95	0.0 (0.0)	3.67 (0.08)
M07	123.14 (1.82)	30.87 (0.20)	155.47	191.40	0.0 (0.0)	2.60 (0.05)
M08	53.59 (4.00)	46.38 (0.41)	147.74	208.68	0.0 (0.0)	5.16 (0.08)
M09	57.61 (1.54)	38.86 (0.21)	86.09	101.95	0.0 (0.0)	3.67 (0.08)
M10	51.33 (4.16)	47.88 (0.43)	154.96	208.68	0.0 (0.0)	5.16 (0.08)
M11	48.33 (2.18)	42.34 (0.23)	78.92	191.51	0.0 (0.0)	4.88 (0.08)
M12	648.23 (3.80)	19.27 (0.28)	718.57	740.66	4334.9 (84.0)	0.39 (0.06)
M13	114.71 (2.28)	32.15 (0.25)	149.57	244.04	310.5 (2.7)	2.43 (0.07)
M14	274.93 (2.45)	25.15 (0.22)	311.05	329.93	1173.2 (8.1)	1.46 (0.06)
M15	599.03 (5.80)	20.53 (0.25)	674.77	684.00	3208.4 (37.5)	0.49 (0.06)
M16	608.59 (4.07)	19.81 (0.29)	682.51	730.55	4246.5 (105.9)	0.52 (0.07)
M17	1721.33 (18.21)	17.74 (0.27)	2007.20	2125.64	14000.0 (0.0)	0.15 (0.04)
M18	599.07 (4.20)	20.25 (0.29)	679.56	734.39	4545.4 (138.9)	0.62 (0.07)
M19	616.64 (4.34)	19.49 (0.29)	697.14	725.74	4091.9 (91.9)	0.47 (0.07)
M20	639.72 (7.74)	20.42 (0.24)	736.40	784.73	3392.1 (21.0)	0.40 (0.05)
M21	600.59 (5.73)	20.41 (0.25)	675.37	690.25	3213.3 (37.2)	0.45 (0.06)
M22	634.11 (3.82)	21.63 (0.25)	690.19	728.06	2717.2 (16.3)	0.81 (0.06)
M23	2148.36 (9.24)	26.47 (0.25)	2270.18	2408.93	14000.0 (0.0)	1.75 (0.07)
M24	626.52 (3.71)	21.65 (0.25)	686.17	722.11	2774.2 (17.6)	0.84 (0.07)
M25	653.71 (3.96)	21.53 (0.24)	719.23	747.38	2780.2 (15.3)	0.79 (0.06)
Burst						
M01	532.82 (3.15)	54.63 (0.62)	585.06	598.29	0.0 (0.0)	3.55 (0.10)
M02	724.78 (3.37)	46.59 (0.56)	780.92	827.89	0.0 (0.0)	2.55 (0.07)
M03	140.79 (1.77)	82.42 (0.26)	170.17	180.16	0.0 (0.0)	7.82 (0.04)
M04	131.95 (6.16)	86.39 (0.36)	262.48	395.78	0.0 (0.0)	8.14 (0.04)
M05	131.95 (6.16)	86.39 (0.36)	262.48	395.78	0.0 (0.0)	8.14 (0.04)
M06	80.01 (3.14)	88.86 (0.53)	123.07	242.47	0.0 (0.0)	8.51 (0.06)
M07	177.09 (2.35)	81.38 (0.25)	212.55	222.32	0.0 (0.0)	7.79 (0.04)
M08	68.56 (6.93)	94.57 (0.66)	208.89	357.27	0.0 (0.0)	9.22 (0.08)
M09	80.01 (3.14)	88.86 (0.53)	123.07	242.47	0.0 (0.0)	8.51 (0.06)
M10	65.18 (6.86)	96.56 (0.62)	200.54	357.27	0.0 (0.0)	9.43 (0.08)
M11	75.14 (6.38)	94.47 (0.65)	158.30	373.92	0.0 (0.0)	9.24 (0.08)
M12	594.12 (5.00)	30.06 (0.49)	680.09	765.86	3025.9 (50.9)	1.42 (0.07)
M13	169.71 (6.59)	65.37 (0.62)	264.17	487.78	302.0 (2.3)	4.42 (0.08)
M14	355.03 (4.83)	43.47 (0.41)	434.14	524.47	1235.4 (12.3)	2.24 (0.06)
M15	570.91 (5.15)	31.77 (0.47)	650.32	724.50	2567.0 (45.9)	1.47 (0.07)
M16	530.54 (5.42)	33.25 (0.55)	622.73	738.13	2485.2 (47.2)	1.82 (0.08)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\text{sep}}(T)$ (SE)
M17	1949.53 (21.51)	24.20 (0.58)	2262.61	2385.86	13980.1 (19.9)	0.43 (0.06)
M18	528.90 (5.51)	34.23 (0.60)	618.39	738.06	2528.0 (52.9)	1.96 (0.08)
M19	543.51 (5.37)	32.25 (0.53)	618.30	738.97	2583.8 (48.2)	1.72 (0.08)
M20	746.45 (7.14)	29.30 (0.41)	861.62	903.97	3514.4 (21.3)	0.73 (0.07)
M21	583.21 (4.87)	31.35 (0.49)	658.83	730.68	2531.6 (42.7)	1.49 (0.07)
M22	546.34 (5.51)	42.48 (0.55)	642.43	751.22	1999.9 (14.4)	2.71 (0.09)
M23	2757.76 (10.98)	54.26 (0.34)	2938.87	3067.95	13975.3 (24.7)	1.99 (0.08)
M24	564.09 (6.81)	42.17 (0.54)	703.97	854.85	2041.6 (14.7)	2.48 (0.07)
M25	572.09 (5.25)	42.08 (0.55)	663.31	752.59	2118.2 (13.9)	2.60 (0.08)
Membership boundary						
M01	482.45 (3.95)	35.90 (0.66)	545.01	647.49	0.0 (0.0)	2.38 (0.08)
M02	656.30 (4.14)	32.01 (0.71)	719.21	798.41	0.0 (0.0)	2.01 (0.09)
M03	115.74 (2.10)	52.29 (0.43)	149.82	165.12	0.0 (0.0)	4.06 (0.05)
M04	113.09 (2.11)	52.66 (0.42)	145.27	168.13	0.0 (0.0)	4.05 (0.05)
M05	113.09 (2.11)	52.66 (0.42)	145.27	168.13	0.0 (0.0)	4.05 (0.05)
M06	83.32 (2.20)	58.09 (0.28)	115.21	151.43	0.0 (0.0)	4.53 (0.05)
M07	225.66 (3.15)	48.34 (0.54)	272.91	310.47	0.0 (0.0)	3.59 (0.06)
M08	53.57 (5.09)	65.93 (0.33)	80.13	511.92	0.0 (0.0)	5.18 (0.04)
M09	83.32 (2.20)	58.09 (0.28)	115.21	151.43	0.0 (0.0)	4.53 (0.05)
M10	53.57 (5.09)	65.93 (0.33)	80.13	511.92	0.0 (0.0)	5.18 (0.04)
M11	61.60 (1.70)	59.63 (0.19)	90.14	112.51	0.0 (0.0)	4.85 (0.04)
M12	598.50 (7.11)	21.72 (0.59)	691.05	724.84	3790.9 (71.1)	1.16 (0.06)
M13	136.93 (2.50)	42.98 (0.56)	175.15	207.99	409.5 (6.1)	2.96 (0.07)
M14	287.33 (5.47)	31.60 (0.60)	373.44	435.27	1643.2 (23.5)	1.90 (0.07)
M15	558.76 (8.45)	23.67 (0.64)	679.28	727.25	3597.1 (70.3)	1.22 (0.07)
M16	544.13 (8.73)	22.65 (0.66)	652.86	677.14	3613.5 (77.7)	1.30 (0.07)
M17	1885.22 (19.81)	17.05 (0.30)	2167.29	2272.99	14000.0 (0.0)	0.99 (0.03)
M18	531.78 (9.14)	22.74 (0.67)	645.06	673.07	3549.1 (79.0)	1.32 (0.07)
M19	560.95 (8.37)	22.57 (0.63)	671.19	721.55	3712.4 (78.4)	1.24 (0.07)
M20	677.19 (9.67)	21.83 (0.59)	806.34	935.17	4234.9 (42.2)	1.14 (0.06)
M21	576.63 (7.66)	23.07 (0.60)	689.07	731.96	3588.4 (65.0)	1.20 (0.07)
M22	566.67 (5.49)	31.73 (0.59)	630.49	662.60	2634.7 (32.7)	2.07 (0.06)
M23	2472.24 (8.98)	27.47 (0.30)	2609.15	2719.73	14000.0 (0.0)	1.50 (0.05)
M24	604.72 (5.76)	30.60 (0.59)	669.83	703.85	2692.8 (29.4)	2.05 (0.06)
M25	589.58 (5.64)	31.66 (0.60)	660.21	673.98	2734.7 (32.5)	2.05 (0.06)
Many close means						
M01	341.20 (2.40)	77.73 (0.16)	380.00	397.55	0.0 (0.0)	7.94 (0.06)
M02	373.97 (1.93)	78.30 (0.14)	409.09	412.87	0.0 (0.0)	8.10 (0.04)
M03	188.60 (3.68)	78.39 (0.13)	256.12	297.02	0.0 (0.0)	8.28 (0.06)
M04	190.01 (4.97)	81.00 (0.03)	263.64	364.69	0.0 (0.0)	9.00 (0.00)
M05	190.01 (4.97)	81.00 (0.03)	263.64	364.69	0.0 (0.0)	9.00 (0.00)
M06	202.27 (4.22)	80.90 (0.05)	278.83	309.45	0.0 (0.0)	9.00 (0.00)
M07	267.46 (3.28)	76.42 (0.20)	319.59	358.47	0.0 (0.0)	7.57 (0.07)
M08	169.80 (5.71)	81.16 (0.02)	275.48	355.32	0.0 (0.0)	9.00 (0.00)
M09	202.27 (4.22)	80.90 (0.05)	278.83	309.45	0.0 (0.0)	9.00 (0.00)
M10	169.80 (5.71)	81.16 (0.02)	275.48	355.32	0.0 (0.0)	9.00 (0.00)
M11	169.45 (5.22)	81.14 (0.02)	258.90	319.02	0.0 (0.0)	9.00 (0.00)
M12	247.81 (3.00)	76.38 (0.17)	306.35	327.68	6000.0 (0.0)	7.74 (0.07)
M13	183.17 (4.49)	80.86 (0.05)	263.88	302.51	478.9 (3.7)	8.96 (0.02)
M14	213.12 (3.75)	78.69 (0.11)	284.13	308.33	1891.4 (9.8)	8.25 (0.07)
M15	250.25 (3.24)	76.52 (0.17)	306.40	345.45	5066.0 (14.6)	7.82 (0.06)
M16	269.14 (3.44)	76.02 (0.19)	334.15	359.81	9773.3 (34.9)	7.71 (0.06)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\text{sep}}(T)$ (SE)
M17	271.22 (3.39)	75.84 (0.20)	335.10	358.46	12000.0 (0.0)	7.69 (0.07)
M18	269.74 (3.47)	75.85 (0.19)	337.53	358.46	11723.6 (57.1)	7.69 (0.07)
M19	262.75 (3.26)	76.30 (0.18)	325.47	358.82	7161.6 (15.6)	7.77 (0.05)
M20	250.31 (3.22)	76.52 (0.17)	306.40	345.45	5068.5 (14.4)	7.82 (0.06)
M21	258.86 (3.25)	76.60 (0.17)	314.09	363.24	5120.8 (14.7)	7.89 (0.07)
M22	423.55 (3.28)	79.66 (0.12)	470.61	498.30	5771.6 (27.1)	8.64 (0.05)
M23	533.47 (1.17)	80.47 (0.09)	553.14	563.74	12000.0 (0.0)	8.85 (0.04)
M24	394.35 (2.59)	79.87 (0.11)	433.59	454.49	5865.1 (21.3)	8.60 (0.05)
M25	375.91 (3.43)	79.61 (0.12)	427.22	468.71	4529.5 (15.4)	8.48 (0.06)
Late improvement						
M01	364.87 (3.92)	21.75 (0.39)	438.35	475.11	0.0 (0.0)	0.02 (0.01)
M02	460.60 (3.76)	20.89 (0.37)	530.03	570.09	0.0 (0.0)	0.02 (0.01)
M03	126.45 (2.35)	26.97 (0.37)	155.86	210.78	0.0 (0.0)	0.24 (0.04)
M04	101.93 (2.19)	31.88 (0.26)	129.72	192.11	0.0 (0.0)	0.29 (0.05)
M05	101.93 (2.19)	31.82 (0.27)	129.72	192.11	0.0 (0.0)	0.29 (0.05)
M06	83.89 (2.94)	32.52 (0.27)	132.91	191.22	0.0 (0.0)	0.83 (0.04)
M07	196.49 (3.53)	23.71 (0.42)	255.28	319.80	0.0 (0.0)	0.39 (0.05)
M08	70.56 (7.70)	43.62 (0.41)	149.87	677.65	0.0 (0.0)	1.41 (0.07)
M09	83.89 (2.94)	32.52 (0.27)	132.91	191.22	0.0 (0.0)	0.83 (0.04)
M10	67.64 (7.76)	44.37 (0.42)	146.73	677.65	0.0 (0.0)	1.51 (0.08)
M11	63.59 (3.42)	36.66 (0.20)	122.39	230.89	0.0 (0.0)	0.86 (0.04)
M12	471.41 (5.22)	20.35 (0.31)	569.68	646.44	3731.2 (68.7)	0.00 (0.00)
M13	102.84 (3.59)	30.87 (0.32)	155.71	246.30	378.2 (4.0)	0.76 (0.04)
M14	189.22 (4.17)	26.72 (0.30)	263.93	305.29	1400.5 (14.9)	0.47 (0.05)
M15	431.85 (5.47)	21.30 (0.32)	509.62	628.04	3110.8 (48.6)	0.04 (0.02)
M16	447.54 (5.28)	19.59 (0.31)	536.71	629.55	3729.3 (60.4)	0.04 (0.02)
M17	1637.52 (8.25)	17.40 (0.25)	1776.99	1877.79	14000.0 (0.0)	0.00 (0.00)
M18	441.15 (5.30)	19.44 (0.31)	518.65	629.21	4067.5 (69.4)	0.04 (0.02)
M19	452.13 (5.37)	19.73 (0.30)	541.79	638.71	3632.0 (58.3)	0.03 (0.02)
M20	499.53 (7.11)	20.89 (0.29)	627.38	726.15	3411.4 (25.2)	0.04 (0.02)
M21	445.40 (5.29)	20.80 (0.29)	537.53	641.66	3067.3 (44.6)	0.03 (0.02)
M22	456.35 (4.27)	26.18 (0.32)	520.46	609.79	2726.6 (26.4)	0.57 (0.05)
M23	1799.48 (11.83)	22.31 (0.29)	1982.12	2155.58	13999.9 (0.1)	0.03 (0.02)
M24	452.42 (3.82)	26.31 (0.31)	512.33	581.01	2900.4 (28.6)	0.29 (0.05)
M25	470.14 (4.50)	26.27 (0.31)	540.63	646.29	2745.9 (27.2)	0.64 (0.05)
Rare rewards						
M01	146.50 (0.40)	58.77 (0.06)	152.07	154.80	0.0 (0.0)	6.88 (0.04)
M02	152.74 (0.34)	58.77 (0.06)	157.72	159.98	0.0 (0.0)	6.86 (0.04)
M03	102.35 (0.96)	58.87 (0.04)	116.96	123.30	0.0 (0.0)	6.95 (0.03)
M04	81.43 (1.33)	58.93 (0.01)	98.45	110.92	0.0 (0.0)	6.99 (0.01)
M05	81.43 (1.33)	58.93 (0.01)	98.45	110.92	0.0 (0.0)	6.99 (0.01)
M06	60.16 (1.50)	58.94 (0.00)	81.77	105.36	0.0 (0.0)	7.00 (0.00)
M07	78.06 (0.99)	58.94 (0.00)	91.04	105.58	0.0 (0.0)	7.00 (0.00)
M08	50.29 (1.77)	58.94 (0.00)	83.20	101.67	0.0 (0.0)	7.00 (0.00)
M09	60.16 (1.50)	58.94 (0.00)	81.77	105.36	0.0 (0.0)	7.00 (0.00)
M10	50.29 (1.77)	58.94 (0.00)	83.20	101.67	0.0 (0.0)	7.00 (0.00)
M11	50.71 (1.77)	58.94 (0.00)	76.95	115.20	0.0 (0.0)	7.00 (0.00)
M12	109.62 (1.62)	58.74 (0.06)	140.70	155.13	6000.0 (0.0)	6.83 (0.04)
M13	54.34 (1.65)	58.94 (0.00)	78.77	111.39	349.4 (3.2)	7.00 (0.00)
M14	66.79 (1.39)	58.94 (0.00)	85.79	108.95	1346.0 (11.7)	7.00 (0.00)
M15	94.11 (1.31)	58.84 (0.04)	114.34	130.30	3967.3 (25.2)	6.93 (0.03)
M16	116.20 (2.07)	58.66 (0.07)	155.78	168.61	9056.1 (70.0)	6.80 (0.05)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\text{sep}}(T)$ (SE)
M17	150.40 (3.29)	58.66 (0.07)	189.42	223.94	12000.0 (0.0)	6.82 (0.04)
M18	115.75 (2.00)	58.73 (0.06)	146.30	166.94	10253.8 (150.6)	6.83 (0.04)
M19	111.02 (1.83)	58.71 (0.07)	143.21	166.71	7165.0 (11.0)	6.84 (0.04)
M20	94.11 (1.31)	58.84 (0.04)	114.34	130.30	3967.3 (25.2)	6.93 (0.03)
M21	96.75 (1.26)	58.85 (0.04)	116.74	135.77	4047.1 (28.2)	6.93 (0.03)
M22	106.79 (0.84)	58.93 (0.01)	120.91	125.83	5744.9 (0.1)	6.97 (0.02)
M23	112.92 (0.84)	58.92 (0.01)	125.51	134.61	12000.0 (0.0)	6.98 (0.01)
M24	107.91 (0.80)	58.92 (0.01)	121.32	130.56	6000.0 (0.0)	6.96 (0.03)
M25	92.93 (0.99)	58.94 (0.00)	108.69	118.71	3913.6 (23.1)	6.99 (0.01)
Ceiling						
M01	366.96 (1.70)	26.73 (0.45)	392.03	410.90	0.0 (0.0)	2.72 (0.06)
M02	433.71 (1.52)	24.06 (0.40)	460.45	467.64	0.0 (0.0)	2.33 (0.05)
M03	126.06 (1.32)	38.14 (0.55)	147.43	164.24	0.0 (0.0)	4.54 (0.07)
M04	102.01 (1.53)	40.42 (0.55)	130.24	150.43	0.0 (0.0)	4.94 (0.08)
M05	102.01 (1.53)	40.42 (0.55)	130.24	150.43	0.0 (0.0)	4.94 (0.08)
M06	28.42 (1.14)	44.69 (0.41)	48.97	65.44	0.0 (0.0)	5.83 (0.04)
M07	59.24 (1.24)	40.12 (0.55)	79.39	98.80	0.0 (0.0)	5.05 (0.08)
M08	26.46 (2.44)	45.26 (0.36)	90.94	121.84	0.0 (0.0)	5.88 (0.03)
M09	28.42 (1.14)	44.69 (0.41)	48.97	65.44	0.0 (0.0)	5.83 (0.04)
M10	26.46 (2.44)	45.26 (0.36)	90.94	121.84	0.0 (0.0)	5.88 (0.03)
M11	25.56 (1.51)	44.85 (0.39)	56.71	87.01	0.0 (0.0)	5.84 (0.04)
M12	461.98 (4.44)	19.27 (0.29)	530.69	551.13	5396.3 (40.2)	1.78 (0.05)
M13	54.05 (1.54)	41.78 (0.51)	74.96	131.44	237.9 (3.7)	5.23 (0.06)
M14	115.99 (2.17)	34.77 (0.53)	151.20	184.19	1086.7 (15.8)	4.04 (0.08)
M15	242.46 (2.89)	24.48 (0.41)	290.04	315.81	3000.7 (28.0)	2.33 (0.06)
M16	462.79 (4.20)	20.01 (0.33)	529.57	540.97	6240.9 (80.2)	2.00 (0.08)
M17	717.55 (10.91)	19.26 (0.33)	889.12	956.44	11976.2 (23.8)	1.90 (0.08)
M18	475.47 (3.70)	19.90 (0.34)	529.63	556.40	7151.8 (136.1)	2.11 (0.07)
M19	446.66 (4.99)	20.22 (0.34)	526.48	538.53	5757.6 (56.7)	1.94 (0.07)
M20	242.46 (2.89)	24.48 (0.41)	290.04	315.81	3000.7 (28.0)	2.33 (0.06)
M21	245.66 (2.99)	24.28 (0.41)	297.17	317.78	2992.8 (29.2)	2.32 (0.06)
M22	435.69 (3.33)	28.14 (0.39)	487.90	526.93	4311.2 (29.4)	3.41 (0.08)
M23	1264.12 (11.26)	34.85 (0.59)	1392.01	1426.95	11982.9 (17.1)	3.05 (0.09)
M24	507.78 (3.69)	24.98 (0.51)	568.71	588.27	4594.8 (30.9)	2.53 (0.10)
M25	309.53 (2.84)	29.56 (0.38)	363.38	380.55	2873.5 (18.0)	3.83 (0.05)
Larger pool						
M01	702.32 (3.76)	90.04 (0.64)	765.93	781.70	0.0 (0.0)	4.69 (0.09)
M02	924.14 (3.73)	75.62 (0.53)	984.27	999.65	0.0 (0.0)	3.58 (0.09)
M03	189.70 (2.39)	143.00 (0.51)	228.53	251.76	0.0 (0.0)	11.57 (0.06)
M04	179.02 (6.66)	147.91 (0.46)	295.69	443.93	0.0 (0.0)	11.89 (0.06)
M05	179.02 (6.66)	147.89 (0.46)	295.69	443.93	0.0 (0.0)	11.89 (0.06)
M06	121.25 (4.10)	154.61 (0.53)	184.18	319.96	0.0 (0.0)	12.98 (0.08)
M07	254.50 (3.18)	138.02 (0.57)	304.97	333.63	0.0 (0.0)	10.85 (0.08)
M08	98.74 (6.89)	162.38 (0.34)	237.30	385.94	0.0 (0.0)	13.50 (0.06)
M09	121.25 (4.10)	154.61 (0.53)	184.18	319.96	0.0 (0.0)	12.98 (0.08)
M10	98.74 (6.89)	162.38 (0.34)	237.30	385.94	0.0 (0.0)	13.50 (0.06)
M11	109.33 (5.35)	156.42 (0.51)	201.78	388.91	0.0 (0.0)	13.21 (0.08)
M12	843.80 (5.61)	54.10 (0.48)	937.16	1001.22	4280.5 (52.1)	2.32 (0.07)
M13	242.85 (5.87)	114.09 (0.90)	348.23	523.09	401.8 (2.4)	6.66 (0.12)
M14	507.75 (5.57)	75.77 (0.60)	588.16	808.75	1705.9 (12.1)	3.43 (0.07)
M15	803.23 (6.12)	55.78 (0.53)	905.21	947.90	3706.8 (42.7)	2.45 (0.08)
M16	760.49 (5.62)	57.75 (0.58)	855.84	911.66	3963.2 (61.3)	2.92 (0.08)

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Code	Regret (SE)	Area/10^3 (SE)	$q_{.95}(R)$	$\max R$	Service (SE)	$V_{\mathrm{sep}}(T)$ (SE)
M17	2268.16 (20.24)	43.86 (0.51)	2582.93	2829.03	16000.0 (0.0)	1.39 (0.06)
M18	750.91 (5.84)	58.24 (0.57)	848.75	894.04	4063.2 (69.3)	3.05 (0.09)
M19	768.10 (5.38)	56.70 (0.52)	850.94	927.79	3865.7 (53.1)	2.78 (0.08)
M20	1035.45 (7.31)	52.15 (0.49)	1146.20	1250.60	4793.4 (19.8)	1.72 (0.08)
M21	819.01 (5.71)	54.98 (0.51)	912.98	966.43	3683.8 (41.4)	2.41 (0.07)
M22	792.37 (5.42)	71.10 (0.57)	883.25	1008.80	2886.2 (17.2)	3.79 (0.10)
M23	3203.58 (10.06)	75.46 (0.38)	3345.34	3476.06	16000.0 (0.0)	3.79 (0.07)
M24	824.89 (7.08)	70.01 (0.50)	942.13	1109.12	2957.5 (17.1)	3.36 (0.09)
M25	827.16 (5.35)	70.44 (0.57)	916.33	1006.05	3046.1 (19.9)	3.67 (0.10)
Bounded low variance						
M01	365.81 (0.25)	33.69 (0.11)	369.66	370.95	0.0 (0.0)	3.23 (0.04)
M02	482.69 (0.26)	29.65 (0.08)	486.45	487.21	0.0 (0.0)	2.65 (0.05)
M03	111.25 (0.17)	51.66 (0.09)	113.91	114.72	0.0 (0.0)	6.89 (0.03)
M04	88.96 (0.13)	53.22 (0.02)	91.23	92.04	0.0 (0.0)	7.00 (0.00)
M05	88.96 (0.13)	53.22 (0.02)	91.23	92.04	0.0 (0.0)	7.00 (0.00)
M06	62.89 (0.22)	55.87 (0.02)	66.44	68.10	0.0 (0.0)	8.00 (0.00)
M07	171.08 (0.24)	44.35 (0.26)	174.58	176.07	0.0 (0.0)	4.20 (0.04)
M08	37.61 (0.17)	59.27 (0.09)	40.45	42.39	0.0 (0.0)	8.00 (0.00)
M09	62.89 (0.22)	55.87 (0.02)	66.44	68.10	0.0 (0.0)	8.00 (0.00)
M10	37.61 (0.17)	59.27 (0.09)	40.45	42.39	0.0 (0.0)	8.00 (0.00)
M11	47.53 (0.17)	56.32 (0.03)	49.97	51.91	0.0 (0.0)	8.00 (0.00)
M12	604.51 (2.03)	23.56 (0.08)	624.41	631.74	3650.4 (43.0)	1.01 (0.01)
M13	126.57 (0.32)	45.01 (0.09)	131.04	133.09	260.4 (0.5)	6.00 (0.00)
M14	229.79 (0.59)	33.44 (0.05)	238.86	256.61	1238.4 (2.1)	4.00 (0.00)
M15	499.41 (1.61)	26.64 (0.06)	522.59	530.92	2741.4 (5.3)	1.53 (0.05)
M16	591.43 (0.89)	24.01 (0.08)	601.34	606.84	3767.6 (16.2)	1.18 (0.04)
M17	1676.97 (7.51)	18.75 (0.04)	1728.67	1752.68	12000.0 (0.0)	1.65 (0.05)
M18	583.81 (0.69)	24.35 (0.10)	592.60	599.73	4067.6 (22.3)	1.20 (0.04)
M19	596.22 (1.27)	23.86 (0.07)	607.05	610.38	3528.1 (21.6)	1.09 (0.03)
M20	607.48 (1.32)	25.98 (0.07)	625.75	633.76	3116.6 (3.1)	1.43 (0.05)
M21	508.66 (1.53)	26.14 (0.06)	530.16	533.52	2714.1 (5.8)	1.26 (0.05)
M22	570.12 (0.43)	34.05 (0.04)	577.23	580.42	2441.9 (4.6)	4.00 (0.00)
M23	2152.88 (4.12)	25.80 (0.05)	2209.92	2231.76	12000.0 (0.0)	2.01 (0.01)
M24	598.61 (0.43)	32.49 (0.02)	605.89	607.44	2481.0 (2.8)	4.00 (0.00)
M25	587.96 (0.43)	34.17 (0.02)	594.55	600.48	2512.3 (5.0)	4.00 (0.00)

Every cost summary includes report-failure and fallback paths. Standard errors describe independent repetitions within one fixed scenario; sharing streams across methods is used in the paired differences below. Equal weighting of scenario means is descriptive because horizons, arm counts and reporting tolerances differ. Empirical quantiles and maxima are finite-suite summaries, not distribution-wide probability bounds.

S15.4 Paired reward and joint-cost comparisons

Table S5: All 60 prespecified reward comparisons. Differences are F-R minus the named baseline. The interval is an individual two-sided approximate t interval; U_{60} is the one-sided Bonferroni upper endpoint at family level .05. The last column instead tests F-R minus 1.05 times the comparator minus 2. These margins are different.

Scenario	Baseline	Difference	SE	Individual 95% CI	U_{60}	U_{60}^{relaxed}
two_close	UCB1	-23.45	5.09	[-33.56, -13.35]	-6.98	-12.84
two_close	Growing-MOSS	-0.10	4.60	[-9.24, 9.04]	14.80	10.13
two_close	MC-MOSS	-0.10	4.60	[-9.24, 9.04]	14.80	10.13
two_close	Thompson	0.70	3.01	[-5.28, 6.68]	10.45	5.86
two_close	KL-UCB	-10.56	4.78	[-20.06, -1.07]	4.92	-0.31
few_arrivals	UCB1	-26.97	1.03	[-29.02, -24.92]	-23.62	-27.16
few_arrivals	Growing-MOSS	-5.09	0.79	[-6.65, -3.53]	-2.54	-5.05
few_arrivals	MC-MOSS	-5.40	0.55	[-6.50, -4.30]	-3.60	-6.14
few_arrivals	Thompson	-1.84	0.52	[-2.87, -0.82]	-0.17	-2.55
few_arrivals	KL-UCB	-6.89	0.82	[-8.52, -5.26]	-4.24	-6.82
stable	UCB1	-351.25	3.74	[-358.67, -343.82]	-339.14	-360.37
stable	Growing-MOSS	-56.21	2.56	[-61.29, -51.13]	-47.93	-54.76
stable	MC-MOSS	-50.30	2.61	[-55.49, -45.12]	-41.85	-48.39
stable	Thompson	-24.57	2.54	[-29.61, -19.54]	-16.37	-21.61
stable	KL-UCB	-139.11	3.09	[-145.25, -132.98]	-129.11	-139.89
successive_improvements	UCB1	-260.41	4.30	[-268.94, -251.87]	-246.49	-264.01
successive_improvements	Growing-MOSS	-51.85	3.99	[-59.78, -43.93]	-38.93	-46.14
successive_improvements	MC-MOSS	-32.70	3.56	[-39.76, -25.64]	-21.20	-27.43
successive_improvements	Thompson	-4.02	3.34	[-10.65, 2.60]	6.78	1.84
successive_improvements	KL-UCB	-69.55	3.97	[-77.44, -61.67]	-56.70	-64.79
burst	UCB1	-464.26	7.23	[-478.61, -449.91]	-440.86	-469.34
burst	Growing-MOSS	-72.23	6.90	[-85.92, -58.53]	-49.90	-58.91
burst	MC-MOSS	-63.39	4.25	[-71.82, -54.96]	-49.65	-58.06
burst	Thompson	-11.45	4.53	[-20.43, -2.46]	3.20	-3.11
burst	KL-UCB	-108.53	6.81	[-122.05, -95.01]	-86.49	-97.29
membership_boundary	UCB1	-428.88	5.88	[-440.55, -417.21]	-409.85	-435.63
membership_boundary	Growing-MOSS	-62.18	4.85	[-71.80, -52.56]	-46.49	-54.24
membership_boundary	MC-MOSS	-59.52	4.60	[-68.64, -50.40]	-44.65	-52.30
membership_boundary	Thompson	-29.75	4.33	[-38.36, -21.15]	-15.73	-21.93
membership_boundary	KL-UCB	-172.09	5.41	[-182.83, -161.36]	-154.59	-167.67
many_close_means	UCB1	-171.40	5.10	[-181.53, -161.27]	-154.88	-173.95
many_close_means	Growing-MOSS	-18.79	4.78	[-28.27, -9.31]	-3.34	-14.69
many_close_means	MC-MOSS	-20.21	4.12	[-28.38, -12.04]	-6.89	-18.19
many_close_means	Thompson	-32.47	2.94	[-38.30, -26.64]	-22.96	-35.22
many_close_means	KL-UCB	-97.66	5.20	[-107.98, -87.34]	-80.84	-96.13
late_improvement	UCB1	-294.31	8.02	[-310.23, -278.39]	-268.35	-288.38
late_improvement	Growing-MOSS	-55.88	7.33	[-70.44, -41.33]	-32.15	-40.47
late_improvement	MC-MOSS	-31.37	7.36	[-45.99, -16.75]	-7.54	-14.64
late_improvement	Thompson	-13.33	6.35	[-25.94, -0.73]	7.22	0.89
late_improvement	KL-UCB	-125.93	7.78	[-141.36, -110.49]	-100.76	-112.44
rare_rewards	UCB1	-96.21	1.69	[-99.56, -92.85]	-90.74	-100.07
rare_rewards	Growing-MOSS	-52.06	1.73	[-55.50, -48.62]	-46.45	-53.53
rare_rewards	MC-MOSS	-31.14	1.41	[-33.94, -28.35]	-26.58	-32.61
rare_rewards	Thompson	-9.87	1.01	[-11.88, -7.86]	-6.60	-11.59
rare_rewards	KL-UCB	-27.77	1.70	[-31.13, -24.40]	-22.28	-28.15
ceiling	UCB1	-340.50	2.86	[-346.18, -334.82]	-331.24	-351.44
ceiling	Growing-MOSS	-99.60	2.61	[-104.79, -94.42]	-91.15	-99.37
ceiling	MC-MOSS	-75.55	2.52	[-80.55, -70.55]	-67.39	-74.40
ceiling	Thompson	-1.96	1.97	[-5.86, 1.95]	4.42	0.97

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Scenario	Baseline	Difference	SE	Individual 95% CI	U_{60}	U_{60}^{relaxed}
ceiling	KL-UCB	-32.78	2.46	[-37.66, -27.90]	-24.82	-29.73
larger_pool	UCB1	-603.58	7.51	[-618.49, -588.68]	-579.28	-616.15
larger_pool	Growing-MOSS	-90.96	6.53	[-103.92, -78.00]	-69.83	-81.30
larger_pool	MC-MOSS	-80.28	5.91	[-92.01, -68.54]	-61.15	-71.64
larger_pool	Thompson	-22.52	4.06	[-30.58, -14.45]	-9.36	-17.70
larger_pool	KL-UCB	-155.76	6.95	[-169.55, -141.97]	-133.27	-147.87
bounded_low_variance	UCB1	-328.20	0.29	[-328.77, -327.62]	-327.26	-347.52
bounded_low_variance	Growing-MOSS	-73.64	0.17	[-73.98, -73.31]	-73.09	-80.64
bounded_low_variance	MC-MOSS	-51.35	0.15	[-51.65, -51.05]	-50.86	-57.30
bounded_low_variance	Thompson	-25.28	0.16	[-25.60, -24.97]	-24.77	-29.90
bounded_low_variance	KL-UCB	-133.47	0.23	[-133.92, -133.02]	-132.73	-143.26

Table S6: Prespecified joint-cost differences by scenario, candidate minus comparator. Codes identify M15=F-auto-0.6, M17=F-area, M12=CCP-original, M23=ALL-area and M25=ALL-auto-costmatched. Individual two-sided approximate t intervals use 96 paired differences and have no family adjustment.

Pair	Scenario	Metric	Difference	SE	Individual 95% CI
M15–M25	two_close	Regret	0.65	0.34	[-0.02, 1.32]
M15–M25	two_close	Area/10^3	-0.00	0.00	[-0.00, 0.00]
M15–M25	few_arrivals	Regret	-31.96	1.89	[-35.72, -28.21]
M15–M25	few_arrivals	Area/10^3	0.00	0.00	[-0.00, 0.00]
M15–M25	stable	Regret	-51.95	2.99	[-57.89, -46.01]
M15–M25	stable	Area/10^3	-5.50	0.26	[-6.00, -4.99]
M15–M25	successive_improvements	Regret	-54.68	4.13	[-62.88, -46.48]
M15–M25	successive_improvements	Area/10^3	-1.00	0.18	[-1.35, -0.65]
M15–M25	burst	Regret	-1.18	4.44	[-10.00, 7.64]
M15–M25	burst	Area/10^3	-10.31	0.53	[-11.37, -9.25]
M15–M25	membership_boundary	Regret	-30.82	6.41	[-43.53, -18.10]
M15–M25	membership_boundary	Area/10^3	-7.99	0.42	[-8.83, -7.16]
M15–M25	many_close_means	Regret	-125.66	2.57	[-130.76, -120.55]
M15–M25	many_close_means	Area/10^3	-3.09	0.12	[-3.34, -2.85]
M15–M25	late_improvement	Regret	-38.30	3.49	[-45.22, -31.38]
M15–M25	late_improvement	Area/10^3	-4.97	0.22	[-5.41, -4.54]
M15–M25	rare_rewards	Regret	1.18	0.88	[-0.57, 2.94]
M15–M25	rare_rewards	Area/10^3	-0.10	0.04	[-0.18, -0.02]
M15–M25	ceiling	Regret	-67.08	2.90	[-72.83, -61.33]
M15–M25	ceiling	Area/10^3	-5.08	0.35	[-5.78, -4.37]
M15–M25	larger_pool	Regret	-23.93	4.20	[-32.28, -15.58]
M15–M25	larger_pool	Area/10^3	-14.66	0.57	[-15.79, -13.53]
M15–M25	bounded_low_variance	Regret	-88.55	1.59	[-91.71, -85.39]
M15–M25	bounded_low_variance	Area/10^3	-7.54	0.06	[-7.65, -7.42]
M17–M23	two_close	Regret	1.54	1.05	[-0.54, 3.61]
M17–M23	two_close	Area/10^3	-0.00	0.03	[-0.06, 0.06]
M17–M23	few_arrivals	Regret	-56.44	11.55	[-79.37, -33.51]
M17–M23	few_arrivals	Area/10^3	-0.01	0.00	[-0.01, -0.00]
M17–M23	stable	Regret	-238.62	14.11	[-266.64, -210.61]
M17–M23	stable	Area/10^3	-2.97	0.28	[-3.54, -2.41]
M17–M23	successive_improvements	Regret	-427.03	18.98	[-464.71, -389.36]
M17–M23	successive_improvements	Area/10^3	-8.73	0.22	[-9.17, -8.28]
M17–M23	burst	Regret	-808.23	22.04	[-851.99, -764.47]

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Pair	Scenario	Metric	Difference	SE	Individual 95% CI
M17–M23	burst	Area/10^3	-30.06	0.49	[-31.04, -29.08]
M17–M23	membership_boundary	Regret	-587.02	20.81	[-628.33, -545.70]
M17–M23	membership_boundary	Area/10^3	-10.43	0.30	[-11.03, -9.83]
M17–M23	many_close_means	Regret	-262.25	3.46	[-269.11, -255.39]
M17–M23	many_close_means	Area/10^3	-4.63	0.18	[-4.98, -4.27]
M17–M23	late_improvement	Regret	-161.97	9.68	[-181.18, -142.76]
M17–M23	late_improvement	Area/10^3	-4.91	0.24	[-5.38, -4.43]
M17–M23	rare_rewards	Regret	37.48	3.49	[30.54, 44.42]
M17–M23	rare_rewards	Area/10^3	-0.26	0.07	[-0.40, -0.12]
M17–M23	ceiling	Regret	-546.57	13.16	[-572.70, -520.44]
M17–M23	ceiling	Area/10^3	-15.59	0.58	[-16.75, -14.43]
M17–M23	larger_pool	Regret	-935.42	21.23	[-977.56, -893.28]
M17–M23	larger_pool	Area/10^3	-31.60	0.53	[-32.66, -30.55]
M17–M23	bounded_low_variance	Regret	-475.91	8.99	[-493.75, -458.07]
M17–M23	bounded_low_variance	Area/10^3	-7.05	0.05	[-7.14, -6.96]
M15–M12	two_close	Regret	-26.28	1.90	[-30.05, -22.51]
M15–M12	two_close	Area/10^3	0.10	0.08	[-0.06, 0.26]
M15–M12	few_arrivals	Regret	56.08	2.16	[51.80, 60.36]
M15–M12	few_arrivals	Area/10^3	0.06	0.01	[0.04, 0.07]
M15–M12	stable	Regret	-67.81	3.37	[-74.50, -61.11]
M15–M12	stable	Area/10^3	3.31	0.18	[2.96, 3.66]
M15–M12	successive_improvements	Regret	-49.20	5.29	[-59.70, -38.71]
M15–M12	successive_improvements	Area/10^3	1.26	0.11	[1.04, 1.48]
M15–M12	burst	Regret	-23.20	4.09	[-31.32, -15.08]
M15–M12	burst	Area/10^3	1.72	0.21	[1.29, 2.14]
M15–M12	membership_boundary	Regret	-39.74	5.73	[-51.12, -28.37]
M15–M12	membership_boundary	Area/10^3	1.95	0.21	[1.52, 2.37]
M15–M12	many_close_means	Regret	2.44	1.10	[0.26, 4.62]
M15–M12	many_close_means	Area/10^3	0.14	0.04	[0.05, 0.23]
M15–M12	late_improvement	Regret	-39.57	3.35	[-46.21, -32.92]
M15–M12	late_improvement	Area/10^3	0.95	0.09	[0.77, 1.13]
M15–M12	rare_rewards	Regret	-15.50	1.32	[-18.12, -12.88]
M15–M12	rare_rewards	Area/10^3	0.09	0.04	[0.01, 0.18]
M15–M12	ceiling	Regret	-219.52	3.87	[-227.21, -211.84]
M15–M12	ceiling	Area/10^3	5.21	0.24	[4.73, 5.69]
M15–M12	larger_pool	Regret	-40.57	3.19	[-46.90, -34.24]
M15–M12	larger_pool	Area/10^3	1.68	0.27	[1.14, 2.22]
M15–M12	bounded_low_variance	Regret	-105.10	2.79	[-110.64, -99.57]
M15–M12	bounded_low_variance	Area/10^3	3.08	0.04	[2.99, 3.16]

The family of 60 reward comparisons was specified before testing. Of these, 59 mean differences are strictly negative, 52 one-sided family-adjusted zero-margin upper endpoints are negative, and 54 pass the separate relaxed criterion. These statements concern different targets. The individual intervals are approximate Student t intervals; no exact finite-sample Monte Carlo coverage is asserted. In particular, the positive F-R minus Thompson point estimate for two close arms remains in all tables and figures.

Table S7: Service diagnostics: proposed rate pooled over busy rounds, and equal-weight scenario means of actual successful service pulls, denied proposals and final boundary-separated unresolved counts. The proposed rate is not the realized service fraction.

Code	Proposed busy rate	Service pulls	Denials	$V_{\mathrm{sep}}(T)$
M08	0.0000	0.0	0.0	6.18
M13	0.0250	314.9	0.0	4.21
M14	0.1012	1276.0	0.0	3.08
M15	0.2749	3170.5	295.7	2.24
M16	0.6747	5036.2	3461.0	2.23
M18	1.0000	5776.9	6816.6	2.28
M17	1.0000	12595.5	0.0	1.93
M12	0.5000	4395.6	1904.6	2.09
M25	0.2665	2828.3	529.8	3.03
M23	1.0000	12593.3	0.0	2.66

S15.5 Service diagnostics and all failures

F-auto-0.6 has one erroneous-label trajectory in the ceiling scenario, associated with replication index 75. The full stored trajectory is retained. A persistent empty-intersection failure changes all later reports to all unresolved, so an observed zero label-error count alone does not show successful maintenance. The raw failure records, including their scenario, replication and method keys, accompany the release. Methods sharing latent streams are not independent reliability trials.

S16 Development, ablations, and implementation audit

S16.1 Distinct studies and prespecified controls

The first CCP study and the subsequent frontier/Shared study are development predecessors with their own frozen protocols. Their tests contain 13,440 and 12,672 records, respectively. They preserve unsuccessful candidates and negative comparisons; they are not added to the F-CCP win counts. The earlier checkpoint comparison is retained in Supplementary Section S17. Its 5,760 observations, four environments and resource measurements remain separate.

Within the F-CCP study, the three valid development rounds contain 2,640, 5,616 and 5,760 trajectories (14,016 in total). The initial development compilation failure generated no valid trajectory; its protocol and failure logs remain in the source archive. The final temperature and reward guard were selected using the recorded development rule, before the test. Development and test seeds and numerical environments differ, although their task types overlap. These archives document selection rather than provide independent repetitions of the main test.

Cold-TS-unprotected removes the reward guard while retaining temperature .5; Warm-TS-guard changes the temperature schedule. The fixed-rate variant, the uncapped certification variant and the temperature-one variant each preserve the settings stated in the frozen method map. Their full scenario results include unfavorable comparisons. These are configuration comparisons, not evidence that an untested mechanism caused the observed gain. No configuration was retuned during this manuscript reanalysis.

Table S8: Runs with at least one label error, coverage failure or empty intersection, out of 1,152 per method. Failures across methods can share the same latent reward stream and are dependent. Superseded M22 is retained as a diagnostic.

Method code	Label error	Coverage failure	Empty intersection
M01	1	7	1
M02	0	7	1
M03	1	6	1
M04	1	6	1
M05	1	6	1
M06	0	5	1
M07	1	7	1
M08	0	6	1
M09	0	5	1
M10	0	5	1
M11	0	5	1
M12	1	7	1
M13	1	6	1
M14	1	7	1
M15	1	7	1
M16	0	7	1
M17	0	8	0
M18	0	7	1
M19	1	7	1
M20	1	7	1
M21	1	7	1
M22	0	7	1
M23	0	8	0
M24	0	7	1
M25	1	7	1

S16.2 Comparator correction and numerical qualifications

The original adaptive ALL planner returned the proposed arm together with zero as its estimated sampling cost. That second return value is irrelevant when the service rate is fixed or equals one, but changes the interior adaptive controller. The corrected planner returns the gap between the largest projected mean and the proposed arm's projected mean, exactly as in F-CCP. The comparator correction was separately frozen after the audit and rerun on the same 12 scenarios and 96 seeds. No F-CCP or reward-baseline parameter changed. Both versions remain in the archive; the corrected version replaces the original in primary comparisons and is not averaged with it.

The mathematical phase rule uses exact arithmetic. The frozen implementation has 1,035 runs in which floating-point accumulation delays a service attempt by one slot at an integer boundary. Across the stored main and corrected records, the phase residual $\sum p_t -$ attempts ranges from 0.000000000000 to 1.000000002899. These records were not changed retrospectively. The guard comparisons allow 10^{-12}, planning uses the frozen regularization, and confidence inversion uses outward numerical roots. Such diagnostics do not establish exact arithmetic guarantees for all machine inputs. Changing those tolerances would define a new implementation version and require new experiments.

The interactive interface owns a NumPy Generator, whereas the frozen batch engine uses Numba

Table S9: Distinct numerical generations. Similar scenario or method names do not imply identical environments, reporting tolerances, implementations or replication counts. Development trajectories are not pooled with any frozen test.

Generation	Scenarios	Methods	Repetitions	Records
Earlier checkpoint	4	9	160	5,760
CCP development study	10	21	64	13,440
Frontier/Shared study	12	22	48	12,672
F17 original test	12	24	96	27,648
F17 corrected comparator	12	1	96	1,152

Table S10: Prespecified implementation variants, averaged equally over the 12 scenario means. M22 is superseded and is shown only to document its controller input; it is not evidence for an allocation advantage. Scenario-specific results and uncertainty for every row are in Table S4.

Code	Configuration	Regret	Area / 10^3
M08	F-R	60.73	62.22
M10	Cold-TS-unprotected	60.16	62.57
M11	Warm-TS-guard	63.46	59.22
M15	F-auto-0.6	392.99	31.70
M19	F-fixed-0.6	427.66	30.66
M20	F-auto-0.6-no-service-cap	455.24	30.93
M21	F-auto-0.6-TS	400.94	31.42
M17	F-area	1210.02	27.12
M22	ALL-auto-0.6	440.82	36.54
M25	ALL-auto-costmatched	435.68	36.72

randomization. Same-backend compatibility requires the same state, draw order and random stream; equal seeds across these backends do not imply identical paths. The batch production entry is `flexible_v2.choose_dual`; the corrected ALL driver uses `flexible_matched.choose_dual`. The earlier single-ledger development routine is retained for provenance and is not the production entry.

The interactive entry `FlexibleCCP` accepts newly arriving labels and one observed reward per physical round. It accepts no future arrival schedule, true means or evaluation horizon. Priority may change only after the pending observation, and changing it retains counts, intervals, phase and both ledgers. `None` disables a guard; zero is a zero budget. Thus changing priority to one does not remove a pre-existing finite certification cap. The classification preset instead uses no cap from initialization.

The direct implementation examines pairs of arms for comparison, planning and reporting, giving quadratic per-round pair work, together with spectral-bin loops and numerical confidence inversion. Its machine-integer checkpoints and fixed 64-bin spectrum representation are implementation limits; they do not impose corresponding finite-horizon assumptions on the mathematical model. No linear-complexity claim for the older checkpoint method is transferred to F-CCP.

S16.3 Resources and scope of reproducibility

The original F-CCP implementation audit records Python 3.13.5 on Linux x86-64, NumPy 2.3.5 and Numba 0.65.1. The environment file records pandas 2.2.3, SciPy 1.17.0 and matplotlib 3.10.8.

The archive does not identify a benchmark CPU, so we report none. Per-run timing fields and worker logs are retained, but their compilation, concurrency and timing boundaries do not define one comparable total wall-clock runtime. The Windows configuration and 13.5-second measurement in the earlier checkpoint study are not F-CCP measurements.

The source archive reports internal diagnostics for 50,000 same-backend compatibility actions, 800 future-padding checks, 126 interval inversions, 10,000 service-rate formula checks and 4,000 rounds with preference changes. Those are historical diagnostics, not independently rerun probability proofs. The manuscript revision separately performs read-only endpoint reanalysis and short API checks in a copied workspace, with their actual commands and environment saved in the release. Neither form of checking constitutes independent peer review or formal verification.

All results use simulated data. The accompanying local release includes the frozen code dependency closure, both protocols, all test records, development and failure archives, summaries, limited saved paths, source hashes and the read-only table/figure generator. A public repository or permanent accession has not been established. Each main-test scenario saves only its first four diagnostic paths on a 200-round grid; no figure presents them as a 96-replication learning curve. The two main figures are computed entirely from terminal records.

S17 Earlier checkpoint comparison

This earlier, separate experiment compares reward allocation and reliable reporting at a fixed physical budget. They ask how the service rule and the precision of its reporting evidence affect regret and unresolved area. Its four environments are not the restricted family used for the two-endpoint separation result, so they do not test its asymptotic rates.

S17.1 Protocol and competing rules

All rewards are Bernoulli. We fix $T = 20{,}000$, $\varepsilon = .30$, $\eta = .10$, and $\delta = .05$, with eight arms by the end of each run. There are 160 independent replications per environment. Within a replication, each method receives the same potential reward stream for each arm: its nth pull of arm i reveals the same $Y_{i,n}$. The shared streams do not force the methods to take the same actions. Arrivals precede the action, and the report follows the observation. All methods retain previous observations. The configuration records the arrival times, means, seeds, and policy parameters; a policy receives current arrivals and observed history, without future arrivals, true means, or the evaluation horizon.

The stable-benchmark environment starts with means $.65, .50$ and adds three batches of inferior arms. The successive-improvement environment has benchmark levels $.45, .70, .90, .98$, so a later arrival can change an unresolved membership question. The burst environment introduces six arms at one time, including a new best arm. In the fourth environment arrivals stop at pull 4037, leaving nearly 16,000 pulls for recovery. Noninitial batches arrive 36 pulls after a 100-pull admission grid; the short-period baseline is therefore not evaluated only at its most favorable alignment.

We use two FIFO configurations of Checkpoint-UCB, denoted CP-FIFO in the tables and CP in the figures, with service every second pull. The finer choice $\rho = .04$ satisfies $2\rho < \eta$. Once

every currently available arm has completed its checkpoint, all η-separated arms are resolved on the coverage event. This is not the $\rho = \eta/4 = .025$ calibration used for the worst-case area bound. The coarser choice $\rho = .12$ retains reliable labels but does not guarantee completeness for arbitrary environments at $\eta = .10$. Their eight-arm quota totals are 73,083 and 8,123, respectively. In particular, the finer configuration cannot complete all quotas within the physical budget.

Three further rules retain the coarser checkpoint report and change allocation: CP-period8 serves the FIFO queue every eighth pull; CP-confidence uses confidence-driven service every second pull; and UCB-checkpoint uses full-history UCB without designated service. The confidence-driven rule samples the widest interval among unresolved arms and the arm with largest upper endpoint, breaking ties by sample count and arrival order. It stops designated service when every report is resolved, and a completed checkpoint keeps its fixed interval. These are separately identified configurations; the FIFO guarantee is not automatically a guarantee for every allocation variant.

Three methods combine the full-prefix confidence report with a reward or sampling rule. UCB-prefix admits every arm immediately and uses the same full-history UCB index as Checkpoint-UCB. Growing-MOSS uses

$$\hat{r}_i(n) + \sqrt{\frac{\max\{\log(t/(K_t n)), 0\}}{n}},$$

with infinite index at $n = 0$. We do not transfer the fixed-horizon MOSS minimax theorem to this growing-arm rule. ALL-ε adaptation uses confidence-driven service every second pull and UCB otherwise. It retains history and can revise positive reports after arrivals. It is a natural adaptation of confidence-based all-good-arm sampling, rather than an implementation of FAREAST or $(\mathrm{ST})^2$.

Batch100-UCB admits waiting arms at pulls $1, 101, 201, \ldots$, applies UCB to admitted reward-active arms, and retires an arm from reward sampling when its upper bound falls below the largest active lower bound. Observations are retained. Its report still covers every physically arrived arm; those awaiting admission have interval $[0, 1]$. Statistical exclusion, admission, and reward retirement therefore have different meanings in this comparison.

Table S11 displays six methods with means and Monte Carlo standard errors. Figure S1 includes all nine methods, with error bars of 1.96 Monte Carlo standard errors. The complete allocation table, additional paths, configuration mapping, and regret accounting are in Appendix S17.3. The run records contain paired differences, unresolved paths on the saved time grid, service pulls, and the regret charged to service.

S17.2 Finite-budget findings and their limits

In all 640 finer-checkpoint runs, every available arm remains unresolved at every time step. Five to eight arms contribute to the final η-unresolved count because the metric exempts the boundary band; the remaining arms have not been classified. Its zero-error record therefore does not establish useful classification at this budget. The charged unresolved area ranges from 74,820 to 111,784. This is a limitation of the conservative construction at the chosen budget. The confidence-based adaptation resolves every η-separated arm by the endpoint in all replications of these four environments.

The coarser checkpoints reduce area relative to UCB alone, but can incur much larger regret.

Table S11: Earlier checkpoint comparison: final cumulative dynamic regret and cumulative η-unresolved arm-time at $T = 20{,}000$. Entries are means (Monte Carlo standard errors), from 160 paired replications. Area is in units of 10^3. The two checkpoint precisions have different completeness guarantees.

Scenario	Method	Regret	Area / 10^3
Stable benchmark	CP-FIFO ($\rho = .04$)	1360.5 (1.5)	111.78 (0.00)
	CP-FIFO ($\rho = .12$)	3629.9 (0.7)	49.72 (0.23)
	UCB-prefix	441.7 (2.3)	83.97 (0.46)
	Growing-MOSS	77.3 (0.8)	92.43 (0.00)
	ALL-ε adapt.	1367.8 (11.6)	11.31 (0.09)
	Batch100-UCB	209.3 (2.6)	85.27 (0.52)
Successive improvements	CP-FIFO ($\rho = .04$)	5297.4 (1.2)	74.82 (0.00)
	CP-FIFO ($\rho = .12$)	1194.7 (0.0)	14.74 (0.10)
	UCB-prefix	246.2 (1.7)	18.45 (0.14)
	Growing-MOSS	53.1 (0.8)	33.65 (0.11)
	ALL-ε adapt.	2905.8 (2.4)	9.53 (0.13)
	Batch100-UCB	238.8 (2.1)	22.85 (0.09)
Burst	CP-FIFO ($\rho = .04$)	2798.0 (1.6)	75.89 (0.00)
	CP-FIFO ($\rho = .12$)	3392.2 (0.2)	28.21 (0.17)
	UCB-prefix	354.1 (2.2)	41.75 (0.39)
	Growing-MOSS	69.3 (0.9)	57.03 (0.01)
	ALL-ε adapt.	2017.4 (9.8)	10.31 (0.09)
	Batch100-UCB	278.9 (2.7)	55.87 (0.15)
Arrivals stop	CP-FIFO ($\rho = .04$)	4759.0 (1.3)	89.86 (0.00)
	CP-FIFO ($\rho = .12$)	2645.9 (0.3)	25.14 (0.01)
	UCB-prefix	248.8 (1.5)	45.96 (0.56)
	Growing-MOSS	52.0 (0.6)	71.59 (0.01)
	ALL-ε adapt.	3027.5 (4.3)	12.07 (0.07)
	Batch100-UCB	214.2 (1.8)	55.28 (0.21)

Under the stable benchmark, CP-FIFO has mean regret 3629.9 and area 49,718, compared with 1367.8 and 11,306 for ALL-ε adaptation. The paired differences, CP-FIFO minus the adaptation, are 2262.1 (standard error 11.7) for regret and 38,411.7 (standard error 241.9) for area. The adaptation improves both quantities in the burst environment as well. These findings do not support a general numerical advantage of checkpoint FIFO over a strong history-retaining identification rule.

Under successive improvements, CP-FIFO occupies a different cost point: regret 1194.7 and area 14,743, compared with 2905.8 and 9525 for the adaptation. The paired regret difference is -1711.0 (standard error 2.4), and the area difference is 5217.7 (standard error 141.1). This comparison does not isolate task expiration. The adaptation uses all 10,000 designated service pulls, whereas coarse CP uses about 5,786; some remaining questions fall inside the unpenalized boundary band. The regret difference can therefore partly reflect different stopping rules for reporting work. Identifying the mechanism requires a comparison with the same boundary tolerance and service requirement. These observed cost points do not estimate an optimal frontier.

Growing-MOSS has the smallest mean regret in all four environments, between 52.0 and 77.3, while leaving substantially larger unresolved areas. Small regret alone provides little evidence of

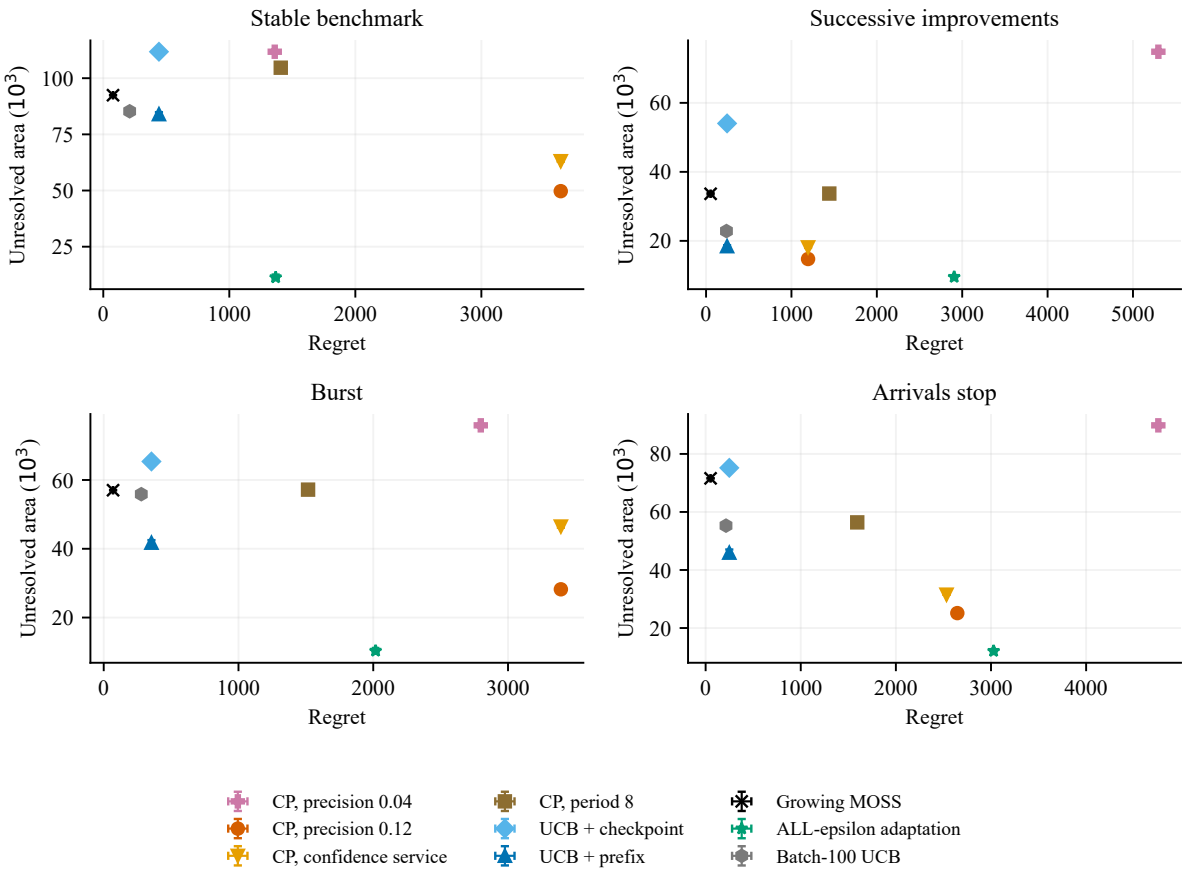


Figure S1: Regret and cumulative η-unresolved area for all nine methods. Lower values on both axes are preferable. CP denotes the Checkpoint-UCB configurations described in the text. Error bars quantify Monte Carlo uncertainty. Neither checkpoint precision nor its allocation variants uniformly dominate the confidence-based adaptation.

timely set maintenance; it does not rule out adding an effective certification layer to that allocation rule. Explicit service pays for observations of inferior arms. Reducing service from every second to every eighth pull generally reduces regret and enlarges unresolved area. Confidence-driven checkpoint service does not uniformly improve either criterion.

For Batch100-UCB, the admission-delay contributions to regret are 0, 33.92, 16.00, and 22.40 in the four environments. No historical retirement deficit occurred in these runs. Appendix S17.3 defines this decomposition and distinguishes it from learning within the reward-active set.

There were no incorrect positive or negative reports in any of the 5,760 runs. For one fixed method and environment, zero failures in 160 replications gives an exact one-sided 95% upper confidence bound below 0.0186 on its failure probability. This finite experiment does not establish uniform reliability over all environments and times. Nor do the eight-arm comparisons validate the asymptotic lower bounds, settle the critical logarithmic gap, or establish checkpoint optimality. They quantify the finite-budget cost of the conservative quotas and show why reward and maintenance must be reported together.

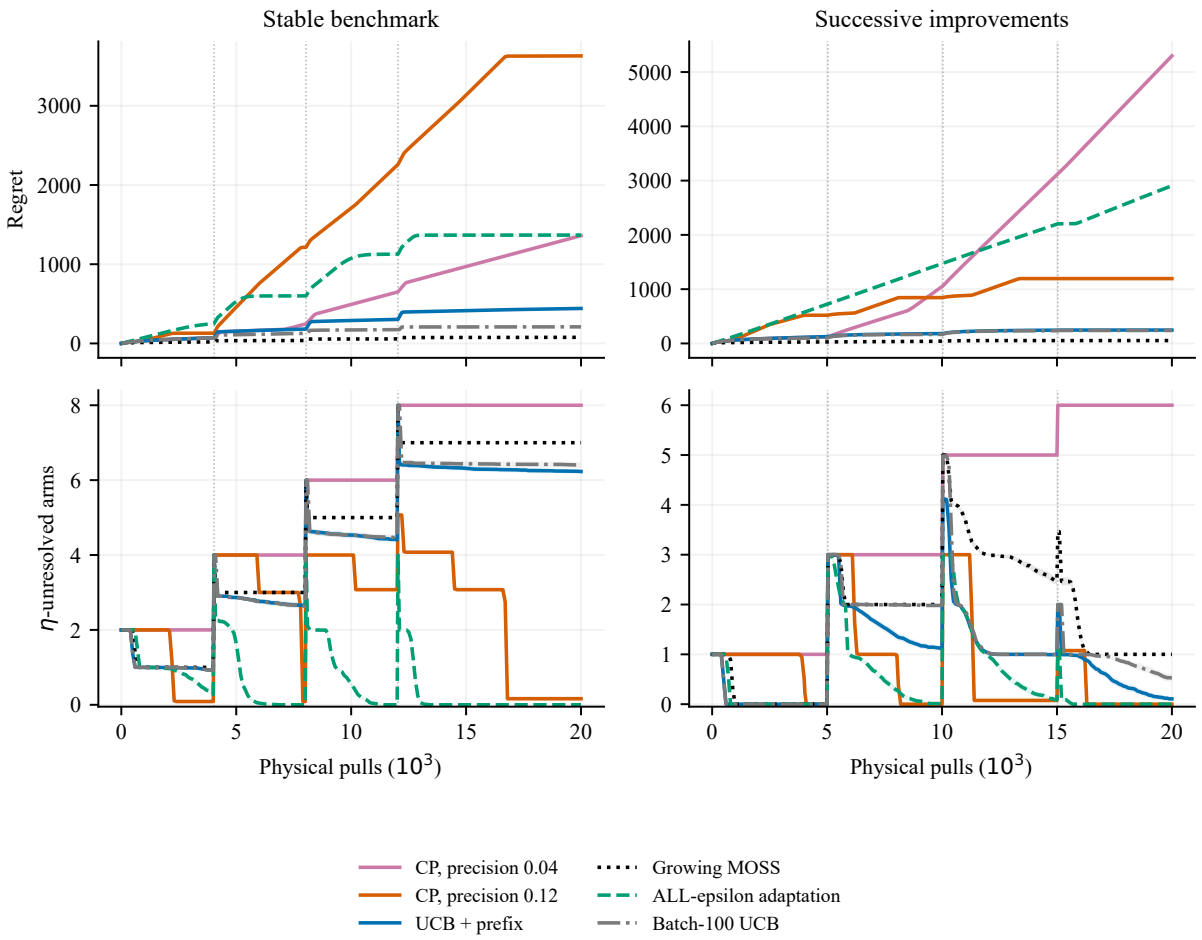


Figure S2: Mean regret and mean η-unresolved arm counts at the physical sampling clock. Bands are 1.96 Monte Carlo standard errors; vertical dotted lines mark arrivals. Stable and improving benchmarks are shown here, with the other two environments in Appendix S17.3. The six methods in Table S11 are displayed. The saved results retain all nine methods and their paths on the common observation grid.

S17.3 Configuration, records, and implementation names

This earlier numerical study has four environments, nine methods, and 160 paired replications per environment, for 5,760 runs of 20,000 physical pulls. The six methods in Table S11 appear in both path figures. The cost–area figure and Table S12 include all nine.

The local reproducibility materials contain the complete configuration, run-level records, saved paths, summary statistics, and paired differences in `experiments/results/`. The configuration gives every arrival time and mean, each method's parameters, and the seed mechanism. For replication seed s in environment index j, the generator is NumPy PCG64 initialized with `SeedSequence([s,j])`, where $s = 202610150, \ldots, 202610309$ and $j = 0, \ldots, 3$. The resulting Bernoulli potential streams are shared across methods. The policy uses only observed rewards and current arrivals. Array preallocation is simulation storage and does not expose future arms or the evaluation endpoint to a

Table S12: Allocation diagnostics. Each cell gives mean forced service pulls and mean final η-unresolved arms. UCB and MOSS have zero forced service.

Method	Stable	Improvements	Burst	Stop
CP-FIFO ($\rho = .04$)	10000, 8.00	10000, 6.00	10000, 5.00	10000, 5.00
CP-FIFO ($\rho = .12$)	7340, 0.16	5786, 0.00	5991, 0.00	5737, 0.00
CP-confidence	7144, 0.16	4653, 0.00	6092, 0.00	5322, 0.00
CP-period8	2500, 6.74	2500, 0.20	2225, 3.00	2500, 2.00
UCB-checkpoint	0, 8.00	0, 3.00	0, 4.00	0, 4.00
UCB-prefix	0, 6.23	0, 0.11	0, 1.69	0, 1.71
Growing-MOSS	0, 7.00	0, 1.00	0, 4.00	0, 4.00
ALL-ε adapt.	3696, 0.00	10000, 0.00	7794, 0.00	10000, 0.00
Batch100-UCB	0, 6.40	0, 0.53	0, 4.00	0, 2.89

policy.

The reporting configuration is named Checkpoint-UCB in the article. For reproducibility, its unchanged reference interface is `CheckpointUCBAA` in `reference_cp_ucb_aa.py`. The experiment driver is `run_experiments.py`. The data IDs `CP-FIFO-fine`, `CP-FIFO`, `CP-confidence`, and `CP-period8` retain their existing meaning. In particular, the first two are FIFO service every second pull with $\rho = .04$ and .12; the latter two change allocation as specified in this section. The remaining IDs are `UCB-checkpoint`, `UCB-prefix`, `Growing-MOSS`, `ALL-epsilon-adapt`, and `Batch100-UCB`. The ALL-ε rule is a specified retained-history adaptation, not a reproduction of FAREAST or $(\mathrm{ST})^2$.

`runs.csv` records one row per environment, seed, and method, including full-run label errors, both unresolved-area totals, service counts and regret, final per-arm counts, and the regret components below. `paths.npz` contains all nine methods' regret, unresolved counts, cumulative area, and total unresolved counts on a common grid, with every arrival included. This is a saved time grid, not a per-round action log. `summary.csv` and `paired_differences.csv` provide the numerical summaries; the two table source files are generated from these records. `figures/gen_fig_performance.py` is the plotting and tabulation entry point.

S17.4 Admission and reward-retirement accounting

Let $\mathcal{A}_t^{\mathrm{adm}} \subseteq \mathcal{A}_t$ be the arms admitted for reward selection, and let $B_t \subseteq \mathcal{A}_t^{\mathrm{adm}}$ be its nonempty reward-active subset. Assume $I_t \in B_t$. Adding and subtracting the admitted and active best means at each physical pull gives the pathwise identity

$$\begin{aligned} R_T = &\sum_{t \le T} \big(\max_{i \in B_t} r_i - r_{I_t}\big) \\ &+ \sum_{t \le T} \big(r^*(t) - \max_{i \in \mathcal{A}_t^{\mathrm{adm}}} r_i\big) + \sum_{t \le T} \big(\max_{i \in \mathcal{A}_t^{\mathrm{adm}}} r_i - \max_{i \in B_t} r_i\big). \end{aligned} \tag{S21}$$

The components are within-active learning, admission delay, and the historical-retirement deficit. They are all nonnegative. In Batch100-UCB, admission occurs at pulls $1, 101, 201, \ldots$, before action selection. Reward retirement uses the confidence comparison among currently admitted active arms

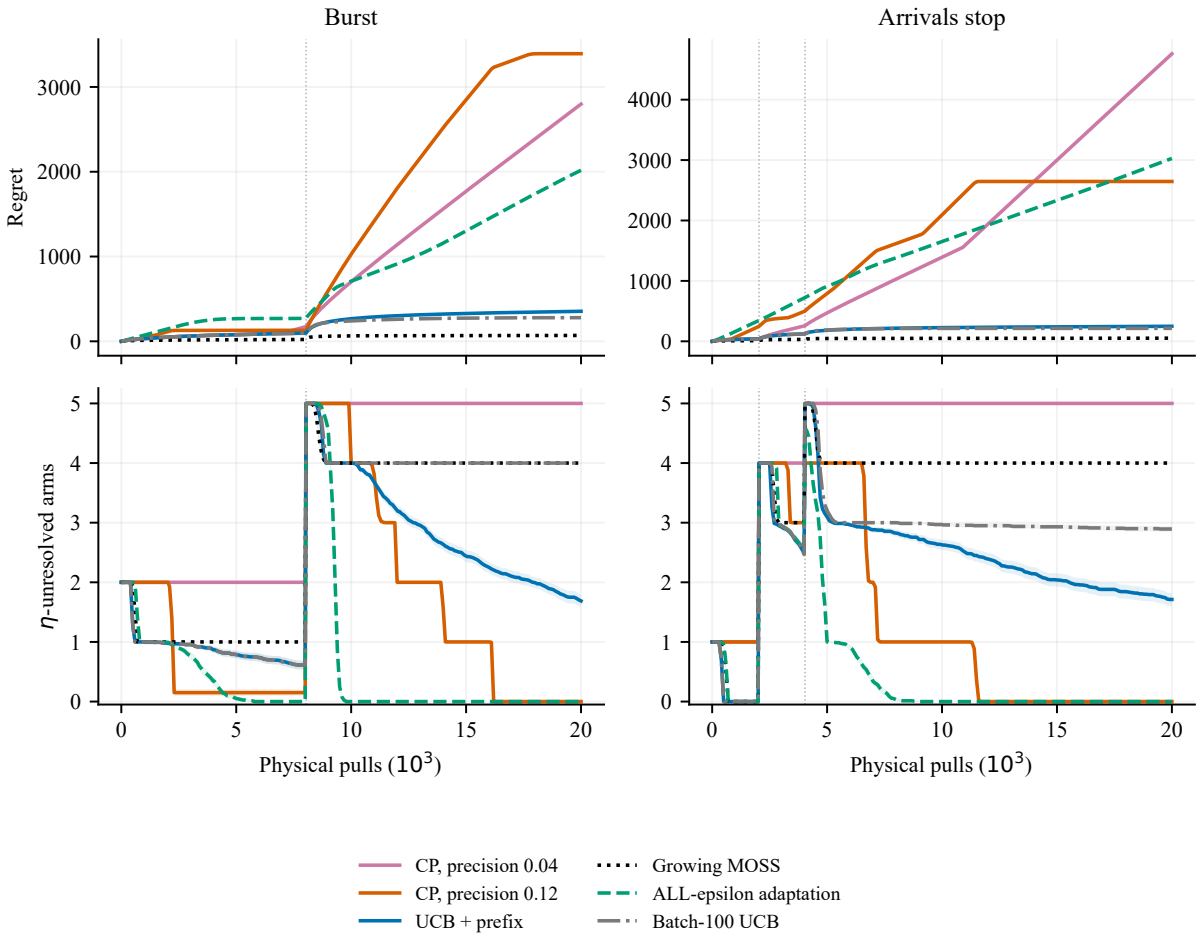


Figure S3: Additional paths for the burst and stopped-arrival environments. The six displayed policies, common potential reward streams, 160 replications, and 1.96 Monte Carlo standard error bands are the same as in Figure S2. Vertical lines mark actual arrivals. The full run-level data and saved paths retain all nine policies.

after the current observation. Reporting continues to cover all physically arrived arms, with $[0, 1]$ intervals before admission. Neither waiting for admission nor reward retirement is itself a certificate of membership or nonmembership in the near-optimal set.

For Checkpoint-UCB, every arrived arm is immediately reward-eligible, and negative reporting does not remove it from reward selection. Thus $B_t = \mathcal{A}_t^{\mathrm{adm}} = \mathcal{A}_t$, so the last two components vanish identically. The same statement applies to the other immediate-admission methods that retain every arm for reward selection.

In the Batch100-UCB records, the admission contributions are 0, 33.92, 16.00, and 22.40 for the stable, improving, burst, and stopped-arrival environments, respectively. No retirement deficit occurred. Across all 5,760 records, the largest absolute difference between regret and the sum in (S21) is below 3.1×10^{-12}.

S17.5 Verification and computational resources

The production manifest records Python 3.11.7, NumPy 1.26.4, Numba 0.59.0, Windows 10, and 13.5 seconds for production generation and recording, excluding initial compilation and subsequent plotting. The resource snapshot records 20 physical CPU cores, 28 logical cores, and 31.71 GB system memory. The small physical-round loops use compiled CPU execution. A detected GPU was not needed. Dependencies and hashes of the reference implementation, driver, configuration, and numerical records accompany the materials.

The saved production parity check compared the compiled FIFO procedure with the reference interface for 20,000 consecutive rounds, checking actions, service indicators, available-arm counts, and reports. That is a historical implementation check. During the first revision, separate short deterministic checks compare the same interfaces around arrivals, quota crossings, frozen intervals, and negative reports. That revision also recomputed the 24 legacy-table rows, the complete allocation table, and the 640 all-unresolved finer-checkpoint records from the saved data. These operations leave the simulation records and policy code unchanged. They are implementation and accounting checks, rather than new method comparisons or probability certificates.

The local materials support these resource and reproducibility descriptions. A public repository or permanent release identifier has not been verified for this version.

S17.6 Historical all-prefix report

The all-prefix reporter in this earlier comparison uses the following Hoeffding intervals. These are separate from F-CCP's mixed evidence and its persistent reporting-failure rule. With

$$b_i(n) = \sqrt{\frac{\log(8i^2n^2/\delta)}{2n}},$$

the union of all potential-prefix failures has probability at most $\delta\zeta(2)^2/4 < \delta$. Intersecting the prefix intervals with $[0, 1]$ therefore gives simultaneous coverage. On that event each endpoint differs from its mean by at most $2b_i(n)$. No conditioning on a previously selected empirical mean is used. Empty

intersections on failure paths are handled by exclusion-first reporting; the reward policy remains defined on all paths.

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